Principles of online decision-making
Prof. Matthias Grossglauser

Luigi Ieracitano

November 6, 2026

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0.1 Introduction to Online Decision-Making

0.1.1 What is online (sequential) decision-making?

Online, or sequential, decision-making refers to settings in which an *agent* repeatedly decides on an action, receives feedback from the environment, and then decides on another action, and so on. Usually, the sequence of decisions coincides with time.

Three ingredients characterize this class of problems:

- **Sequential**: the agent takes a decision, observes feedback, takes another decision, etc.
- **Decision**: a decision is *irrevocable* – once taken, it cannot be undone.
- **Uncertainty**: the environment is not known in advance, so the agent must *probe* it; the future is not known either, so the agent must rely on statistical assumptions.

[Teaching a new class] Consider a class being taught for the first time. The instructor has some prior information (student background, curriculum, goals) and a finite horizon (e.g., 14 weeks of lectures). Uncertainties include the students' background, how content is best acquired, and extraneous effects (other courses, external events). Every lecture reveals new information (questions asked, homework performance, midterm results), based on which the instructor takes actions: tweaking the lectures, and students may drop or join the class.

0.1.2 Three classes of models

This course studies three families of sequential decision problems, in increasing order of complexity.

- **Optimal stopping** (the secretary problem, covered this week): the horizon is finite and the agent must commit to a choice.
- **Bandits** (about 6 weeks): the agent repeatedly decides which *arm* (option) to play; outcomes are random, and there is an optimal arm that is not known a priori. The agent must learn the value of the different options over time.
- **Reinforcement learning** (about 6 weeks): the agent learns to interact with an environment that has a *state* (i.e., memory); outcomes of actions are random and rewards can be delayed.

Example: optimal stopping – finding a parking spot

Consider the problem of finding a parking spot close to a destination, with limited lookahead: if the destination is reached without having found a spot, this is a failure. If a spot is found, the reward equals minus the distance walked to the destination. There is a clear tradeoff: parking too early means a lot of walking, while parking too late risks missing the destination entirely (or arriving too late). How should one park optimally?

Example: multi-armed bandits

In a multi-armed bandit problem, the set of actions is called the set of *arms*. The agent takes a sequence of decisions on which action (arm) to pull, and receives a reward after each action according to some reward function. In a *stochastic* bandit, the reward is random, and is independent of everything except the choice of action taken. The name traces back to Thompson [1933].

[Clinical trial] Consider running a clinical trial without unnecessarily harming patients. The gold standard for clinical trials is to randomize: randomly give each patient the drug or a placebo. There is always a nonzero probability that the medicine does not work, or is even more harmful than the placebo – so when should we decide that this is unlikely enough that we should stop giving placebos?

The classical statistical approach fixes a confidence criterion (a p-value) beforehand, which determines the size n of the trial (how many patients are needed): more patients give higher confidence. But if the medicine works very well (or seems worse than the placebo), many people stay sick needlessly while the trial runs to completion. Ideally, we should be able to abort the trial early once the outcome becomes clear – but what if this apparent trend was only a fluke?

The question is therefore: instead of assigning treatments purely at random, how do we decide, for each new patient arriving sequentially, whether to give the drug or the placebo, using all information available so far?

Remark 1 (The importance of the horizon)**.**
Suppose every arm a generates a reward distributed like X_a, independent of everything else. How would you play if $n = 1$? How would you play if $n \to \infty$? This is analogous to portfolio choice: stocks have higher expected return but also higher volatility, while bonds have lower return and lower volatility – investment advice differs for a 20-year-old versus a 50-year-old. The horizon over which the agent needs to perform well is a key parameter of the problem.

Example: reinforcement learning – Atari Pong

Consider learning to play a computer game (e.g., Atari Pong). We do not (or cannot) program the rules of the game and then compute the optimal move analytically. Instead, an algorithm should observe and master the game as quickly as possible by playing and observing. Feedback comes as reward signals ($+1$ if the agent scores, -1 if the adversary scores). The environment has a *state* (e.g., ball position and velocity, own position), and the right action depends on this state. A key additional difficulty is that signals arrive *with delay*.

0.1.3 Probability tools we will need

Concentration results

Concentration results describe the limiting behavior of probability distributions at different "zoom levels". Suppose $X_1, \ldots, X_n$ is a sequence of i.i.d. random variables, and let $S_n = \sum_{i=1}^{n} X_i$.

- **Law of Large Numbers**: $\dfrac{S_n}{n} \to \mu$ (the average converges to the mean). This is an *asymptotic* statement ($n \to \infty$) and gives no information for a specific finite n.
- **Central Limit Theorem**: $\dfrac{S_n - n\mu}{\sqrt{n}} \to \mathcal{N}(0, \sigma^2)$, characterizing the "typical" (standard) deviation from the mean.
- **Chernoff bound**: $\mathbb{P}\left(\dfrac{S_n}{n} \geq \mu + \epsilon\right) \leq \exp(-f(\epsilon)n)$, giving a more precise tail behavior for a specific finite n, i.e., the probability of very large departures from the mean.

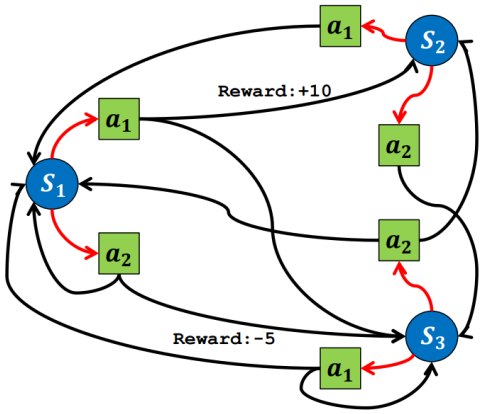
Markov chains

We are familiar with sequences of i.i.d. random variables $X_1, \ldots, X_n$ (LLN, CLT, etc.). To add "a bit of dependence", consider (X_1, X_2, X_3) with $X_1 \perp X_3 \mid X_2$: this defines a *Markov chain*. Now let the number of variables go to infinity, assume homogeneity (the probabilistic relationship between neighboring variables is the same everywhere) and discrete random variables. The chain is then described by a transition matrix P, with $P_{ij} = \mathbb{P}(X_n = j \mid X_{n-1} = i)$.

Depending on P, sets of states can have different long-term behaviors (absorbing, transient, periodic). Under certain conditions on P, the long-term behavior of the chain converges (*ergodicity*); in this ergodic regime, one is interested in the *stationary distribution* π, the normalized solution of the eigenvector problem $\pi = \pi P$.

Markov Decision Processes (MDPs)

An MDP formalizes decision-making in an environment with state. A *policy* π maps a state to an action. The goal is to maximize the reward over a long time horizon.



[Learning to walk] Model: physics of limbs, gravitation, friction with the ground. Actions: move joints. Rewards: getting to some place. Key challenges: large, continuous state and action spaces. This topic will not be covered in detail in this course.

0.1.4 Reinforcement learning vs. supervised learning

In classical supervised learning: labeled data is collected, a model is trained to give good predictions, and at test time the model is deployed and used.

In the real world, data keeps accruing during test/deployment time, which can be used to refine the model; moreover, the system might not be stationary, requiring adaptation (e.g., periodic retraining). Updating the model may also influence the future data distribution: e.g., in a recommender system, people consume the items the system recommends, so future data becomes biased toward recommended items – a potential positive feedback loop.

Reinforcement learning integrates "training" (*explore*) with "testing", i.e., using the model to accomplish something (*exploit*).

0.1.5 Key concept: the explore-exploit tradeoff

Definition 1 (Explore-exploit tradeoff)**.**
Explore: collect samples to learn about the environment/model – somewhat like the training set in supervised learning; the agent does not attempt to take good decisions, but to understand the problem.
Exploit: try to take good actions/decisions – somewhat like the test set in supervised learning; the agent does try to take good decisions, even though the model is still noisy.

This tradeoff recurs across all the examples seen so far:

- **Parking**: explore (look for a better spot) vs. exploit (park now).
- **Clinical trial**: explore (give a random treatment) vs. exploit (give the treatment believed best).
- **Pong**: explore (random move) vs. exploit (best-guess move).

This course develops probabilistic models for such situations, algorithms to handle them, and analysis of the optimality of these algorithms.

0.1.6 Prerequisites

This is essentially a probability class; introductory probability is enough, although mathematical maturity helps. It is not a bad idea to take *Modèles stochastiques* in parallel, since there is some overlap (mostly Markov chains) – however, this course (PODM) is self-contained. Solid prerequisites in basic probability are required.

0.2 Optimal Stopping: the Secretary Problem

0.2.1 Problem statement

Definition 2 (Secretary problem, a.k.a. "apartment hunting problem")**.**
You are looking for an apartment to rent on a hot market: you must take it or leave it forever, on the spot. You know the number of options in advance: n apartments will be viewed, and you need shelter after the last one. Every apartment has a quality level x_i, drawn i.i.d. from some *unknown* law – equivalently, one may simply assume that the apartments are viewed in a uniformly random order, with no information about the underlying probability law. The only information available at step i is the values $x_1, \dots, x_i$ seen so far (or, more precisely, their relative ranking).
Decisions are irrevocable: sign the lease immediately, or that apartment is gone forever. The goal is to maximize the probability of selecting the overall best apartment among the n.

This model captures many practical situations with the same flavor, e.g.:

- Resource allocation: how much effort to spend searching and evaluating before hiring a candidate or selecting a vendor.
- Online auctions: updating one's own bid upon observing other bids.
- Acquiring friendships/social contacts over a lifetime.

The problem is also known under other names: the *marriage problem*, the *sultan's dowry*, or the *googol game*.

0.2.2 Optimal policy: threshold structure

Theorem 1 (Form of the optimal policy)**.**
The optimal policy is necessarily of the following form, for some constant threshold k:
(1) If x_i is not *a new maximum (relative to $x_1, \ldots, x_{i-1}$), continue.*
(2) If x_i is *a new maximum, stop (commit) if $i \geq k$; otherwise continue.*

Justification. **Why (1)?** Stopping on a value that is not a new maximum is guaranteed to fail (the true best has already appeared, or will appear later and we have already committed to a non-optimal value) – this is never a good idea.

Why (2)? We claim the optimal policy continues before k and stops at or after k upon seeing a new maximum, i.e., the policy is *monotonic*: if we decide to stop upon seeing a new maximum-so-far at position i, we must also decide to stop upon seeing a new maximum-so-far at any $j > i$.

To see this, condition on observing the maximum over $x_1, \ldots, x_i$ seen so far (i.e., $x_i > \max(x_1, \ldots, x_{i-1})$). The probability of success if we commit to x_i is exactly i/n: the overall maximum is uniformly distributed over $\{1, \ldots, n\}$, so this conditional probability of success is *increasing* in i. On the other hand, the (unconditional) probability of seeing a new maximum-so-far at step i is *decreasing* in i. Hence, if the tradeoff (value of stopping vs. probability of the opportunity arising) is favorable at i, it is even more favorable at any $j > i$. □

Small cases: $n = 1, 2, 3$

- $n = 1$: nothing to do, commit to the only option.
- $n = 2$: the best is first with probability $1/2$ and second with probability $1/2$. Either commit to the first, or (implicitly) commit to the second (=last) if not; $\mathbb{P}(\text{success}) = 1/2$.
- $n = 3$: the smallest n for which we actually "make a decision". With $k = 2$, we use the comparison between the first and second position to decide whether to take the second or the third apartment.

The table below enumerates all $3! = 6$ orderings of the ranks $\{1, 2, 3\}$ (1 = best) and the resulting success probability for each candidate threshold $k \in \{1, 2, 3\}$.

	$k = 1$	$k = 2$	$k = 3$
	1 2 3	1 2 3	1 2 3
	1 3 2	1 3 2	1 3 2
	2 1 3	2 1 3	2 1 3
	2 3 1	2 3 1	2 3 1
	3 1 2	3 1 2	3 1 2
	3 2 1	3 2 1	3 2 1
$\mathbb{P}(\text{success})$	1/3	1/2	1/3

The threshold $k = 2$ is optimal for $n = 3$, giving $\mathbb{P}(\text{success}) = 1/2$.

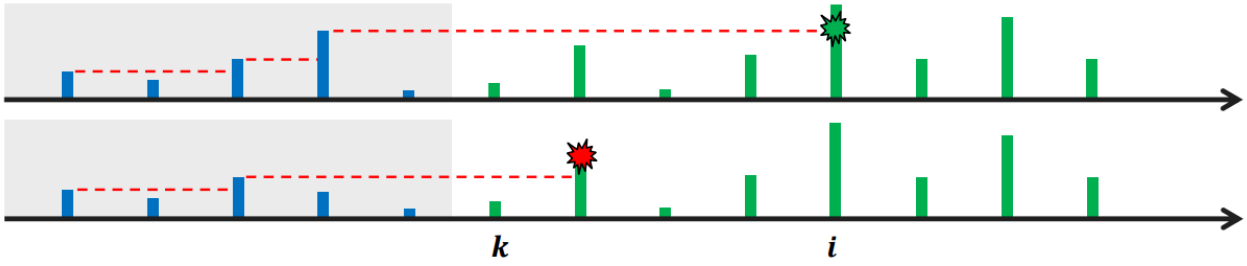
0.2.3 General optimal threshold k

Let U denote the index (position) of the true overall maximum; note that $U \sim \text{Unif}\{1, \dots, n\}$. Let Z denote the output of the policy (its guess for U), and let E be the event "success", i.e., the policy chooses index i and i is indeed the best. Then:

$$\mathbb{P}(E) = \sum_{i=k}^{n} \mathbb{P}(Z = i \cap U = i) = \sum_{i=k}^{n} \mathbb{P}(Z = i \mid U = i)\, \mathbb{P}(U = i), \qquad \mathbb{P}(U = i) = \frac{1}{n}.$$

Computing $\mathbb{P}(Z = i \mid U = i)$. The event $\{Z = i \mid U = i\}$ holds when i is the overall best (so rule (2) is triggered with certainty at step i: success), *provided* the policy has not stopped earlier. This requires that all previous best-so-far values occur *before* k. Conditional on $\{U = i\}$, where is the best value among $x_1, \dots, x_{i-1}$ (not necessarily the second-best overall, which could occur after i – just the best *so far*)? All we know is that it is not at i, so its position is uniformly distributed over $\{1, \dots, i-1\}$. Hence:

$$\mathbb{P}(Z = i \mid U = i) = \mathbb{P}\big(\text{Unif}(i-1) \in \{k-1\}\big) \cdot (k-1) = \frac{k-1}{i-1}.$$



Finite n

Combining the two pieces:

$$\mathbb{P}(E) = \sum_{i=k}^{n} \frac{k-1}{i-1} \cdot \frac{1}{n} = \frac{k-1}{n} \sum_{i=k}^{n} \frac{1}{i-1} = \frac{k-1}{n}\big(H_{n-1} - H_{k-2}\big),$$

where $H_m = \sum_{j=1}^{m} \frac{1}{j}$ is the m-th harmonic number. Optimizing over k for each n gives:

n	1	2	3	4	5	6	7	8	9	10	20	30	40	50	100
k	1	1	2	2	3	3	3	4	4	4	7	11	15	18	37
$\mathbb{P}(E)$	1	0.5	0.5	0.458	0.433	0.428	0.414	0.410	0.406	0.399	0.384	0.379	0.376	0.374	0.371

Remarkably, $\mathbb{P}(E)$ appears to converge to a positive constant as $n \to \infty$. Also note that neither k nor $\mathbb{P}(E)$ depends on the distribution of the x_i – only on n.

Large n: the $1/e$ rule

Set $x = k/n$. As $n \to \infty$, the sum becomes an integral:

$$\mathbb{P}(E) \xrightarrow{n\to\infty} x \int_{y=x}^{1} \frac{1}{y}\, dy = -x \ln x.$$

Optimizing over x: setting $\dfrac{d\,\mathbb{P}(E)}{dx} = 0$ gives $x = e^{-1} \approx 37\%$.

Theorem 2 (The $1/e$ rule)**.**
As $n \to \infty$, the optimal threshold is $k \approx n/e$, and the optimal success probability converges to $1/e \approx 0.368$. Informally: "look at the first 37% of the candidates without committing, then leap at the next candidate that beats everything seen so far."

0.2.4 Random order vs. i.i.d. random variables

An important assumption underlying the whole analysis is that we have *no information about future values*. Two ways to phrase this assumption, and why both matter:

- **I.i.d. random variables**: the player does not know the underlying distribution. *Counterexample if this fails*: if we knew the distribution of the x_i, we could directly compute the probability that x_i is the overall maximum; if this probability is already large, there would be no need to wait until k.
- **Rank order**: only the partial order (ranking) of the items seen so far (up to i) is available. *Counterexample if this fails*: if x_i revealed its total rank among all n items, we could simply wait for an item ranked #1 at any time and stop immediately.

The need to "pay" with some samples in order to discover the distribution and correctly judge future samples is what makes the problem interesting, and is baked into these assumptions.

0.2.5 Does the goal matter? Success probability vs. expected value

The optimal policy derived above is specific to the particular criterion adopted, namely maximizing $\mathbb{P}(Z = U)$: it never accepts anything below the maximum-so-far.

Suppose instead we wanted to maximize the *expected value* of the chosen apartment, $\mathbb{E}[x_Z]$ (i.e., rent a *good* apartment, without it being catastrophic if it is not the very best). Does the optimal policy change?

Remark 2.
Yes, it does. *Counterexample*: suppose we are close to n and we observe the second-best value seen so far – it is very unlikely that this can still be beaten, so we should take it. In fact, under this new goal, we should accept anything above the *median* of the values seen so far near the end of the sequence. This policy guarantees eventual failure under the previous (probability-of-best) criterion, but performs much better in terms of expected score.

This illustrates a general lesson: the goal (objective function) fundamentally shapes what the optimal policy looks like.

0.2.6 Explore-exploit tradeoff, revisited

The explore-exploit tradeoff introduced earlier is the key underlying concept of this course:

- **Explore**: make potentially bad decisions (with respect to some specified goal) in order to estimate parameters of the system, helping make better decisions in the future.
- **Exploit**: based on these estimates, make good decisions with respect to the specified goal.

In the secretary problem, explore and exploit are *sharply separated*: there are two phases, with an a-priori threshold (k) at which the policy switches from one to the other. In bandit problems and in reinforcement learning, exploration and exploitation will instead meld together more gradually:

- **Bandits**: gradually improve estimates of the rewards of different arms, stop playing some arms earlier than others, and gradually optimize the rewards obtained.
- **Reinforcement learning**: gradually exclude states and actions that are unlikely or costly to lead to rewards, driving the system towards good states.

0.3 Multi-Armed Bandits

0.3.1 A bit of history and motivation

The name "bandit" comes from the slot machine: pulling an arm yields a reward (minus the investment), and $\mathbb{E}[\text{payoff}] < 0$ – hence a "bandit". In the old days, a slot machine used a mechanical random number generator; different machines ("one-armed bandits") can behave subtly differently. The fundamental question is: *which arm to pull*? In particular, if we play for a long time but know nothing at the beginning, how do we use the information gathered along the way optimally?

The field was initiated by William R. Thompson [*Biometrika*, 1933], motivated by clinical trials: sequential decision-making under uncertainty, aiming to minimize unnecessary harm to patients. Bandit models have many applications in online services, where there is a lot of data, a lot of decisions to make, and human intervention is generally not feasible.

Remark 3.
The objective of this course is to extract *principles*: clean "vignettes" of often messier real-world situations, chosen for their mathematical tractability. The algorithms studied are not necessarily directly applicable to all real-world situations, but the underlying principles are tools that guide the design of real-world solutions.

Real-world examples

- **A/B-testing web interfaces**: automate design choices for web/online interfaces; reward = user traffic, activity, engagement time, etc.
- **Ad placement**: place ads at the right location and show the right ad at the appropriate time; reward = ad revenue.
- **News recommendations**: serve news stories of interest to every user; reward = clicks, engagement time, etc.
- **Dynamic pricing**: optimize price over time so as to sell the full inventory while maximizing revenue; reward = total revenue.
- **Monte Carlo Tree Search (MCTS)**: used in recent successes of ML approaches to game playing (Atari games, chess, Go, …). A tree captures positions already explored; the algorithm generates a path down the known tree, expands a leaf, plays randomly to the end, and propagates the reward back up. Conceptually, every node contains a bandit algorithm that learns rewards "further down" based on the choice of child node. Over time, higher-reward paths get chosen more often, while uncertainty is still handled explicitly.

Back to the clinical trial example: what might go wrong?

With a purely randomized test, each patient randomly receives the drug or the placebo. At the end of the trial, effectiveness is assessed, e.g., via a hypothesis test: fix a p-value (e.g., 0.05) and a criterion (e.g., probability of recovery twice as high as without treatment); this determines the necessary number of samples. The trial is carried out, then evaluated – success or failure.

The issue: nothing is done *during* the trial. What if, after only a short time, the data already show a strongly imbalanced outcome between the two arms?

This motivates a *sequential* approach, where decisions can be updated as data comes in, rather than fixed in advance.

Reminder: p-value

A hypothesis test is a probabilistic proof by contradiction. We fix a **null hypothesis** H_0 (the status quo, e.g. "the new treatment is no better than the old one") and an **alternative hypothesis** H_1 (what we want to show, e.g. "the new treatment is better").
Given a test statistic T (e.g. the number of cured patients) with observed value t_{obs}, the **p-value** is the probability, *assuming* H_0 *is true*, of observing a result at least as extreme as the one we observed:

$$p = \mathbb{P}\left(T \geq t_{\text{obs}} \mid H_0\right)$$

Decision rule. We fix a significance level α *before* looking at the data (typically $\alpha = 0.05$):

- If $p \leq \alpha$: we **reject** H_0, and the result is statistically significant.
- If $p > \alpha$: we **fail to reject** H_0. This does *not* mean that H_0 is true, only that the data does not provide enough evidence against it.

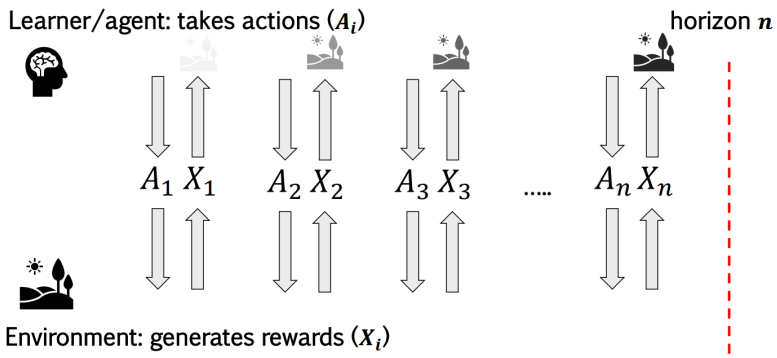
Errors.

- **Type I error** (false positive): rejecting H_0 while it is true. By construction, this happens with probability α.
- **Type II error** (false negative): not rejecting H_0 while it is false. This happens with probability β, and $1 - \beta$ is called the **power** of the test.

0.3.2 Stochastic bandits: definitions, assumptions, vocabulary

The basic interaction loop is as follows: the **learner/agent** takes actions A_i, and the **environment** generates rewards X_i, over a **horizon** n:

$$A_1, X_1, A_2, X_2, A_3, X_3, \ldots, A_n, X_n.$$



Definition 3 (History)**.**
The history up to time t is $H_t = (A_1, X_1, A_2, X_2, \ldots, A_t, X_t)$. It contains all the information the learner has so far about the environment.

Definition 4 (Policy)**.**
A policy π is a map from history to action: $A_t = \pi(H_{t-1})$. More specifically, the action can be random: π_t is a conditional distribution over A_t given $A_1, X_1, A_2, X_2, \ldots, A_{t-1}, X_{t-1}$ (but it cannot depend on the future).

Definition 5 (Environment)**.**
The environment is a map from $A_1, X_1, A_2, X_2, \ldots, A_t$ to X_t (possibly random).

Goal. Maximize the cumulative reward up to the horizon n: $S_n = \sum_{i=1}^{n} X_i$.

Remark 4 (Key assumption)**.**
The learner does *not* know the environment at the beginning, although it might know some *class* of possible environments.

Definition 6 (Regret)**.**
The regret of the learner relative to a policy π^* is the expected reward under π^* minus the expected reward achieved by the learner – the "cost of ignorance". Usually, π^* is the optimal policy for the environment, i.e., a policy that knows the environment perfectly.

0.3.3 Unstructured vs. structured bandits

Definition 7 (Unstructured bandit)**.**
Intuitively, an unstructured bandit is one where pulling one arm provides no information on the reward distribution of other arms. Formally, the environment class is

$$\mathcal{E} = \{\nu = (P_a : a \in \mathcal{A}) : P_a \in \mathcal{M}_a \text{ for all } a \in \mathcal{A}\},$$

where $\mathcal{M}_a$ is a family (set) of possible distributions P_a for action a. In other words, $\mathcal{E} = \bigtimes_{a\in\mathcal{A}} \mathcal{M}_a$ is a *product space*: any combination of elements from the $\mathcal{M}_a$'s is possible. This ensures that knowing anything about P_a provides no information about $P_{a'}$ for $a' \neq a$.

Remark 5.
The objects in this definition live at different levels:

- P_a is **one** distribution: the true (unknown) reward distribution of arm a, e.g. $\mathrm{Ber}(0.7)$.
- $\mathcal{M}_a$ is a **set** of distributions: all candidates that P_a could be, i.e. what we assume about arm a before playing. For instance, $\mathcal{M}_a = \{\mathrm{Ber}(p) : p \in [0,1]\}$ (binary rewards), $\mathcal{M}_a = \{\mathcal{N}(\mu, 1) : \mu \in \mathbb{R}\}$ (Gaussian with known variance), or all distributions supported on $[0,1]$ (bounded rewards).
- $\nu = (P_a : a \in \mathcal{A})$ is a **tuple** of distributions, one per arm: it is *one* specific environment, e.g. $\nu = (\mathrm{Ber}(0.7), \mathrm{Ber}(0.4))$.
- $\mathcal{E}$ is a **set** of environments: all those consistent with our assumptions.

The learner does not know ν (this is what it must learn), it only knows that $\nu \in \mathcal{E}$. Learning amounts to progressively narrowing down the candidates in each $\mathcal{M}_a$ until P_a is identified well enough. The guarantees of an algorithm hold only if the true P_a indeed belongs to the assumed $\mathcal{M}_a$.

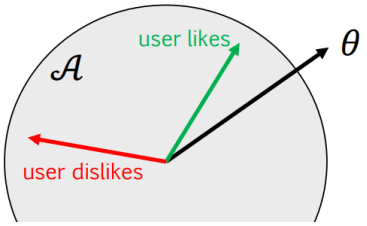
Examples of unstructured bandit classes

Name	Parametric	Definition
Bernoulli	Yes	$\mathcal{B}(\mu_i)$, $\mu_i \in [0,1]$
Uniform	Yes	$\mathcal{U}(a_i, b_i)$, $a_i \leq b_i \in \mathbb{R}$
Gaussian, known variance	Yes	$\mathcal{N}(\mu_i, \sigma^2)$
Gaussian, unknown variance	Yes	$\mathcal{N}(\mu_i, \sigma_i^2)$
Finite variance	No	$\mathbb{V}(P_i) \leq \sigma^2$
Bounded support	No	$\mathrm{Supp}(P_i) \subseteq [a, b]$
Subgaussian	No	P_i is σ-subgaussian

Definition 8 (Structured bandit)**.**
Intuitively, a structured bandit has an environment class where the reward of one arm provides information on the reward distribution of other arms.

Examples.

- *"Complementary Bernoulli" bandit*: $\mathcal{A} = \{1, 2\}$, $\mathcal{E} = \{(\mathcal{B}(\theta), \mathcal{B}(1-\theta)) : \theta \in [0, 1]\}$. One could learn everything by playing only one arm forever. Example: two mutually exclusive options to click on a webpage.
- *Stochastic linear bandit*: $\mathcal{A} \subset \mathbb{R}^d$ (an uncountably infinite action space), $\theta \in \mathbb{R}^d$, $\mathcal{E} = \{\nu_\theta : \theta \in \mathbb{R}^d\}$ with $\nu_\theta = \{\mathcal{N}(\langle a, \theta \rangle, 1) : a \in \mathcal{A}\}$. Example: a recommender system, where θ captures unknown user preferences and a captures the feature vector of an item; $\langle a, \theta \rangle = a^T\theta$ is the inner product measuring how much the user likes item a.



Thought experiment: the case of no noise

In this idealized setting, we assume that pulling an arm reveals its *exact* expected reward, rather than a noisy sample drawn around it: $r_t = \mu_a$ instead of $r_t = \mu_a + \varepsilon_t$. In other words, a single pull gives perfect information about the arm's true mean, with no randomness to average out. This is of course unrealistic (real feedback is always noisy), but it is a useful thought experiment to understand how much easier the bandit problem becomes once uncertainty is removed.

- **Unstructured bandit**: try each of the k arms once, observe a (deterministic) reward, then play the best arm forever – no more regret is incurred afterwards; the cumulative regret is constant.
- **Structured bandit**: e.g., in the complementary Bernoulli case, there is one permanent winner and one permanent loser, so pulling once suffices to be perfect forever. In the stochastic linear case, getting exact inner products (i.e., projections) requires that the set of actions spans $\mathbb{R}^d$; then d pulls suffice to estimate θ perfectly.

This noiseless setting is not very interesting, nor realistic in most practical settings. **The main topic in bandit theory is dealing with noise in the reward**, which gives rise to a non-zero expected regret "forever".

0.3.4 Review of some probability notions

CDF, CCDF, density

For a continuous random variable A:

$$F_A(a) = \mathbb{P}(A \leq a), \qquad F'_A(a) = \mathbb{P}(A > a) = 1 - F_A(a) \text{ (CCDF)}, \qquad f_A(a) = \frac{dF_A(a)}{da} \geq 0.$$

Independence

Two random variables are independent iff $F_{A,B}(a,b) = F_A(a)F_B(b)$ for all a, b: information on one variable provides no information on the other.

A set of random variables is **pairwise independent** iff every pair is independent. A set of random variables is **mutually independent** iff $F_{A,B,C}(a,b,c) = F_A(a)F_B(b)F_C(c)$ (for three variables; analogously for more): any information on all the other variables provides no information on a target variable.

Remark 6.
Mutual independence is *strictly stronger* than pairwise independence!

Let $A, B \sim \mathcal{B}(1/2)$ i.i.d., and let $C = A \oplus B$ (XOR). Contrast the two joint laws below (left: A, B, C constructed this way; right: $A, B, C \sim \mathcal{B}(1/2)$ i.i.d. and mutually independent):

ABC	$\mathbb{P}$
000	1/4
001	0
010	0
011	1/4
100	0
101	1/4
110	1/4
111	0

ABC	$\mathbb{P}$
000	1/8
001	1/8
010	1/8
011	1/8
100	1/8
101	1/8
110	1/8
111	1/8

In the XOR construction, every *pair* (A, B), (A, C), (B, C) is uniform on $\{0,1\}^2$ with all four combinations having probability $1/4$ – i.e., every pair is independent. However, the three variables are *not* mutually independent: knowing any two of them determines the third exactly.

Conditional independence

Definition 9.
A and B are conditionally independent given C iff, in the discrete case,

$$\mathbb{P}(A \le a, B \le b \mid C = c) = \mathbb{P}(A \le a \mid C = c)\,\mathbb{P}(B \le b \mid C = c) \quad \text{for all } a, b, c \ (\mathbb{P}(C = c) > 0),$$

or, in the continuous case, $f_{AB|C}(a, b \mid c) = f_{A|C}(a \mid c)\, f_{B|C}(b \mid c)$ for all a, b, c (with $f_C(c) > 0$).

Intuitively: if C is known, then A provides no further information about B (and vice versa). The notions of pairwise and mutual independence also extend naturally to conditional independence when there are more than two variables.

Expectation

$$\mathbb{E}[X] = \mu_X = \begin{cases} \sum_{x_i} x_i\, \mathbb{P}(X = x_i) & \text{discrete} \\ \int x f_X(x)\, dx & \text{continuous} \end{cases}$$

Some important properties:

- **Linearity**: $\mathbb{E}[A + B] = \mathbb{E}[A] + \mathbb{E}[B]$ for any A, B.
- **Product**: $\mathbb{E}[AB] = \mathbb{E}[A]\,\mathbb{E}[B]$ if $A \perp B$, but this is *not* true in general.
- For $A \ge 0$: $\mathbb{E}[A] = \int_0^\infty F'_A(x)\, dx$ (integral of the CCDF).
- $\mathbb{E}[\mathbf{1}\{E\}] = \mathbb{P}(E)$, where $\mathbf{1}(E)$ is the indicator of event E.

Conditional expectation

Definition 10.
$\mathbb{E}[A \mid B = b] = \sum_a a\,\mathbb{P}(A = a \mid B = b) =: f(b)$, and $\mathbb{E}[A \mid B] = f(B)$. Note that this is a *random variable* in B.

Let $B \sim \mathcal{U}(0,1)$ and $A \sim \mathcal{B}(B)$ (i.e., a Bernoulli whose parameter is itself the random variable B). Then $\mathbb{E}[A \mid B] = B$.

Further properties:

- If $A \perp B$, then $\mathbb{E}[A \mid B] = \mathbb{E}[A]$.
- "Pulling out what is known": $\mathbb{E}[A\,g(B) \mid B] = g(B)\,\mathbb{E}[A \mid B]$, since conditional on $B = b$, $g(B) = g(b)$ is a constant and can be pulled in front of the sum/integral.
- **Tower property (law of total expectation)**: $\mathbb{E}[\mathbb{E}[A \mid B]] = \mathbb{E}[A]$.

[Using the tower property] Let $C \sim \text{Unif}(-1, 1)$, and let $A = \mathcal{N}(C, 1)$, $B = \mathcal{N}(C, 1)$, with $A \perp B \mid C$ (i.e., $A - C - B$ forms a Markov chain). Note that A and B are *dependent* (even though conditionally independent given C). Suppose we need to compute $\mathbb{E}[AB]$. Rather than computing the joint density $f_{AB}(a, b)$ and integrating $\iint ab\, f_{AB}(a, b)\, da\, db$, we can use the tower property:

$$\mathbb{E}[AB] = \mathbb{E}\big[\mathbb{E}[AB \mid C]\big].$$

By conditional independence, $\mathbb{E}[AB \mid C = c] = \mathbb{E}[A \mid C = c]\,\mathbb{E}[B \mid C = c] = c^2$. Hence

$$\mathbb{E}\big[\mathbb{E}[AB \mid C]\big] = \mathbb{E}[C^2] = \int_{-1}^{1} \frac{1}{2} c^2\, dc = \frac{1}{3}.$$

0.3.5 Back to bandits: regret

Definition 11 (Expected and maximum reward)**.**
For a stochastic bandit given by $\nu = (P_a : a \in \mathcal{A})$, the expected reward of arm a is

$$\mu_a(\nu) = \int_{-\infty}^{+\infty} x\, dP_a(x) = \mathbb{E}[X], \quad X \sim P_a,$$

and the maximum expected reward is $\mu^*(\nu) = \max_{a \in \mathcal{A}} \mu_a(\nu)$.

Definition 12 (Regret)**.**
The regret of policy π on bandit instance ν is

$$R_n(\pi, \nu) = n\mu^* - \mathbb{E}\left[\sum_{t=1}^{n} X_t\right].$$

Note that the regret is *not* a random quantity (the randomness has already been integrated out via the expectation).

Lemma 1.

(a) $R_n(\pi, \nu) \geq 0$ for all policies π.
(b) The (oracle) policy π choosing $A_t \in \arg\max_a \mu_a$ for all t satisfies $R_n(\pi, \nu) = 0$.
(c) If $R_n(\pi, \nu) = 0$ for some policy π, then $\mathbb{P}(\mu_{A_t} = \mu^*) = 1$ for all $t \in [n]$.

0.3.6 Objective of the learner: sublinear regret

A policy incurring **linear regret** $\Theta(n)$ is trivial to achieve and provides no interesting guarantee – it does not require learning anything, so it is "too easy". A slightly more ambitious objective is **sublinear regret**: we look for policies such that, for all $\nu \in \mathcal{E}$,

$$\lim_{n\to\infty} \frac{R_n(\pi,\nu)}{n} = 0 \qquad \text{(also written } R_n = o(n)\text{)}.$$

This means that, in the long run, the learner chooses the optimal action almost always. (Usually, we are interested in even more ambitious objectives, e.g., controlling the rate at which R_n grows.)

In general, the learner needs to discover the arm (or, for structured bandits, some subset of arms) with the largest mean efficiently. For this, it needs to pull each arm (unstructured case) – or some number of arms (structured case) – a number of times to estimate the environment's parameters. This is exactly the **explore-exploit tradeoff**: exploring (playing all arms to improve the estimators) versus exploiting (playing the best arm(s) to improve the regret).

0.3.7 Decomposing the regret

Definition 13 (Suboptimality gap)**.**
The suboptimality gap of action a is $\Delta_a(\nu) = \mu^*(\nu) - \mu_a(\nu)$.

Definition 14 (Action count)**.**
The action count of action a by time t is $T_a(t) = \sum_{s=1}^{t} \mathbf{1}\{A_s = a\}$. Note that $T_a(t)$ is a *random variable*, since it depends on the random rewards observed before t (through the policy's choices).

Lemma 2 (Regret decomposition)**.**
For any policy π and environment ν, the regret satisfies

$$R_n(\pi,\nu) = \sum_{a\in\mathcal{A}} \Delta_a \, \mathbb{E}[T_a(n)].$$

Proof. We rewrite the n-step regret, splitting the sum over time into a sum over arms and using the indicator of which arm is played at each step:

$$R_n = n\mu^* - \mathbb{E}\left[\sum_{t=1}^{n} X_t\right] = \sum_{a\in\mathcal{A}} \sum_{t=1}^{n} \mathbb{E}\big[(\mu^* - X_t)\,\mathbf{1}(A_t = a)\big].$$

By the tower property (conditioning on A_t):

$$= \sum_{a\in\mathcal{A}} \sum_{t=1}^{n} \mathbb{E}\Big[\mathbb{E}\big[(\mu^* - X_t)\,\mathbf{1}(A_t = a) \mid A_t\big]\Big].$$

Now, conditional on A_t, the indicator $\mathbf{1}(A_t = a)$ is a constant and can be pulled out:

$$\mathbb{E}\big[(\mu^* - X_t)\,\mathbf{1}(A_t = a) \mid A_t\big] = \mathbf{1}(A_t = a)\,\mathbb{E}[\mu^* - X_t \mid A_t] = \mathbf{1}(A_t = a)\,(\mu^* - \mu_{A_t}),$$

using the (by-assumption) fact that $\mathbb{E}[X_t \mid A_t] = \mu_{A_t}$. Since the first factor is nonzero only when $A_t = a$, we may replace μ_{A_t} by μ_a:

$$\mathbf{1}(A_t = a)\,(\mu^* - \mu_{A_t}) = \mathbf{1}(A_t = a)\,(\mu^* - \mu_a) = \mathbf{1}(A_t = a)\,\Delta_a.$$

Putting everything together:

$$R_n = \sum_{a\in\mathcal{A}} \sum_{t=1}^{n} \mathbb{E}\big[\mathbf{1}(A_t = a)\big]\,\Delta_a = \sum_{a\in\mathcal{A}} \Delta_a \sum_{t=1}^{n} \mathbb{P}(A_t = a) = \sum_{a\in\mathcal{A}} \Delta_a\,\mathbb{E}[T_a(n)],$$

since $\sum_{t=1}^{n} \mathbf{1}(A_t = a) = T_a(n)$. □

Remark 7 (Intuition)**.**
The learner should pull arms with low (or zero) suboptimality gap, and minimize the expected number of pulls of arms with high suboptimality gap. This is not exactly shocking – the challenge of the course is devising algorithms that achieve this efficiently, *without* knowing the gaps Δ_a in advance.

Lecture 3 — 2026-09-23: Concentration of Measure and Explore-then-Commit

0.4 Concentration of Measure

0.4.1 Motivation: sums of i.i.d. random variables

To analyze bandit algorithms, we have to deal with sums of random rewards (hence of random regrets). The building block is the following setting.

Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables whose mean and variance exist:

$$\mu = \mathbb{E}[X], \qquad \sigma^2 = \mathbb{V}[X] = \mathbb{E}\big[(X - \mathbb{E}[X])^2\big].$$

A natural *estimator* of μ is the empirical mean

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} X_i =: \frac{S_n}{n}.$$

It satisfies:

- $\mathbb{E}[\hat{\mu}] = \mu$: $\hat{\mu}$ is an *unbiased* estimator;
- $\mathbb{V}[\hat{\mu}] = \mathbb{E}\big[(\hat{\mu} - \mu)^2\big] = \dfrac{\sigma^2}{n}$: the variance of the estimator decreases with n.

The question of this chapter is: *how close to μ is $\hat{\mu}$, with what probability, and for a given finite n?* We go through increasingly precise answers: law of large numbers, central limit theorem, Markov/Chebyshev inequalities, and finally exponential (Chernoff-type) bounds valid for finite n.

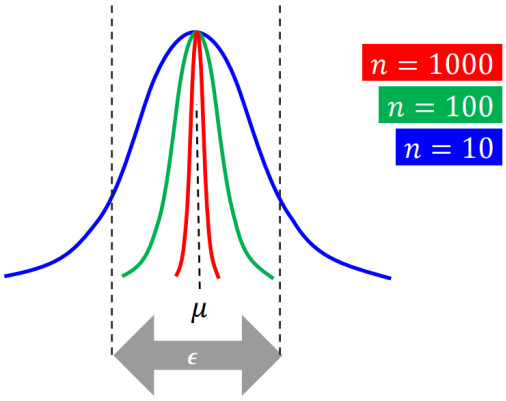
0.4.2 Large n limits

Law of large numbers (LLN)

Theorem 3 (Law of large numbers)**.**
Let $X_1, \dots, X_n$ be i.i.d. with mean μ.

- ***Weak law**: for every $\epsilon > 0$, $\lim_{n\to\infty} \mathbb{P}\big(|S_n - n\mu| \geq n\epsilon\big) = 0$.*
- ***Strong law**: $\mathbb{P}\left(\lim_{n\to\infty} \dfrac{S_n}{n} = \mu\right) = 1$.*

Remark 8 (Two notions of convergence)**.**
The weak law is a statement of *convergence in probability*, the strong law a statement of *almost sure convergence*. The latter is strictly stronger.
Example of a (non i.i.d.) sequence that converges to 0 in probability but not almost surely: let the Y_n be independent with $Y_n = 1$ with probability $1/n$ and $Y_n = 0$ otherwise. Then $\mathbb{P}(Y_n = 1) = 1/n \to 0$, so $Y_n \to 0$ in probability; but since $\sum_n 1/n = \infty$, almost every trajectory ω contains infinitely many 1's, so $Y_n(\omega)$ does not converge to 0.



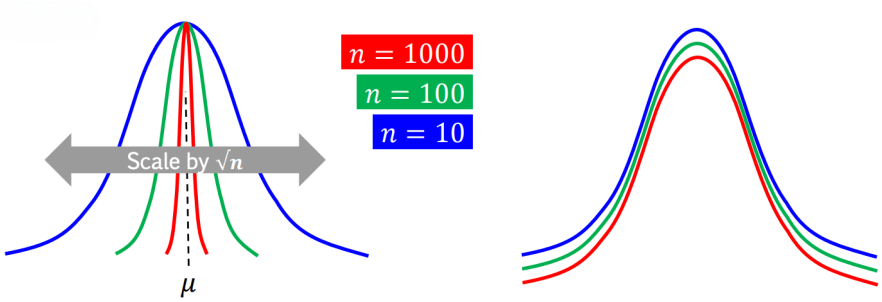
Note that the LLN is an *asymptotic* statement ($n \to \infty$): it says nothing about the probability of a deviation for a specific finite n.

Central limit theorem (CLT)

The LLN says that S_n/n concentrates around μ. The CLT tells us at which *scale*: rescale $\hat{\mu} - \mu$ by the inverse of its standard deviation $(1/\sqrt{n})$, i.e., look at $S_n - n\mu$ divided by $\sqrt{n}$:

$$\frac{S_n - n\mu}{\sqrt{n}} \xrightarrow{n\to\infty} \mathcal{N}(0, \sigma^2).$$

Equivalently, for large n, $\mathbb{P}\big(S_n - n\mu \geq n\epsilon\big) \to \mathbb{P}\big(\mathcal{N}(0,\sigma^2) \geq \epsilon\sqrt{n}\big)$.
This is the primary reason why the Gaussian law is so central (in addition to its stability under addition).



0.4.3 Markov and Chebyshev inequalities

Theorem 4 (Markov's inequality)**.**
For any random variable X and any $\epsilon > 0$,

$$\mathbb{P}(|X| \geq \epsilon) \leq \frac{\mathbb{E}|X|}{\epsilon}.$$

Proof. Set $Y = |X| \geq 0$ with density f. Then

$$\mathbb{E}[Y] = \int_0^\infty y f(y)\, dy \geq \int_\epsilon^\infty y f(y)\, dy \geq \epsilon \int_\epsilon^\infty f(y)\, dy = \epsilon\, \mathbb{P}(Y \geq \epsilon).$$

□

Theorem 5 (Chebyshev's inequality)**.**
For any random variable X with finite variance and any $\epsilon > 0$,

$$\mathbb{P}\big(|X - \mathbb{E}X| \geq \epsilon\big) \leq \frac{\mathbb{V}[X]}{\epsilon^2}.$$

Proof. Apply Markov's inequality to the non-negative random variable $(X - \mathbb{E}X)^2$ with threshold ϵ^2. □

Applied to the empirical mean (whose variance is σ^2/n):

$$\mathbb{P}\big(|\hat{\mu} - \mu| \geq \epsilon\big) \leq \frac{\sigma^2}{n\epsilon^2}.$$

- **Pro**: a nice, very general bound: only the mean and the variance are needed, and it holds for every finite n.
- **Con**: the bound is loose, it only decreases as $1/n$.

We need something much stronger for a meaningful analysis of bandit algorithms.

0.4.4 Gaussian tail from the CLT

By the CLT, $\dfrac{S_n - n\mu}{\sqrt{n\sigma^2}} \to \mathcal{N}(0,1)$. What if the normalized sum were *exactly* Gaussian? Let $Z \sim \mathcal{N}(0,1)$; we want to bound

$$\mathbb{P}(Z \geq z) = \int_z^\infty \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) dx.$$

This integral has no closed form, so we look for a bound. Since $x/z \geq 1$ on the domain of integration $(x \geq z > 0)$,

$$\int_z^\infty \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx \leq \frac{1}{\sqrt{2\pi}} \int_z^\infty \frac{x}{z}\, e^{-x^2/2}\, dx = \frac{1}{\sqrt{2\pi}\, z} \exp\left(-\frac{z^2}{2}\right).$$

Now, using $\hat{\mu} - \mu = \dfrac{S_n - n\mu}{n}$ and taking $z = \epsilon\sqrt{n/\sigma^2}$, the CLT heuristic gives

$$\mathbb{P}(\hat{\mu} - \mu \geq \epsilon) \;\longrightarrow\; \mathbb{P}\left(Z \geq \epsilon\sqrt{\tfrac{n}{\sigma^2}}\right) \leq \sqrt{\frac{\sigma^2}{2\pi n\epsilon^2}}\, \exp\left(-\frac{n\epsilon^2}{2\sigma^2}\right).$$

- **Pro**: much tighter than Chebyshev in general – *exponential* in n versus $1/n$.
- **Con**: this is an *asymptotic* result: the CLT does not allow us to bound the probability for a finite n.

We will need to pay for this limitation by making additional assumptions on the noise: **subgaussianity**.

0.4.5 Subgaussian random variables

Definition 15 (Subgaussian random variable)**.**
A random variable X is *σ-subgaussian* if its moment-generating function $M_X(\lambda) = \mathbb{E}[e^{\lambda X}]$ satisfies, for all $\lambda \in \mathbb{R}$,

$$M_X(\lambda) = \mathbb{E}\big[e^{\lambda X}\big] \leq \exp\left(\frac{\lambda^2\sigma^2}{2}\right).$$

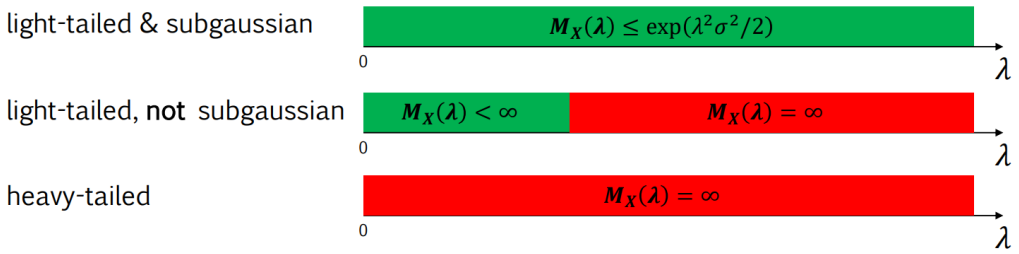
Equivalently, in terms of the cumulant-generating function,

$$\psi_X(\lambda) = \log M_X(\lambda) \leq \frac{1}{2}\lambda^2\sigma^2$$

for all $\lambda \in \mathbb{R}$.

Remark 9.

- If X is σ-subgaussian, then it is also σ'-subgaussian for any $\sigma' > \sigma$.
- Interpretation: the right-hand side $\exp(\lambda^2\sigma^2/2)$ is exactly the MGF of $\mathcal{N}(0, \sigma^2)$. A subgaussian variable has an MGF dominated by that of a Gaussian, i.e., its tail is at least as light as a Gaussian tail: "nice" noise with "no big excesses".
- Three cases can be distinguished: *heavy-tailed* variables ($M_X(\lambda) = \infty$ for all $\lambda \neq 0$), *light-tailed but not subgaussian* ($M_X(\lambda) < \infty$ on a neighborhood of 0 but $M_X(\lambda) = \infty$ for large $|\lambda|$, or M_X not dominated by the Gaussian one), and *light-tailed and subgaussian* ($M_X(\lambda) \leq \exp(\lambda^2\sigma^2/2)$ for all λ).



Properties of subgaussian random variables

Lemma 3.
Assume X is σ-subgaussian. Then:

(a) $\mathbb{E}[X] = 0$ and $\mathbb{V}[X] \leq \sigma^2$;
(b) for any $\alpha \in \mathbb{R}$, αX is $|\alpha|\sigma$-subgaussian;
(c) if X_1 is σ_1-subgaussian, X_2 is σ_2-subgaussian and $X_1 \perp X_2$, then $X_1 + X_2$ is $\sqrt{\sigma_1^2 + \sigma_2^2}$-subgaussian.

Remark 10 (Abuse of notation)**.**
We sometimes refer to subgaussian random variables that are *not* zero-mean; by this we mean that $X - \mathbb{E}X$ is subgaussian.

Proof of (a), sketch. Compare the Taylor expansions of both sides of $\mathbb{E}[e^{\lambda X}] \leq \exp(\lambda^2\sigma^2/2)$ around $\lambda = 0$. On the left,

$$e^{\lambda X} = 1 + \lambda X + \frac{\lambda^2 X^2}{2} + o(\lambda^2) \implies \mathbb{E}\left[e^{\lambda X}\right] = 1 + \lambda\,\mathbb{E}X + \frac{\lambda^2\,\mathbb{E}X^2}{2} + o(\lambda^2),$$

and, once we know $\mathbb{E}X = 0$ (the linear term in λ must vanish, otherwise the inequality would fail for small λ of the appropriate sign), $\mathbb{E}X^2 = \mathbb{V}X$, so the left side equals $1 + \frac{\lambda^2}{2}\mathbb{V}X + o(\lambda^2)$. On the right, $\exp(\lambda^2\sigma^2/2) = 1 + \frac{\sigma^2\lambda^2}{2} + o(\lambda^2)$. Hence

$$\mathbb{V}X + \frac{o(\lambda^2)}{\lambda^2/2} \leq \sigma^2 + \frac{o(\lambda^2)}{\lambda^2/2},$$

and taking $\lambda \to 0$ gives $\mathbb{V}X \leq \sigma^2$.
(b) follows since $\mathbb{E}[e^{\lambda\alpha X}] \leq \exp(\lambda^2\alpha^2\sigma^2/2)$; (c) follows from independence: $\mathbb{E}[e^{\lambda(X_1+X_2)}] = \mathbb{E}[e^{\lambda X_1}]\,\mathbb{E}[e^{\lambda X_2}] \leq \exp\left(\lambda^2(\sigma_1^2 + \sigma_2^2)/2\right)$. □

0.4.6 Tail of a subgaussian random variable

Theorem 6 (Tail bound)**.**
If X is σ-subgaussian, then for all $\epsilon > 0$,

$$\mathbb{P}(X \geq \epsilon) \leq \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right).$$

Proof. Let $\lambda > 0$. By monotonicity of exp, then Markov's inequality, then the definition of subgaussianity,

$$\begin{aligned}\mathbb{P}(X \geq \epsilon) &= \mathbb{P}\big(\exp(\lambda X) \geq \exp(\lambda\epsilon)\big) \\ &\leq \mathbb{E}\big[\exp(\lambda X)\big]\ \exp(-\lambda\epsilon) \\ &\leq \exp\left(\frac{\lambda^2\sigma^2}{2} - \lambda\epsilon\right).\end{aligned}$$

This holds for all $\lambda > 0$, so we may choose λ freely: taking $\lambda = \epsilon/\sigma^2$ (the minimizer of the exponent) gives $\exp\big(-\epsilon^2/(2\sigma^2)\big)$. □

The same argument gives the corresponding inequality for the left tail, so that, combining both tails (union bound),

$$\mathbb{P}(|X| \geq \epsilon) \leq 2\exp\left(-\frac{\epsilon^2}{2\sigma^2}\right).$$

Confidence intervals

Inverting this relationship gives *confidence intervals* for X. Fix a small probability (error tolerance) $\delta \in (0, 1)$. Solving $\exp(-\epsilon^2/(2\sigma^2)) = \delta$ for ϵ:

$$\mathbb{P}\left(X \geq \sqrt{2\sigma^2 \ln(1/\delta)}\right) \leq \delta.$$

For the two-sided version, we use the **union bound** $\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B)$, which yields the factor 2:

$$\mathbb{P}\left(|X| \geq \sqrt{2\sigma^2 \ln(2/\delta)}\right) \leq \delta.$$

So, with probability at least $1 - \delta$, X concentrates in the confidence interval

$$\left[-\sqrt{2\sigma^2 \ln \tfrac{2}{\delta}},\ +\sqrt{2\sigma^2 \ln \tfrac{2}{\delta}}\right].$$

We will apply such confidence intervals to characterize the estimators of per-arm reward in different bandit algorithms.

0.4.7 Large deviation bound for finite n

Theorem 7 (Concentration of the empirical mean)**.**
Assume that $X_i - \mu$, $i = 1, \dots, n$, are independent σ-subgaussian random variables. Then for any $\epsilon > 0$,

$$\mathbb{P}(\hat{\mu} \geq \mu + \epsilon) \leq \exp\left(-\frac{n\epsilon^2}{2\sigma^2}\right).$$

Proof. $\hat{\mu} - \mu = \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)$ is $\sigma/\sqrt{n}$-subgaussian (by properties (b) and (c): the sum is $\sqrt{n}\sigma$-subgaussian, and scaling by $1/n$ gives $\sigma/\sqrt{n}$). Apply the tail bound of the previous theorem with parameter $\sigma/\sqrt{n}$. □

Equivalently, $\mathbb{P}(S_n - n\mu \geq n\epsilon) \leq \exp\big(-n\epsilon^2/(2\sigma^2)\big)$. This is a bound similar to what the CLT suggests, but this time it is a true bound valid for every finite n – at the price of applying only to a (much smaller) class of variables. On a semi-log plot, the bound is a straight line in n with slope $-\epsilon^2/(2\sigma^2)$.

Confidence intervals for the mean. Inverting as before:

$$\mathbb{P}\left(\mu \geq \hat{\mu} + \sqrt{\frac{2\sigma^2 \ln(1/\delta)}{n}}\right) \leq \delta, \qquad \mathbb{P}\left(\mu \leq \hat{\mu} - \sqrt{\frac{2\sigma^2 \ln(1/\delta)}{n}}\right) \leq \delta,$$

and, two-sided,

$$\mathbb{P}\left(\mu \in \left[\hat{\mu} - \sqrt{\frac{2\sigma^2 \ln(2/\delta)}{n}},\ \hat{\mu} + \sqrt{\frac{2\sigma^2 \ln(2/\delta)}{n}}\right]\right) \geq 1 - \delta.$$

The width of the interval shrinks as $1/\sqrt{n}$ and grows only as $\sqrt{\ln(1/\delta)}$ when we demand more confidence.

0.4.8 Examples of subgaussian random variables

- **Gaussian is subgaussian**: $X \sim \mathcal{N}(0, \sigma^2)$ has $M_X(\lambda) = \exp(\lambda^2\sigma^2/2)$, hence is σ-subgaussian (with equality in the definition).
- **Bounded is subgaussian**: if $X \in [a, b]$ almost surely, then X is $\frac{b-a}{2}$-subgaussian (Hoeffding's lemma).
  In particular, if $|X| \leq a$ almost surely, then X is a-subgaussian. For instance, a Bernoulli variable is $\frac{1}{2}$-subgaussian.
- **Exponential is *not* subgaussian** (its tail is not light enough!): for $X \sim \mathrm{Exp}(1/\mu)$ (mean μ),

$$M_X(\lambda) = \int_0^\infty \mu^{-1} e^{-x/\mu} e^{\lambda x}\, dx = \begin{cases} (1 - \lambda\mu)^{-1} & \lambda < 1/\mu \\ \infty & \text{otherwise.} \end{cases}$$

  The MGF is infinite for $\lambda \geq 1/\mu$, whereas the subgaussian condition requires it to be finite (and bounded by a Gaussian MGF) for *all* λ.

0.5 The Explore-then-Commit Algorithm

0.5.1 Idea and definitions

Recall the secretary problem: pure exploration for a fixed number of steps, then pure exploitation. We can try a similar policy for bandits: invest some predetermined number of samples to estimate the mean reward of each arm, then play the empirically best arm until the horizon n.

Definition 16 (Estimated reward and number of pulls)**.**
For an arm i and a time t,

$$T_i(t) = \sum_{s=1}^{t} \mathbb{I}\{A_s = i\}, \qquad \hat{\mu}_i(t) = \frac{1}{T_i(t)} \sum_{s=1}^{t} \mathbb{I}\{A_s = i\}\, X_s.$$

$T_i(t)$ is the number of pulls of arm i up to time t and $\hat{\mu}_i(t)$ is the estimated (empirical mean) reward of arm i.

Definition 17 (Explore-then-Commit (ETC))**.**
Let k be the number of arms and $m \in \mathbb{N}$ a parameter: the number of exploration pulls *per arm* (so mk pulls overall).
Explore-then-Commit(m):
for $t = 1, \dots, n$:

$$A_t = \begin{cases} (t \bmod k) + 1 & \text{if } t \le mk \quad \text{(explore)} \\ \arg\max_i \hat{\mu}_i(mk) & \text{if } t > mk \quad \text{(commit)} \end{cases}$$

Remark 11.

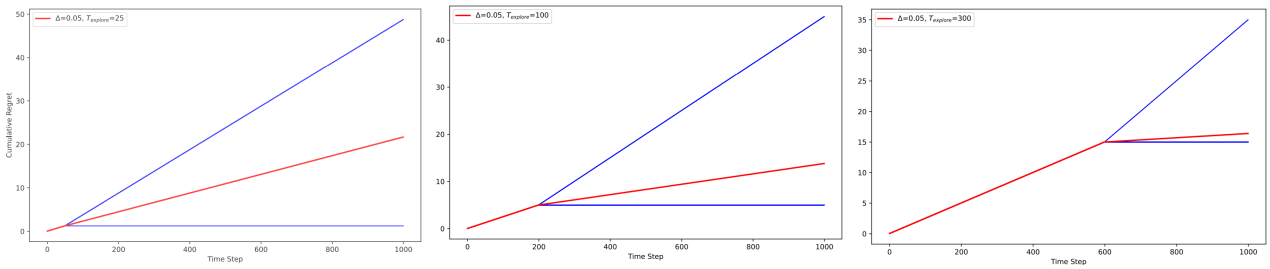
- The only parameter to set is the number of exploration steps: m per arm, mk overall (the duration of the explore phase).
- The *order* of pulls in the explore phase does not matter, it just needs to be m per arm. One could pull arm 1 m times back-to-back, then arm 2, etc.
- The algorithm itself is generic, but the *analysis* will need to assume subgaussian rewards for all arms.
- After time mk the estimates are frozen, so the arm played from $mk + 1$ to n is the same one at every step (ties broken arbitrarily).

0.5.2 Simulations: Bernoulli bandit with $k = 2$

To build intuition, we look at simulations on a Bernoulli bandit with $k = 2$ arms and suboptimality gap $\Delta = \mu_1 - \mu_2$. In the plots, the **blue** curves are individual trajectories of $\sum_{a \in \mathcal{A}} \Delta_a T_a(t)$, the trajectory-dependent (random) regret, while the **red** curve is $R_n = \sum_{a \in \mathcal{A}} \Delta_a \, \mathbb{E}[T_a(t)]$, the expected regret.

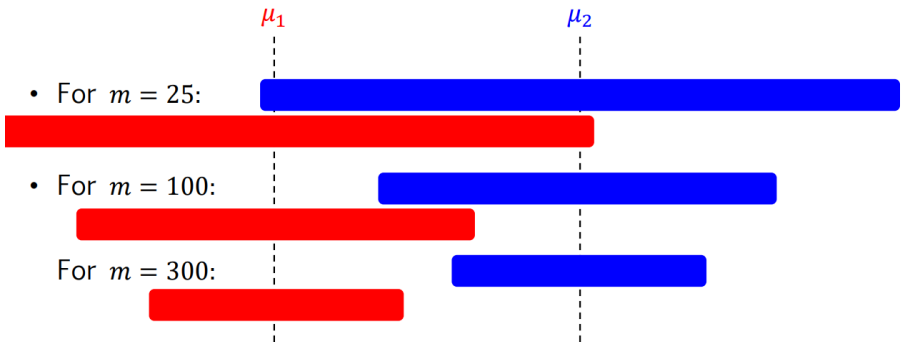
Small gap $\Delta = 0.05$

- **Small** $m = 25$: exploration is cheap but the estimates are noisy, so many trajectories commit to the wrong arm and keep paying linear regret afterwards.
- **Medium** $m = 100$: fewer trajectories fail.
- **Large** $m = 300$: the expected regret is close to its minimum: most trajectories find the right arm and incur no further regret after $t = mk$. The confidence intervals of μ_1 and μ_2 have almost no overlap.



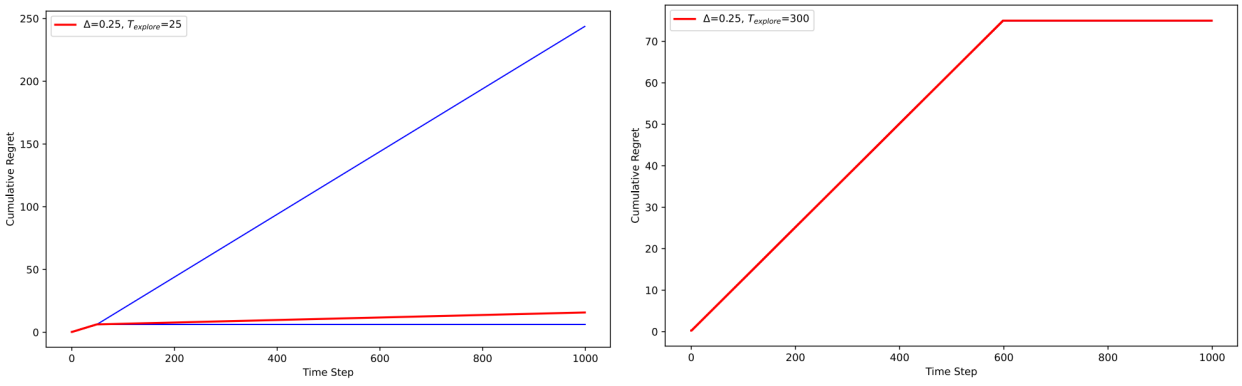
Confidence intervals of the per-arm reward

Recall the confidence interval $\left[\hat{\mu} - \sqrt{\frac{2\sigma^2 \ln(2/\delta)}{m}},\ \hat{\mu} + \sqrt{\frac{2\sigma^2 \ln(2/\delta)}{m}}\right]$ after m pulls. As m grows ($m = 25$, 100, 300), the intervals around μ_1 and μ_2 shrink until they no longer overlap: the algorithm can then tell the arms apart reliably.



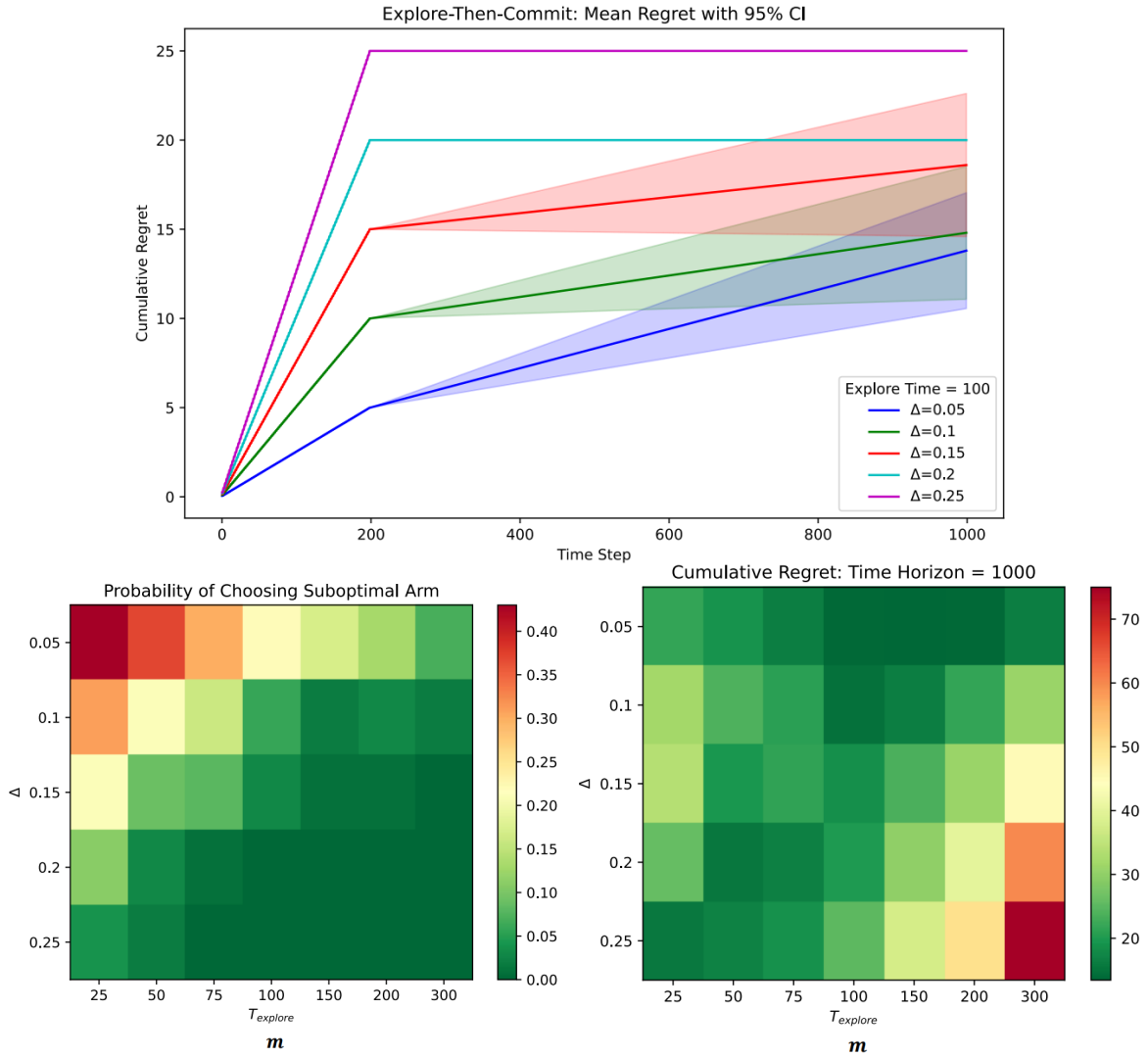
Large gap $\Delta = 0.25$

- **Small** $m = 25$: a larger Δ means the winning arm is more likely to be the better arm, even with few samples.
- **Large** $m = 300$: no more failures in the simulation.



Effect of Δ and m

- **Cumulative regret R_n for different Δ (fixed $m = 100$)**: a larger Δ means more costly exploration (each pull of the bad arm costs Δ), but less risky exploitation (the wrong arm is rarely selected).
- $\mathbb{P}$(**wrong arm at time n**) **as a function of Δ and m**: increasing m helps finding the optimal arm; this is made harder by a smaller Δ.
- **Cumulative regret R_n as a function of Δ and m**: there is a tradeoff. For small Δ one needs a large m; for large Δ a small m suffices. Intermediate Δ gives a higher regret than both the small and the large Δ regimes.



0.5.3 Regret bound for ETC with 1-subgaussian rewards

Theorem 8 (Regret of ETC)**.**
For ETC on a 1-subgaussian bandit (all arms have 1-subgaussian reward noise), for $n \geq mk$,

$$R_n \leq \underbrace{m \sum_{i=1}^{k} \Delta_i}_{\text{exploration}} + \underbrace{(n - mk) \sum_{i=1}^{k} \Delta_i \exp\left(-\frac{m\Delta_i^2}{4}\right)}_{\text{exploitation}}.$$

Proof. Assume without loss of generality that $\mu_1 = \mu^* = \max_i \mu_i$ (so $\Delta_1 = 0$). By the regret

decomposition lemma (previous lecture),

$$R_n = \sum_{i=1}^{k} \Delta_i \, \mathbb{E}[T_i(n)].$$

Arm i is pulled exactly m times during exploration; afterwards, it is pulled at each of the $n - mk$ remaining steps if and only if it was selected at time $mk + 1$. Hence

$$\begin{aligned}\mathbb{E}[T_i(n)] &= m + (n - mk)\,\mathbb{P}\big(A_{mk+1} = i\big) \\ &\le m + (n - mk)\,\mathbb{P}\Big(\hat{\mu}_i(mk) \ge \max_{j \ne i} \hat{\mu}_j(mk)\Big).\end{aligned}$$

Now, for $i \neq 1$, the event that i beats *all* other arms is contained in the event that it beats arm 1:

$$\begin{aligned}\mathbb{P}\Big(\hat{\mu}_i(mk) \ge \max_{j \ne i} \hat{\mu}_j(mk)\Big) &\le \mathbb{P}\big(\hat{\mu}_i(mk) \ge \hat{\mu}_1(mk)\big) \\ &= \mathbb{P}\Big(\big(\hat{\mu}_i(mk) - \mu_i\big) - \big(\hat{\mu}_1(mk) - \mu_1\big) \ge \Delta_i\Big).\end{aligned}$$

The empirical mean $\hat{\mu}_i(mk)$ is computed from m i.i.d. samples: $m\hat{\mu}_i - m\mu_i$ is a sum of m independent 1-subgaussian variables, hence $m\hat{\mu}_i(mk)$ is $\sqrt{m}$-subgaussian, so $\hat{\mu}_i(mk) - \mu_i$ is $1/\sqrt{m}$-subgaussian. The two arms are sampled independently, so the difference $\big(\hat{\mu}_i - \mu_i\big) - \big(\hat{\mu}_1 - \mu_1\big)$ is $\sqrt{2/m}$-subgaussian. The tail bound then gives

$$\mathbb{P}\Big(\big(\hat{\mu}_i - \mu_i\big) - \big(\hat{\mu}_1 - \mu_1\big) \ge \Delta_i\Big) \le \exp\left(-\frac{\Delta_i^2}{2 \cdot (2/m)}\right) = \exp\left(-\frac{m\Delta_i^2}{4}\right).$$

Plugging everything back into $R_n = \sum_i \Delta_i \, \mathbb{E}[T_i(n)]$ gives the claim. □

Remark 12 (Tradeoff)**.**

- m large ⇒ exploration is costly (first term grows linearly in m);
- m small ⇒ potential to bet on the wrong arm forever (second term is large, it is multiplied by $n - mk$).

This is exactly the explore-exploit tradeoff, now quantified.

0.5.4 Analysis for $k = 2$

Assume again that the first arm is optimal: $\Delta_1 = 0$, and write $\Delta_2 = \Delta$. The bound for general k becomes

$$R_n \le m\Delta + (n - 2m)\Delta \exp\left(-\frac{m\Delta^2}{4}\right) \le m\Delta + n\Delta \exp\left(-\frac{m\Delta^2}{4}\right).$$

Which m minimizes this quantity? Differentiating the right-hand side with respect to m:

$$\Delta - n\Delta \cdot \frac{\Delta^2}{4} \exp\left(-\frac{m\Delta^2}{4}\right) = 0 \iff \exp\left(-\frac{m\Delta^2}{4}\right) = \frac{4}{n\Delta^2},$$

so (ignoring rounding)

$$m = \frac{4}{\Delta^2} \ln\left(\frac{n\Delta^2}{4}\right).$$

The resulting regret bound is of the form

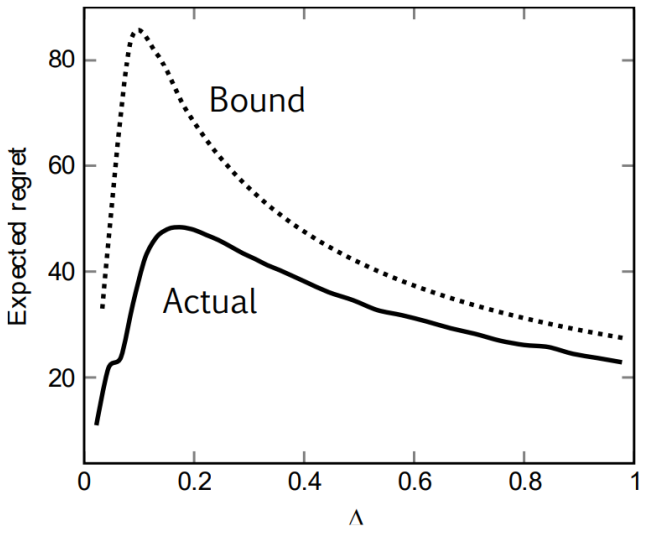
$$R_n \le \Delta + C\sqrt{n},$$

for a universal constant C.

- Without an assumption on Δ, we cannot avoid potentially infinite regret (a single bad pull can cost arbitrarily much). Within a bandit class such that $\Delta \leq 1$: $R_n \leq 1 + C\sqrt{n}$.
- Replacing Δ by a value result in an example of a **worst-case** (or **problem-independent**) bound: it does not depend on the specific environment instance ν, only on the environment class (here: subgaussian and $\Delta \leq 1$) and on the horizon n.
- **But**: to set the correct m, we do need to know Δ "ahead of time". Can we have a universal algorithm that does well for unknown Δ, i.e., that decides *adaptively* when we have explored enough? (This motivates the algorithms of the next lectures.)

0.5.5 Comparison of the bound with simulations

The bound is relatively loose (due to approximations in the proof) for intermediate suboptimality gaps Δ, but pretty good for larger Δ. It slightly underestimates the critical Δ (the gap at which the actual regret peaks).



Lecture 4 — 2026-09-30: Adaptive Explore-then-Commit and the Upper Confidence Bound Algorithm

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At the end of the previous lecture, we saw that Explore-then-Commit (ETC) reaches a good regret only if the exploration length m is tuned with the knowledge of the suboptimality gap Δ, which is unknown in practice. This lecture answers the question raised there: *can we decide adaptively, from the data, when we have explored enough?* We first make ETC adaptive using confidence intervals, extend it to $k > 2$ arms (sequential elimination), and then introduce the main algorithm of the lecture: the **Upper Confidence Bound (UCB)** algorithm, which is based on the principle of *optimism in the face of uncertainty.*

0.6 Recap

0.6.1 Multi-armed bandits

- **Problem setup**: k arms, each arm generates random rewards. Arm i has reward distribution P_i with mean μ_i; the best arm has mean $\mu^* = \max_i \mu_i$. The reward distributions are *unknown* to the learner at the beginning.
- **Algorithm**: at time t, take action $A_t \in \{1, 2, \dots, k\}$, observe the reward $X_t \sim P_{A_t}$, and continue for $t = 1, 2, \dots, n$. This is a sequential decision-making process: A_t may depend on all the past actions and rewards.
- **Goal**: achieve *sublinear* regret, i.e., eventually pull the best arm (almost) every time.

0.6.2 Regret

- **Instantaneous regret** at round t: $r_t = \mu^* - \mu_{A_t} = \Delta_{A_t}$. It measures the (expected) suboptimality of arm A_t: bad arm $\Leftrightarrow$ large regret, good arm $\Leftrightarrow$ small regret, best arm $\Leftrightarrow$ zero regret.
- **Cumulative regret**: $R_n = \sum_{t=1}^{n} \mathbb{E}[r_t]$.
- **Regret decomposition**: $R_n = \sum_{t=1}^{n} \mathbb{E}[r_t] = \sum_{a \in \mathcal{A}} \Delta_a \, \mathbb{E}[T_a(n)]$.
- Sublinear cumulative regret $\Leftrightarrow$ vanishing instantaneous regret.

Hence, to minimize regret, we must eliminate bad arms quickly and eventually identify (and keep pulling) the best arm(s). The difficulty: due to randomness, we can *never* identify good and bad arms with certainty.

0.6.3 Explore-then-Commit

- **Overview**: explore (round robin) until a fixed time mk; this gives an estimate $\hat{\mu}_i$ of μ_i for every arm i. If $\hat{\mu}_i \approx \mu_i$, picking $i^* = \arg\max_i \hat{\mu}_i$ identifies the best arm. Then commit to (keep pulling) i^* for rounds $mk + 1$ to n.
- **Intuition**: if the best arm is correctly identified, there is no regret during the commit phase, so $R_n \propto m$: great! If a suboptimal arm is chosen, we pay a constant positive regret at all remaining rounds, so $R_n \propto n$: bad!
- **Tradeoff**: very short exploration $\Rightarrow |\hat{\mu}_i - \mu_i|$ large $\Rightarrow$ likely to choose a suboptimal arm $\Rightarrow$ large regret. Very long exploration $\Rightarrow$ excessive exploration $\Rightarrow$ large regret.

0.7 Adaptive Explore-then-Commit

0.7.1 Idea: the exploration length as a stopping time

Key idea: think of m as a **stopping time**.

- m is now *random* and can depend on past data.
- It signals when to stop exploring and start exploiting.
- Ideally, we stop as soon as we are sufficiently confident about which arm is optimal.

With a *fixed* m, two things can go wrong:

- we may still not be sure about the optimal arm at time mk (**under-exploration**);
- or we may have been very sure about the optimal arm much earlier (**over-exploration**).

Idea: decide when to stop by looking at the data. Keep pulling all arms equally often and computing the $\hat{\mu}_i$, until we are (reasonably) certain that one arm is better than all the others. How do we quantify certainty? With **confidence intervals**!

0.7.2 Confidence intervals

Let $X_1, X_2, \ldots, X_n$ be i.i.d. samples of a random variable with unknown mean μ, and $\hat{\mu}_n = \frac{1}{n}\sum_i X_i$ the estimate of μ. Typically, $\hat{\mu}_n$ is close to μ, but not exactly equal to it. What we can say is

$$|\hat{\mu}_n - \mu| \leq \epsilon \quad \text{with probability at least } 1-\delta,$$

or equivalently, $\mu \in [\hat{\mu}_n - \epsilon, \hat{\mu}_n + \epsilon]$ with high probability. There are two quantities in this statement: the **confidence interval** (half-width ϵ) and the **confidence level** (error probability δ).

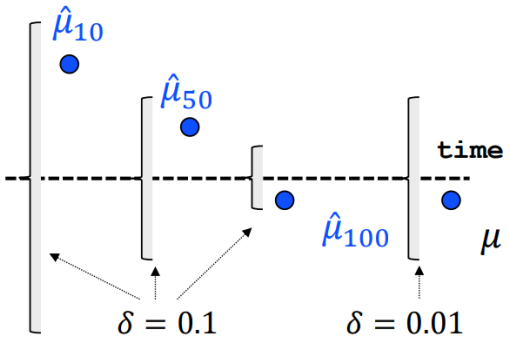
If the $X_i - \mu$ are σ-subgaussian (with the abuse of notation of the previous lecture, we simply say that the X_i are σ-subgaussian), the concentration inequality of the previous lecture gives

$$\mathbb{P}\big(|\hat{\mu}_n - \mu| \geq \epsilon\big) \leq 2\exp\left(-\frac{n\epsilon^2}{2\sigma^2}\right) =: \delta.$$

- Given ϵ, we can compute δ with this inequality; alternatively, we fix δ and compute ϵ:

$$\epsilon = \sigma\sqrt{\frac{2\ln(2/\delta)}{n}}, \qquad \text{or equivalently} \qquad n = \frac{2\sigma^2\ln(2/\delta)}{\epsilon^2}.$$

- For fixed δ, ϵ decreases as n increases (as $1/\sqrt{n}$).
- For fixed n, ϵ decreases as δ increases (we accept a larger error probability in exchange for a tighter interval).
- ϵ grows linearly with the subgaussian parameter σ: noisier rewards need more samples ($n \propto \sigma^2$) for the same precision.

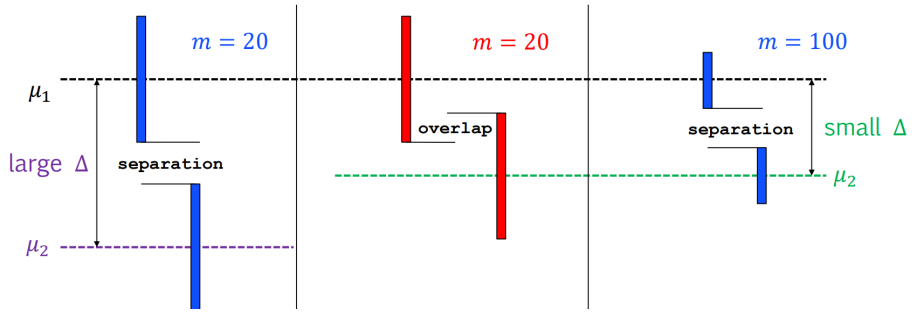


0.7.3 Using confidence intervals to decide when to commit

How can we confidently decide when one arm is better than the other?

- Pretend that each arm's mean reward lies in its confidence interval *with certainty.*
- If the confidence intervals **overlap**, the best arm is undecided.
- If the confidence intervals are **disjoint**, the best arm is clear!

Strategy: keep exploring until the confidence intervals separate.



This makes m **data-dependent**:

- With sufficient exploration, the confidence intervals eventually separate (their width goes to 0 as $1/\sqrt{m}$).
- When $\mu_1 - \mu_2$ is large, the confidence intervals separate early, so $\mathbb{E}[m]$ is small.
- When $\mu_1 - \mu_2$ is small, the confidence intervals separate only after a long time, so $\mathbb{E}[m]$ is large.

The exploration time is chosen from the data, not fixed a priori.

Definition 18 (Upper and lower confidence bounds)**.**
For an arm with n samples and empirical mean $\hat{\mu}_n$, starting from $\mathbb{P}\big(|\hat{\mu}_n - \mu| \geq \epsilon\big) \leq 2\exp\big(-n\epsilon^2/(2\sigma^2)\big)$ and solving for ϵ:

$$UCB(n,\delta) = \hat{\mu}_n + \sigma\sqrt{\frac{2\ln(2/\delta)}{n}}, \qquad LCB(n,\delta) = \hat{\mu}_n - \sigma\sqrt{\frac{2\ln(2/\delta)}{n}}.$$

UCB is the **Upper Confidence Bound** ($\hat{\mu} + \epsilon$), LCB the **Lower Confidence Bound** ($\hat{\mu} - \epsilon$). With probability at least $1 - \delta$, $\mu \in [LCB(n,\delta), UCB(n,\delta)]$.

0.7.4 Algorithm

Definition 19 (Adaptive ETC (ETC with confidence intervals))**.**
Hyperparameters: confidence level $\delta \in (0,1)$ and a minimum number of pulls per arm m_0.

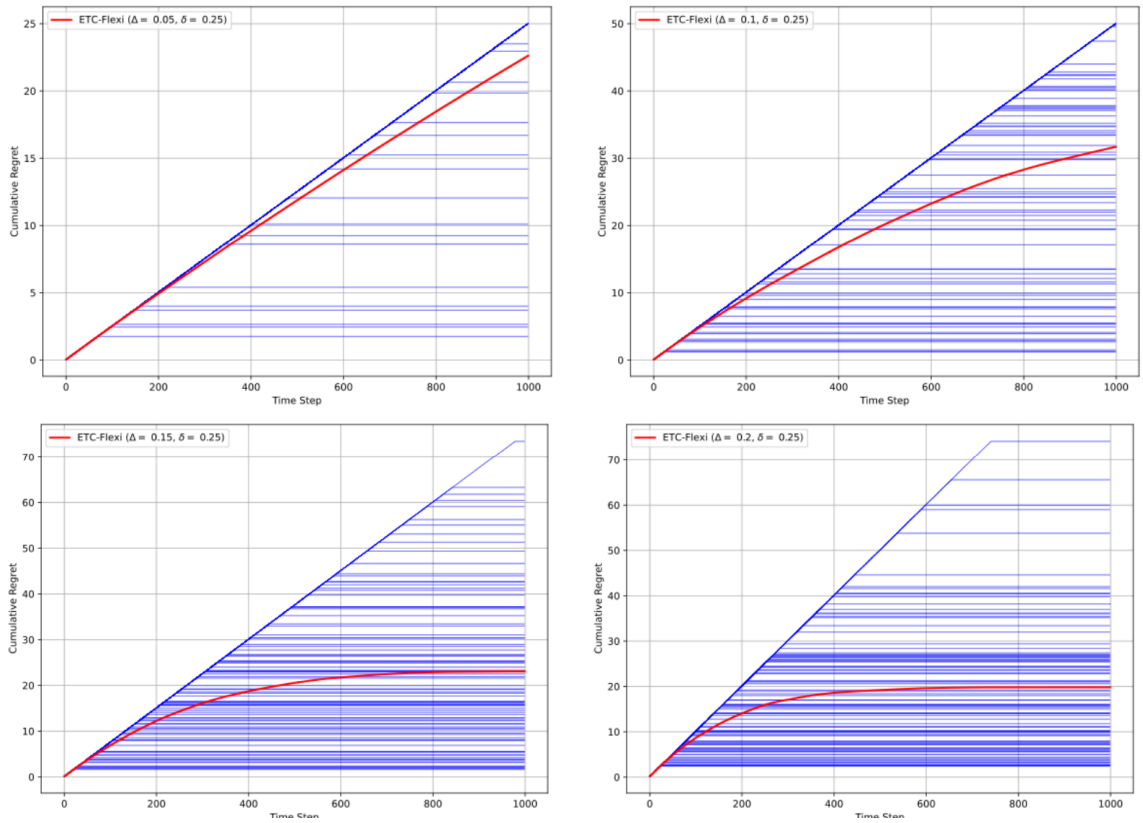
Initialize: $T_a(0) = 0$, $\hat{\mu}_a(0) = 0$ for all arms $a \in \mathcal{A}$; `committed_arm` = `None`.
for $t = 1, 2, \ldots, n$ **do**:
  if `committed_arm` $\neq$ `None`: $A_t =$ `committed_arm`
  else: $A_t = (t \bmod k) + 1$
    check for commitment if $t \geq m_0 k$:
      compute $\hat{\mu}_a(t)$ for all $a \in \mathcal{A}$, and $a_t^* = \arg\max_a \hat{\mu}_a(t)$;
      compute $UCB_a(t,\delta)$ and $LCB_a(t,\delta)$ for all $a \in \mathcal{A}$;
      if $LCB_{a_t^*}(t,\delta) > UCB_a(t,\delta)$ for all $a \neq a_t^*$: `committed_arm` $= a_t^*$.

Here $UCB_a(t,\delta)$ and $LCB_a(t,\delta)$ are the bounds of arm a computed with its $T_a(t)$ samples.

Remark 13.

- The commitment rule is exactly "the confidence interval of the empirically best arm is entirely above the confidence intervals of all other arms".
- m_0 forces every arm to be sampled a minimum number of times before any commitment is even considered.
- Compared to ETC, the only parameter is now δ (and m_0); no knowledge of Δ is needed.

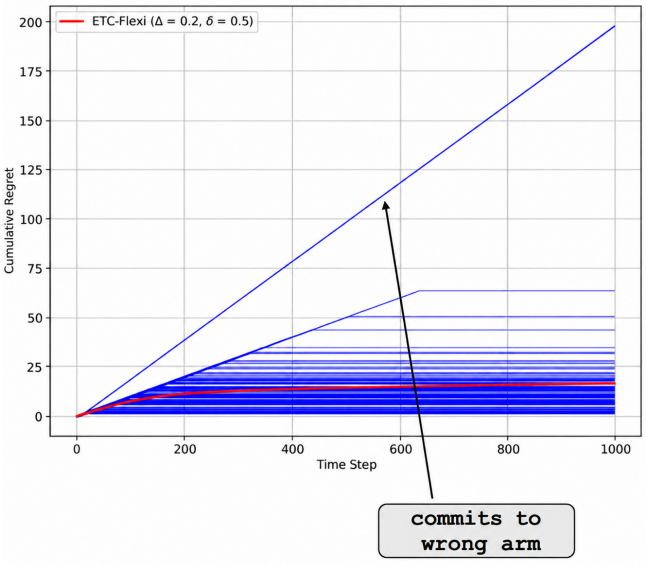
0.7.5 Simulations



Salient features:

- The exploration period is random (each trajectory commits at a different time).
- The expected cumulative regret is sublinear.
- Larger $\Delta \Rightarrow$ the algorithm commits earlier.
- As for fixed ETC, the regret is highest for moderate Δ (here $\Delta = 0.1$): small Δ makes mistakes cheap, large Δ makes the arms easy to distinguish.

Importance of the hyperparameter δ



- **Very small** $\delta \Rightarrow$ wide intervals, too conservative: over-exploration.
- **Very large** $\delta \Rightarrow$ narrow intervals, too optimistic: some trajectories commit to the wrong arm, and then pay linear regret.
- **Rule of thumb**: the larger the horizon n, the smaller δ should be, because a mistaken commitment costs regret at each of the (many) remaining rounds.

Remark 14.
A wrong commitment costs up to $\Delta \cdot n$, and happens with a probability of order δ. If δ stays fixed while n grows, this term is linear in n; this is why δ must decrease with n (we will see the choice $\delta = 1/n^2$ for UCB).

0.8 More than two arms: Sequential Elimination

0.8.1 ETC for $k > 2$ arms

How can we extend the (adaptive) ETC principle to multi-armed bandits? As for two arms, we need to pull each arm a few times to estimate its reward: exploration of all arms is necessary. With more exploration, the confidence interval of each arm keeps shrinking; eventually the intervals become disjoint and the best arm becomes clear.

There are two different interpretations of ETC for $k > 2$:

- **View 1**: keep exploring until the best arm is identified. Commit to arm i once its confidence interval dominates the confidence intervals of *all* other arms.
- **View 2**: keep eliminating arms that are clearly not the best. If the confidence interval of some arm i dominates that of arm j (i.e. $LCB_i > UCB_j$), stop pulling arm j.

The two views are **equivalent with two arms**, but **different in the multi-armed case**!

0.8.2 From ETC to sequential elimination

Algorithm from View 1 (adaptive ETC): all arms are pulled equally often until the commit time mk. Likely behavior: the worse arms clearly separate out long before the competitive ones do, but we keep pulling them anyway. This is wasteful exploration on bad arms, hence a large regret.

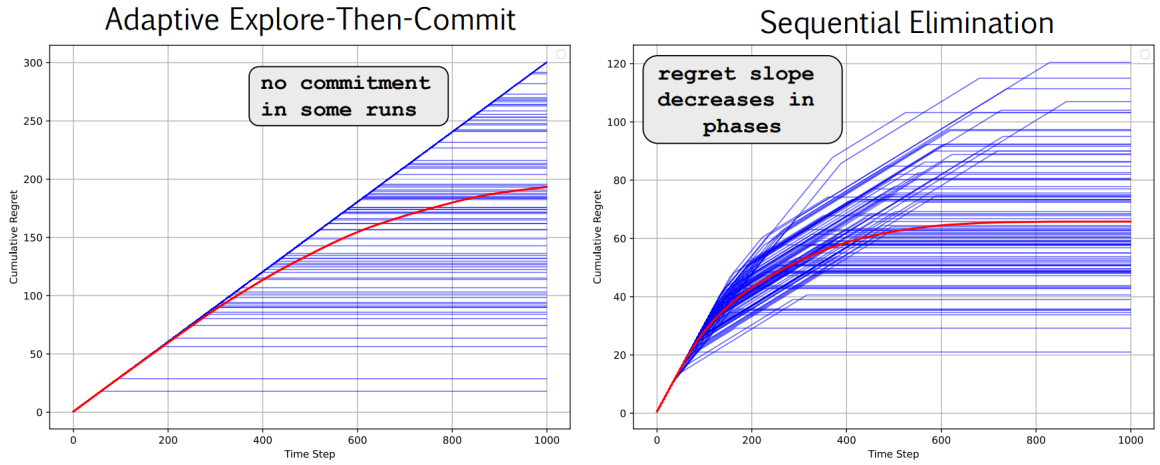
Algorithm from View 2: **Sequential Elimination (SE)**.

- Initially, all arms are pulled in round robin.
- The algorithm proceeds in *phases*, eliminating one arm per phase: a phase ends when the confidence interval of some remaining arm becomes dominated by that of another remaining arm; the dominated arm is removed and the round robin continues on the remaining ones.
- The length of each phase is random.
- The instantaneous regret decreases from phase to phase (the worst arms are removed first).
- Eventually, we are left with only the best arm.

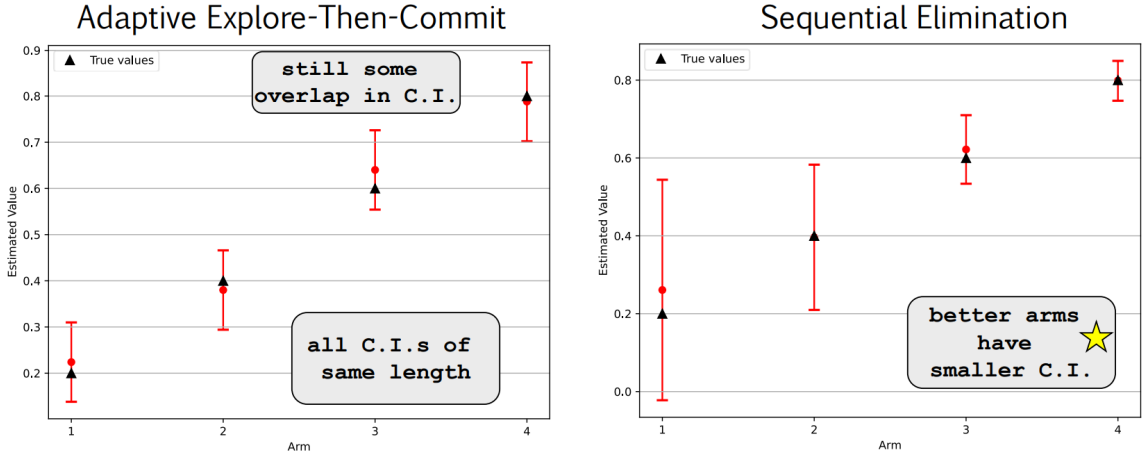
Sequential elimination is the "right" extension of ETC to multi-armed bandits.

0.8.3 Simulations: adaptive ETC versus sequential elimination

Setting: 4-armed Bernoulli bandit with mean rewards $[0.2, 0.4, 0.6, 0.8]$.



- **Regret**: with adaptive ETC, there is no commitment at all in some runs (within the horizon), so the regret keeps growing linearly for those runs. With SE, the regret slope decreases in phases, each time an arm is eliminated.



- **Confidence intervals**: with adaptive ETC, all confidence intervals have the same length (all arms are pulled equally often), and there can still be some overlap between the competitive arms.

With SE, the better arms have smaller confidence intervals: the remaining samples are spent on the arms that still matter.

0.9 The Upper Confidence Bound (UCB) algorithm

0.9.1 Algorithm

UCB is a very popular algorithm for multi-armed bandits. It uses confidence bounds, like adaptive ETC and sequential elimination, but in a different way, which leads to better regret in practice.

Notation ("at time t" means after t overall pulls):

- $\hat{\mu}_a(t)$: estimate of the mean reward of arm a at time t;
- $T_a(t)$: number of pulls of arm a at time t;
- upper confidence bound of arm a at time t (assuming 1-subgaussian rewards):

$$UCB_a(t,\delta) = \hat{\mu}_a(t) + \sqrt{\frac{2\ln(1/\delta)}{T_a(t)}},$$

with $UCB_a(t,\delta) = \infty$ when $T_a(t) = 0$.

Definition 20.
(UCB algorithm).

for $t = 1, 2, \ldots, n$ **do**:
choose $A_t = \arg\max_a UCB_a(t-1,\delta)$;
observe X_t and update $UCB_a(t,\delta)$.

Remark 15.

- Since $UCB_a = \infty$ initially, every arm is pulled once during the first k rounds.
- Here the bound is *one-sided* (only the upper bound matters), hence $\ln(1/\delta)$ instead of $\ln(2/\delta)$; and $\sigma = 1$. By the concentration inequality, for a fixed number s of samples, $\mathbb{P}\big(\mu \geq \hat{\mu}_s + \sqrt{2\ln(1/\delta)/s}\big) \leq \delta$.
- Unlike ETC, there are no separate phases: the same rule is applied at every round.

0.9.2 Exploration-exploitation tradeoff in UCB

The UCB index is the sum of two terms:

$$UCB_a(t,\delta) = \underbrace{\hat{\mu}_a(t)}_{\text{empirical mean}} + \underbrace{\sqrt{\frac{2\ln(1/\delta)}{T_a(t)}}}_{\text{exploration bonus}}.$$

The algorithm pulls an arm more often if it is

- either **promising**: its empirical mean $\hat{\mu}_a(t)$ is large (exploitation);
- or **not well explored**: $T_a(t)$ is small, so the exploration bonus is large (exploration).

Why is this a reasonable strategy?

- With high probability, $UCB_{a^*}(t,\delta) > \mu_{a^*}$ at all times.
- If a suboptimal arm $a \neq a^*$ is pulled, then $UCB_a(t,\delta) > UCB_{a^*}(t,\delta)$.
- With enough pulls, $UCB_a(t,\delta) \approx \mu_a$, and $\mu_a < \mu_{a^*} < UCB_{a^*}(t,\delta)$, so a stops being chosen.
- Hence suboptimal arms are not pulled too many times! Heuristically, arm a stops being pulled once its bonus is smaller than its gap, $\sqrt{2\ln(1/\delta)/T_a} \lesssim \Delta_a$, i.e. after about $T_a \approx 2\ln(1/\delta)/\Delta_a^2$ pulls. The regret bound below makes this precise.

0.9.3 Optimism in the face of uncertainty: UCB versus ETC

UCB follows the **Optimism in the Face of Uncertainty (OFU)** principle:

- UCB plays assuming the arm rewards are the highest they can reasonably be, i.e., it pretends $\mu_a = UCB_a(t, \delta)$ (the top of the confidence interval).

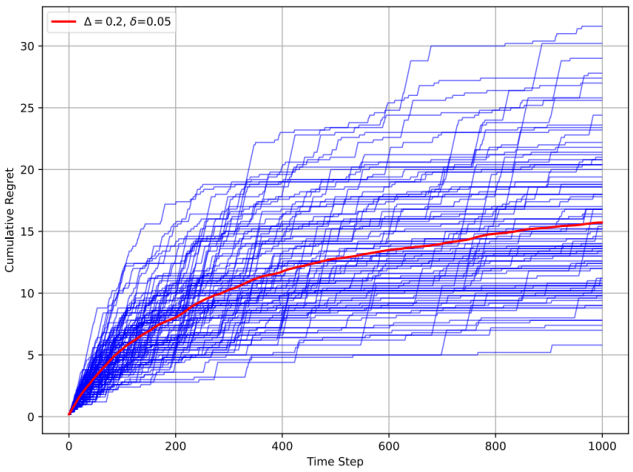
ETC (and sequential elimination) is more **conservative**: it takes a worst-case approach, where arm rewards could be anything in their respective confidence intervals.

- It only commits to the best arm if it is "sure" that it is the best: its confidence interval must dominate all the others.
- Likewise, it only eliminates a bad arm if it is "sure" that this arm is not the best.

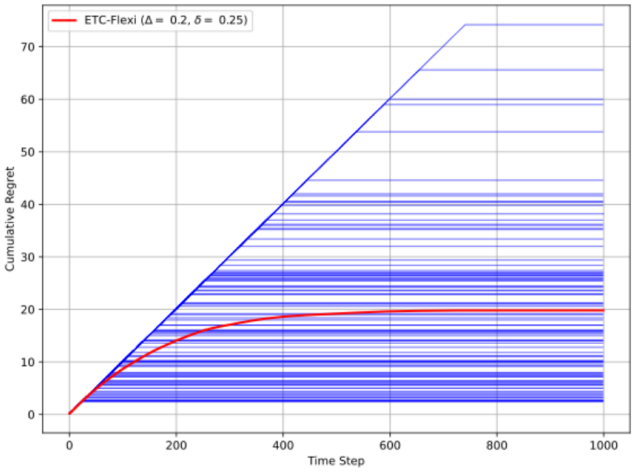
Transition from exploration to exploitation:

- UCB has a **soft** transition: exploration gradually fades out but never stops completely.
- ETC has a **hard** transition: explore phase versus commit phase. If one makes a mistake (commits to or eliminates the wrong arm), it is difficult (impossible) to correct it.

0.9.4 Simulations



UCB Algorithm



ETC Algorithm (with flexible stopping time)

Points to note:

- For each sample path, the regret growth (slope) decays gently with time; contrast with ETC, where each trajectory has a kink at the commitment time and is flat (or linear, if the commitment was wrong) afterwards.
- Exploration never stops completely.
- The mean regret is sublinear. What is its growth rate: $O(\sqrt{n})$? $O(\ln n)$? This is answered by the theorem below: $O(\ln n)$ for a fixed instance.

0.10 Regret bound for UCB

Theorem 9 (Regret of UCB)**.**
Consider the UCB algorithm applied to a stochastic 1*-subgaussian bandit with* k *arms, with error tolerance* $\delta = 1/n^2$*. Then the regret satisfies*

$$R_n \leq 3\sum_{i=1}^{k} \Delta_i + \sum_{i:\Delta_i>0} \frac{16 \ln n}{\Delta_i}.$$

Remark 16.

- The regret is **logarithmic** in n: much better than the $\sqrt{n}$ worst-case behavior of ETC.
- It is a **problem-dependent** bound: it depends on the gaps Δ_i of the instance. It blows up as $\Delta_i \to 0$, but arms with tiny gaps also cost little per pull.
- Unlike ETC, the algorithm does **not** need to know the gaps: the only parameter is δ, which depends only on n.

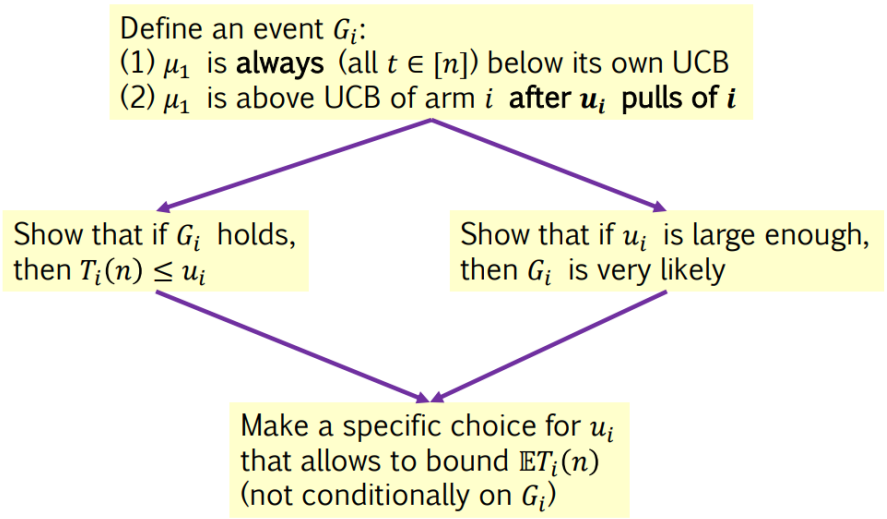
0.10.1 Proof outline

Assume without loss of generality that arm 1 is optimal ($\mu_1 = \mu^*$). By the regret decomposition $R_n = \sum_i \Delta_i \mathbb{E}[T_i(n)]$, it suffices to show that, for every suboptimal arm i,

$$\mathbb{E}[T_i(n)] \leq 3 + \frac{16 \ln n}{\Delta_i^2}.$$

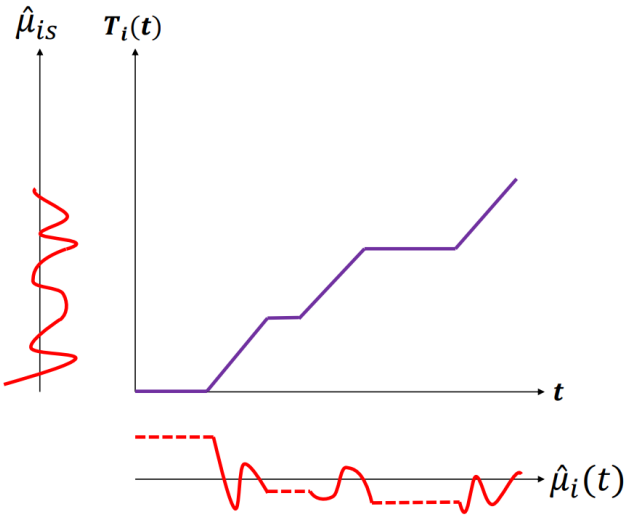
Multiplying by Δ_i and summing over i gives the theorem.

The proof proceeds in four steps:



0.10.2 Notation: $\hat{\mu}_i(t)$ versus $\hat{\mu}_{is}$

- $\hat{\mu}_i(t)$: estimator of the mean of arm i at *time* t.
- $\hat{\mu}_{is}$: estimator after s *pulls* of arm i, i.e., the average of the first s i.i.d. samples of arm i.
- They are related by $\hat{\mu}_i(t) = \hat{\mu}_{iT_i(t)}$. Remember that $T_i(t) \leq t$ for all t.



Remark 17 (Why this distinction matters)**.**
$T_i(t)$ is random and depends on the past rewards, so $\hat{\mu}_i(t)$ is *not* an average of a fixed number of i.i.d. samples, and the concentration inequality cannot be applied to it directly. In contrast, $\hat{\mu}_{is}$, for a *fixed* s, is an average of s i.i.d. 1-subgaussian samples, so $\hat{\mu}_{is} - \mu_i$ is $1/\sqrt{s}$-subgaussian. The proof therefore works with $\hat{\mu}_{is}$, and handles all possible values of s with a union bound.

0.10.3 Step 1: the good event

Definition 21 (Good event)**.**
For a suboptimal arm i and an integer u_i (to be chosen), let

$$G_i = \left\{\mu_1 < \min_{t\in[n]} UCB_1(t,\delta)\right\} \cap \left\{\hat{\mu}_{iu_i} + \sqrt{\frac{2}{u_i}\ln\frac{1}{\delta}} < \mu_1\right\}.$$

The first event says that μ_1 always stays inside its confidence interval (more precisely: below its UCB). The second says that once arm i has been pulled u_i times, its UCB does not exceed μ_1.

Bound on $\mathbb{E}[T_i(n)]$. Splitting on G_i and using $T_i(n) \leq u_i$ on G_i (step 2) and $T_i(n) \leq n$ always:

$$\mathbb{E}[T_i(n)] = \mathbb{E}\big[\mathbb{I}\{G_i\}\, T_i(n)\big] + \mathbb{E}\big[\mathbb{I}\{G_i^c\}\, T_i(n)\big] \leq u_i + \mathbb{P}(G_i^c)\, n.$$

We will need to choose u_i wisely, because there is a tradeoff:

- u_i too large $\Rightarrow$ the first term, hence the regret bound, is too large;
- u_i too small $\Rightarrow$ $\mathbb{P}(G_i^c)$ is too large $\Rightarrow$ the second term, hence the regret bound, is too large.

0.10.4 Step 2: on G_i, $T_i(n) \le u_i$

Lemma 4.
If G_i holds, then $T_i(n) \le u_i$.

Proof. By contradiction: assume G_i holds and $T_i(n) > u_i$. Then arm i was pulled more than u_i times before n, so there must be a round t at which arm i was pulled for the (u_i+1)-th time; at that round, $T_i(t-1) = u_i$. We use this to relate UCB_i to UCB_1:

$$\begin{aligned}
UCB_i(t-1,\delta) &= \hat{\mu}_i(t-1) + \sqrt{\frac{2\ln(1/\delta)}{T_i(t-1)}} \\
&= \hat{\mu}_{iu_i} + \sqrt{\frac{2}{u_i}\ln\frac{1}{\delta}} && \text{since } T_i(t-1) = u_i \\
&< \mu_1 && \text{by definition of } G_i \text{ (second event)} \\
&< UCB_1(t-1,\delta) && \text{by definition of } G_i \text{ (first event).}
\end{aligned}$$

So arm 1 had a strictly larger UCB than arm i at time $t-1$, and UCB could not have chosen $A_t = i$: contradiction. □

0.10.5 Step 3: G_i is likely if u_i is large enough

By De Morgan,

$$G_i^c = \left\{\mu_1 \ge \min_{t\in[n]} UCB_1(t,\delta)\right\} \cup \left\{\hat{\mu}_{iu_i} + \sqrt{\frac{2}{u_i}\ln\frac{1}{\delta}} \ge \mu_1\right\},$$

and we bound the probability of each event separately.

First event: μ_1 exceeds its own UCB at some time

At any time t, $UCB_1(t,\delta) = \hat{\mu}_{1s} + \sqrt{\frac{2}{s}\ln\frac{1}{\delta}}$ with $s = T_1(t)$ (and $UCB_1 = \infty$ if $s = 0$). The minimum over $t \in [n]$ only ranges over the values of s actually reached by $T_1(t)$, whereas the minimum over all $s \in [n]$ ranges over all n values; taking the min over more terms can only lower it. Hence

$$\left\{\mu_1 \ge \min_{t\in[n]} UCB_1(t,\delta)\right\} \subseteq \left\{\mu_1 \ge \min_{s\in[n]} \left(\hat{\mu}_{1s} + \sqrt{\frac{2}{s}\ln\frac{1}{\delta}}\right)\right\} = \bigcup_{s\in[n]} \left\{\mu_1 \ge \hat{\mu}_{1s} + \sqrt{\frac{2}{s}\ln\frac{1}{\delta}}\right\}.$$

Using the union bound, then the concentration inequality for the average of s i.i.d. 1-subgaussian samples (confidence interval with error probability δ):

$$\mathbb{P}\left(\mu_1 \ge \min_{t\in[n]} UCB_1(t,\delta)\right) \le \sum_{s=1}^{n} \mathbb{P}\left(\mu_1 \ge \hat{\mu}_{1s} + \sqrt{\frac{2}{s}\ln\frac{1}{\delta}}\right) \le n\delta.$$

Second event: arm i's UCB after u_i pulls is above μ_1

To bound the second event, we need an assumption on u_i: assume u_i is large enough so that

$$\Delta_i - \sqrt{\frac{2\ln(1/\delta)}{u_i}} \geq c\Delta_i \qquad \text{for some constant } 0 < c < 1.$$

Using $\mu_1 = \mu_i + \Delta_i$,

$$\begin{aligned}\mathbb{P}\left(\hat{\mu}_{iu_i} + \sqrt{\frac{2}{u_i}\ln\frac{1}{\delta}} \geq \mu_1\right) &= \mathbb{P}\left(\hat{\mu}_{iu_i} - \mu_i \geq \Delta_i - \sqrt{\frac{2}{u_i}\ln\frac{1}{\delta}}\right) \\ &\leq \mathbb{P}\big(\hat{\mu}_{iu_i} - \mu_i \geq c\Delta_i\big) \\ &\leq \exp\left(-\frac{u_i c^2 \Delta_i^2}{2}\right),\end{aligned}$$

where the last step is the Chernoff bound for subgaussian sums: $\hat{\mu}_{iu_i}$ is the average of u_i terms, so $\hat{\mu}_{iu_i} - \mu_i$ is $1/\sqrt{u_i}$-subgaussian.

Combining

By the union bound,

$$\mathbb{P}(G_i^c) \leq n\delta + \exp\left(-\frac{u_i c^2 \Delta_i^2}{2}\right).$$

0.10.6 Step 4: choice of u_i and conclusion

Back to the bound of step 1:

$$\mathbb{E}[T_i(n)] \leq u_i + \mathbb{P}(G_i^c)\, n \leq u_i + n\left(n\delta + \exp\left(-\frac{u_i c^2 \Delta_i^2}{2}\right)\right).$$

We set u_i as small as possible given the constraint $\Delta_i - \sqrt{2\ln(1/\delta)/u_i} \geq c\Delta_i$, i.e., $\sqrt{2\ln(1/\delta)/u_i} \leq (1-c)\Delta_i$:

$$u_i = \left\lceil \frac{2\ln(1/\delta)}{(1-c)^2\Delta_i^2} \right\rceil.$$

Note that there is no guarantee that $u_i \leq n$; but if it is not, the bound on $\mathbb{E}[T_i(n)]$ holds trivially since $T_i(n) \leq n$.
Recall $\delta = 1/n^2$, so $n \cdot n\delta = 1$ and $\ln(1/\delta) = 2\ln n$. For the exponential term,

$$\exp\left(-\frac{u_i c^2 \Delta_i^2}{2}\right) \leq \exp\left(-\frac{2\ln(1/\delta)\, c^2}{2(1-c)^2}\right) = \exp\left(-\frac{2\ln(n)\, c^2}{(1-c)^2}\right) = n^{-\frac{2c^2}{(1-c)^2}}$$

Therefore

$$\mathbb{E}[T_i(n)] \leq \left\lceil \frac{2\ln(1/\delta)}{(1-c)^2\Delta_i^2} \right\rceil + 1 + n^{1-\frac{2c^2}{(1-c)^2}},$$

We choose c such that the last term does not blow up with n (we need $2c^2 \geq (1-c)^2$, i.e. $c \geq \sqrt{2}-1$); e.g., $c = 1/2$, for which $\frac{2c^2}{(1-c)^2} = 2$ and the last term is $n^{-1} \leq 1$. With $c = 1/2$, $\frac{2\ln(1/\delta)}{(1-c)^2\Delta_i^2} = \frac{4\ln n}{\Delta_i^2/4} = \frac{16\ln n}{\Delta_i^2}$, so

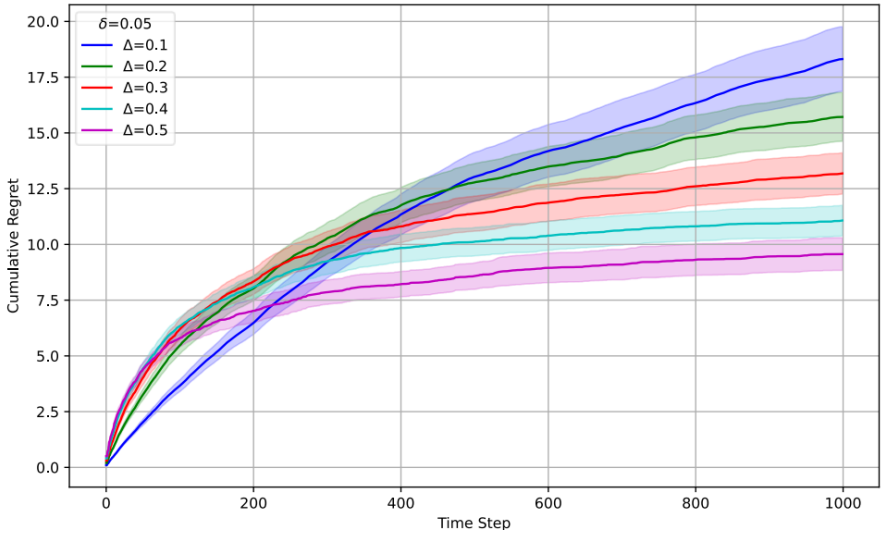
$$\mathbb{E}[T_i(n)] \leq 3 + \frac{16\ln n}{\Delta_i^2}.$$

Plugging into $R_n = \sum_i \Delta_i\, \mathbb{E}[T_i(n)]$ (with $\Delta_1 = 0$) concludes the proof. □

0.10.7 Simulations and comparison with ETC

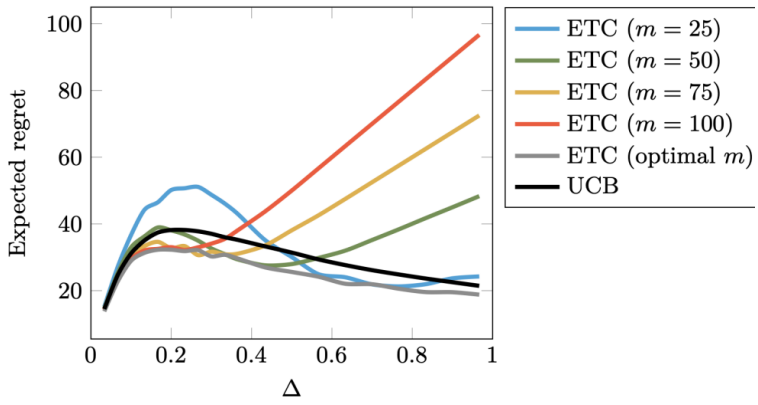
Variation with Δ

Setting: Bernoulli rewards with means $[0.5 - \frac{\Delta}{2}, 0.5 + \frac{\Delta}{2}]$.



- Sublinear regret for all values of Δ, with a *fixed* hyperparameter δ.
- The cumulative regret increases inversely with Δ (consistent with the $\frac{\ln n}{\Delta}$ term of the bound).
- The slope gradually decreases with time, signifying the shift from exploration to exploitation.
- The initial regret slope is higher for large Δ, due to expensive exploration (each pull of the bad arm costs Δ).

UCB versus ETC



UCB performs similarly to ETC with an optimally tuned m, *without* knowing the optimal m (i.e., without knowing the suboptimality gap). Figure from Chapter 7 of Lattimore & Szepesvári.

Remark 18 (Comparison of the bounds for $k = 2$)**.**
For ETC with the optimal $m = \frac{4}{\Delta^2} \ln \frac{n\Delta^2}{4}$ (previous lecture), plugging into $R_n \leq m\Delta + n\Delta \exp(-m\Delta^2/4)$ gives $R_n \lesssim \frac{4}{\Delta}\left(1 + \ln \frac{n\Delta^2}{4}\right)$, also of order $\frac{\ln n}{\Delta}$. UCB achieves the same order, $\frac{16 \ln n}{\Delta} + 3\Delta$, but with a parameter $\delta = 1/n^2$ that does not depend on Δ.

Lecture 5 — 2026-10-07: A Deeper Dive into Unstructured Bandits

In the previous lecture, we introduced the UCB algorithm and proved a logarithmic, *problem-dependent* regret bound for it, assuming the horizon n is known in advance. This lecture digs deeper into unstructured (stochastic, finite-armed) bandits along four directions:

- a **problem-independent** regret bound for UCB, of order $\sqrt{kn \ln n}$, which remains meaningful when the gaps Δ_i are arbitrarily small;
- an **asymptotic (anytime) version of UCB**, which does not need to know the horizon n and can run forever;
- **Thompson Sampling**, a randomized, Bayesian alternative to UCB;
- **best-arm identification**, a *pure exploration* problem where the goal is no longer to minimize regret but to find the best arm, with the **sequential halving** algorithm.

0.11 Recap

0.11.1 The UCB algorithm

Notation:

- $\hat{\mu}_i(t)$: estimate of the mean reward of arm i at time t;
- $T_i(t)$: number of pulls of arm i at time t;
- upper confidence bound of arm i at time t (for 1-subgaussian rewards):

$$UCB_i(t,\delta) = \hat{\mu}_i(t) + \sqrt{\frac{2\ln(1/\delta)}{T_i(t)}}, \qquad UCB_i(t,\delta) = \infty \text{ if } T_i(t) = 0.$$

UCB algorithm:

for $t = 1, 2, \ldots, n$ **do**:
choose $A_t = \arg\max_i UCB_i(t-1,\delta)$;
observe X_t and update $UCB_{A_t}(t,\delta)$.

0.11.2 Regret and the UCB regret bound

- **Instantaneous regret** at round t: $\mu^* - \mu_{A_t} = \Delta_{A_t}$.
- **Cumulative regret**: $R_n = \sum_{t=1}^{n} \mathbb{E}[\mu^* - X_t]$.
- **Regret decomposition**: $R_n = \sum_{i\in[k]} \Delta_i \, \mathbb{E}[T_i(n)]$.
- Sublinear cumulative regret $\Leftrightarrow$ vanishing instantaneous regret.

Theorem 10 (Regret of UCB, recap)**.**
For the UCB algorithm applied to a stochastic 1*-subgaussian bandit with* k *arms and an error tolerance* $\delta = 1/n^2$*,*

$$R_n \le 3\sum_{i=1}^{k} \Delta_i + \sum_{i:\Delta_i>0} \frac{16\ln n}{\Delta_i}.$$

0.11.3 Proof strategy (recap)

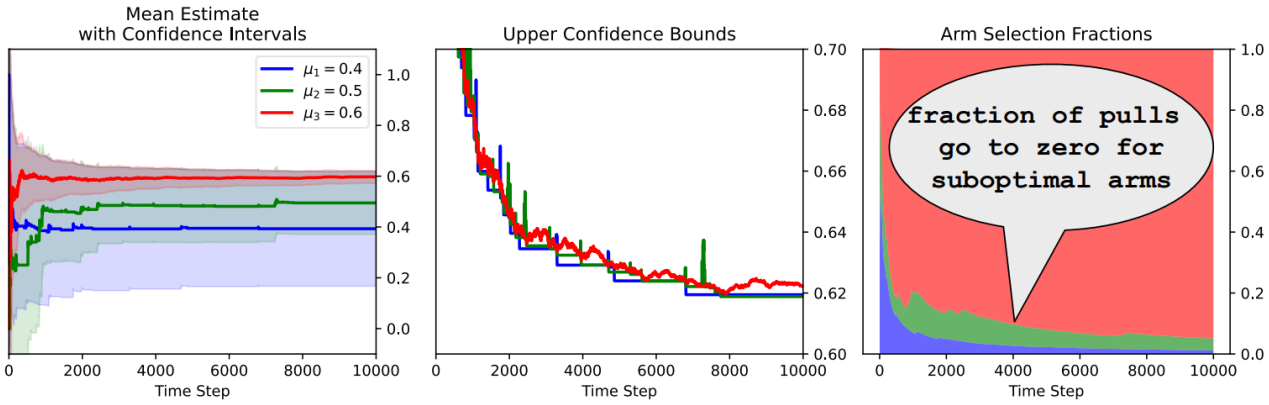
To bound the regret, it suffices to bound $\mathbb{E}[T_i(n)]$ for each suboptimal arm i, since $R_n = \sum_i \Delta_i \, \mathbb{E}[T_i(n)]$. The proof had four steps:

1. Define a **good event** G_i: (1) μ_1 is always below its own UCB, and (2) μ_1 is above the UCB of arm i after u_i pulls of i.
2. Show that if G_i holds, then $T_i(n) \le u_i$.
3. Show that if u_i is large enough, then G_i is very likely.
4. Make a specific choice $u_i = \frac{8 \ln(1/\delta)}{\Delta_i^2}$ (i.e. $c = 1/2$), which gives $\mathbb{E}[T_i(n)] \le 3 + \frac{16 \ln n}{\Delta_i^2}$.

0.12 UCB in practice: confidence intervals and the role of δ

0.12.1 Evolution of the confidence intervals

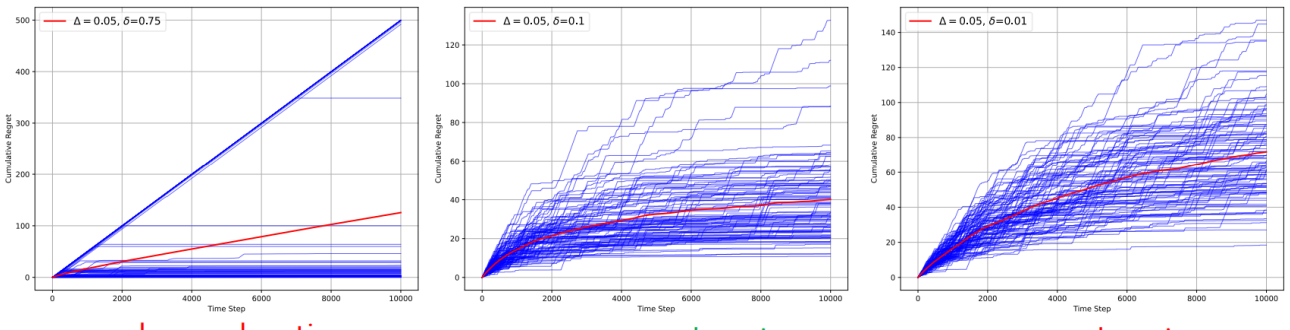
Setting: three-armed Bernoulli bandit with means $[0.4, 0.5, 0.6]$.



- The **upper confidence bounds of the arms are nearly identical throughout**. This is a direct consequence of the algorithm: whenever one arm's UCB is the largest, that arm is pulled, its bonus shrinks and its UCB comes down towards the others. UCB thus keeps all the UCBs "balanced".
- The **worse the arm, the less often it is played**: to have the same UCB as a better arm, a worse arm (lower $\hat{\mu}_i$) needs a larger bonus, i.e. fewer pulls. Worse arms therefore have wider confidence intervals.
- The fraction of pulls $T_i(t)/t$ goes to zero for suboptimal arms (they are pulled only $O(\ln t)$ times).

0.12.2 The effect of δ

Setting: two-armed Bernoulli bandit with means $[0.5 - \Delta/2, \, 0.5 + \Delta/2]$.



- Smaller $\delta \Rightarrow$ larger confidence intervals $\Rightarrow$ more exploration.
- **Too large** δ (under-exploration): narrow intervals, so there is a real chance that the best arm gets unlucky early, its UCB drops below the mean of the other arm, and the algorithm effectively "commits" to the wrong arm $\Rightarrow$ large (linear) regret.
- **Too small** δ (over-exploration): very wide intervals, the bonus dominates the empirical mean for a long time and suboptimal arms are pulled too often $\Rightarrow$ large regret.
- In between, correct exploration gives the smallest regret. This is the same qualitative behavior as for adaptive ETC.

0.13 A problem-independent regret bound for UCB

0.13.1 Motivation

The bound we proved,

$$R_n \le 3\sum_{i=1}^{k} \Delta_i + \sum_{i:\Delta_i>0} \frac{16\ln n}{\Delta_i},$$

is **problem-dependent**: it depends on the gaps of the instance. If some Δ_i is arbitrarily small, the bound blows up and becomes meaningless: it can be much larger than the trivial bound

$$R_n = \sum_i \Delta_i\, \mathbb{E}[T_i(n)] \le \Delta_{\max} \sum_i \mathbb{E}[T_i(n)] = n\,\Delta_{\max}, \qquad \Delta_{\max} = \max_i \Delta_i,$$

which always holds (at most, we pay the largest gap at every round). For instance, with $\Delta_i = 10^{-6}$ and $n = 10^4$, the term $\frac{16\ln n}{\Delta_i} \approx 1.5 \cdot 10^8$, while $n\Delta_{\max} \le 10^4$ for rewards in $[0,1]$. Yet, intuitively, an arm with a tiny gap costs almost nothing per pull: even if it is pulled at every round, it contributes at most $n\Delta_i$ to the regret. Can we get a **problem-independent** (worst-case) bound, that holds uniformly over all instances?

Theorem 11 (Alternate (problem-independent) regret bound for UCB)**.**
For the UCB algorithm applied to a stochastic 1*-subgaussian bandit with* k *arms and an error tolerance* $\delta = 1/n^2$*, the regret satisfies*

$$R_n \le 3\sum_{i=1}^{k} \Delta_i + 8\sqrt{kn\ln n}.$$

Remark 19.

- We trade the term $\sum_{i\in[k]} 1/\Delta_i$ (times $\ln n$) for a term $\sqrt{nk}$ (times $\sqrt{\ln n}$).
- This bound is sublinear for *every* instance, independently of the gaps: $R_n = O(\sqrt{kn\ln n})$. The term $3\sum_i \Delta_i$ does not grow with n (and is at most $3k$ for rewards in $[0,1]$).
- The algorithm is **unchanged**: this is merely a different proof strategy. Both bounds hold simultaneously, and $R_n \le \min(\text{both})$.

0.13.2 Proof

Ingredients

Recall the main ingredients of the previous proof:

- Regret decomposition lemma: $R_n = \sum_i \Delta_i \, \mathbb{E}[T_i(n)]$.
- Bound derived from the concentration inequality: $\mathbb{E}[T_i(n)] \le 3 + \frac{16 \ln n}{\Delta_i^2}$.
- A **trivial bound**: $\mathbb{E}[T_i(n)] \le n$ (for all arms, under all conditions). More precisely, $\sum_i T_i(n) = n$.

When is the trivial bound tighter than the concentration bound? When

$$\frac{16 \ln n}{\Delta_i^2} \ge n \iff \Delta_i \le \sqrt{\frac{16 \ln n}{n}}.$$

The regret from such an arm i is then at most

$$\Delta_i \, \mathbb{E}[T_i(n)] \le \sqrt{\frac{16 \ln n}{n}} \cdot n = O\big(\sqrt{n \ln n}\big).$$

So arms with very small gaps contribute at most $O(\sqrt{n \ln n})$ each, which suggests that the goal $R_n = O(\sqrt{nk \ln n})$ is reachable.

Refined strategy

- Split the arms into two sets, according to a threshold $\Delta > 0$ (a proof parameter, *not* known or used by the algorithm):
  - **nearly optimal arms**: $\Delta_i < \Delta$; use the trivial bound for these arms;
  - **clearly suboptimal arms**: $\Delta_i \ge \Delta$; use the concentration bound for these arms.
- Optimize Δ to get the best expression.

Computation

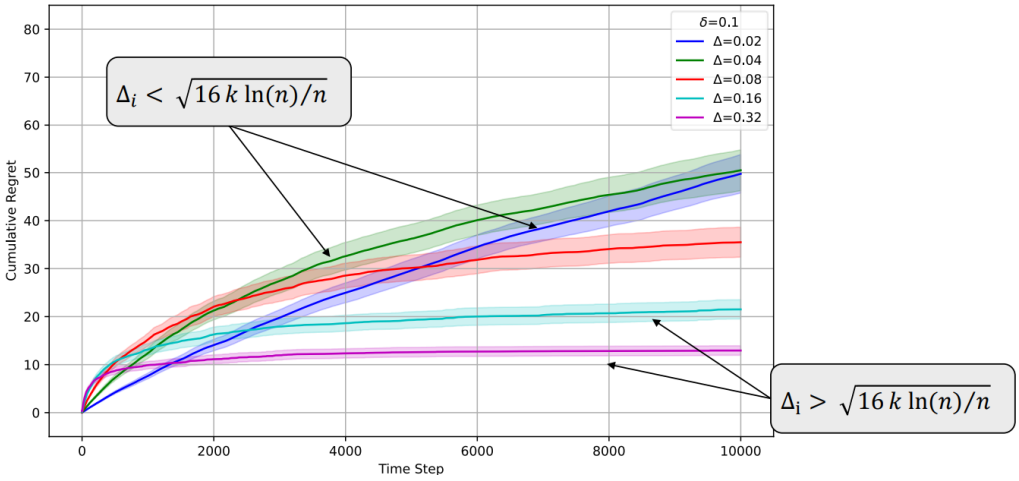
$$\begin{aligned}
R_n &= \sum_i \Delta_i \, \mathbb{E}[T_i(n)] = \sum_{i:\Delta_i<\Delta} \Delta_i \, \mathbb{E}[T_i(n)] + \sum_{i:\Delta_i\ge\Delta} \Delta_i \, \mathbb{E}[T_i(n)] \\
&\le n\Delta + \sum_{i:\Delta_i\ge\Delta} \Delta_i \, \mathbb{E}[T_i(n)] && \text{since } \textstyle\sum_{i:\Delta_i<\Delta} \Delta_i \mathbb{E}[T_i(n)] \le \Delta \sum_i \mathbb{E}[T_i(n)] = n\Delta \\
&\le n\Delta + \sum_{i:\Delta_i\ge\Delta} \Delta_i \left(3 + \frac{16 \ln n}{\Delta_i^2}\right) && \text{concentration bound} \\
&= n\Delta + \sum_{i:\Delta_i\ge\Delta} \left(3\Delta_i + \frac{16 \ln n}{\Delta_i}\right) \\
&\le \sum_i 3\Delta_i + n\Delta + \frac{16k \ln n}{\Delta} && \text{since } \tfrac{1}{\Delta_i} \le \tfrac{1}{\Delta} \text{ and there are at most } k \text{ such arms.}
\end{aligned}$$

Optimizing Δ. The function $\Delta \mapsto n\Delta + \frac{16k \ln n}{\Delta}$ is minimized when its derivative $n - \frac{16k \ln n}{\Delta^2}$ vanishes, i.e. for

$$\Delta = \sqrt{\frac{16k \ln n}{n}}, \qquad \text{giving} \qquad n\Delta + \frac{16k \ln n}{\Delta} = 2\sqrt{16kn \ln n} = 8\sqrt{kn \ln n}.$$

Hence $R_n \le \sum_i 3\Delta_i + 8\sqrt{kn \ln n}$. □

0.13.3 Revisiting the regret plot



- It takes $O\left(\frac{\ln(1/\delta)}{\Delta^2}\right)$ pulls to separate two arms with gap Δ with probability $1-\delta$ (the confidence interval width $\sqrt{2\ln(1/\delta)/T}$ must become smaller than Δ).
- Δ_i **large** ($\Delta_i > \sqrt{16k\ln n/n}$) $\Rightarrow$ confidence intervals separate early (within the horizon) $\Rightarrow$ **logarithmic** regret curves (the regime of the problem-dependent bound).
- Δ_i **small** ($\Delta_i < \sqrt{16k\ln n/n}$) $\Rightarrow$ separation happens late (possibly not before n) $\Rightarrow$ faster-growing (almost linear) regret curves, but with a small slope Δ_i (the regime of the trivial bound).

0.14 Asymptotic UCB: unknown horizon

0.14.1 Motivation: why a fixed δ is not enough

So far, the horizon n was fixed a priori (and used to set $\delta = 1/n^2$). This makes sense in some applications, not in others. What happens when we do not know the horizon? Equivalently: what if we want UCB to run forever?

- The main concern is that we must **eliminate any possibility of persistent error**: any error probability, however small, could lead to a large (infinite!) cumulative regret.
- In the current formulation, the chance of error is about the chance of the concentration event failing, which is about δ.
  - We cannot choose $\delta = 0$: the confidence interval would be infinite.
  - We cannot choose a constant $\delta > 0$: there is then a small (constant) probability of getting into a bad configuration at some time t (e.g. the best arm gets unlucky samples, its UCB drops below the mean of a suboptimal arm, and it is ignored forever) $\Rightarrow$ the regret grows linearly forever, so the expected regret is not sublinear.
- In effect, we need $\delta \to 0$ over time.
- Consequently, we need $T_i(n) \to \infty$ for *all* arms (no arm can be abandoned for good); we just control *how quickly* suboptimal arms are pulled.

0.14.2 Algorithm

The fact that $\delta \to 0$ over time is reflected in the confidence intervals:

- originally: $UCB_i(t, \delta) = \hat{\mu}_i(t) + \sqrt{\frac{2\ln(1/\delta)}{T_i(t)}}$;
- new formulation: $UCB_i(t) = \hat{\mu}_i(t) + \sqrt{\frac{8\ln t}{T_i(t)}}$.

This corresponds to choosing a time-varying confidence level $\delta_t = t^{-4}$, since $2\ln(1/\delta_t) = 2\ln(t^4) = 8\ln t$. The rest of the algorithm remains identical!

Definition 22.
(Asymptotic UCB).

for $t = 1, 2, \ldots$ **do**:

choose $A_t = \arg\max_i \left(\hat{\mu}_i(t-1) + \sqrt{\frac{8\ln t}{T_i(t-1)}} \right)$ (with the index $= \infty$ if $T_i(t-1) = 0$);

observe X_t and update $UCB_i(t)$ for all i.

Remark 20.

- Now *all* the UCBs must be updated at each round, not only the one of the pulled arm: the bonus depends on t through $\ln t$.
- If an arm is not played for a long time, its confidence interval slowly grows (its T_i is frozen but $\ln t$ increases). Its UCB grows (very slowly) without bound, so an arm cannot be neglected forever: each arm is pulled every so often.
- Hence no "error" (mistake in identifying the best arm) is permanent.
- This is **not** the same as $\delta = 1/n^2$ in the proof of last week: that δ was fixed over time (but depended on the known horizon n), whereas here δ_t gradually decreases with the current time t, and no knowledge of n is needed ("anytime" algorithm).

Theorem 12 (Regret of asymptotic UCB)**.**
For any 1*-subgaussian bandit, the regret of the asymptotic UCB algorithm satisfies, for all* n*,*

$$R_n \le \sum_{i:\Delta_i>0} \left(4\Delta_i + \frac{32\ln n}{\Delta_i} \right), \qquad \textit{hence} \qquad \limsup_{n\to\infty} \frac{R_n}{\ln n} \le \sum_{i:\Delta_i>0} \frac{32}{\Delta_i}.$$

0.14.3 Regret bound: proof sketch

As in the fixed-horizon version, the key is to prove that, for every suboptimal arm i,

$$\mathbb{E}[T_i(n)] \le \frac{32\ln n}{\Delta_i^2} + 4,$$

and the theorem then follows from the regret decomposition. We broadly follow the same steps:

- Define good events such that, under the good event ("normal circumstances"), the number of pulls of a suboptimal arm is bounded by some u_i.
- Show that the probability of the bad event is small (concentration inequality).

Differences from the fixed-horizon case:

- the good event now **depends on time** t;
- a careful use of the union bound (over rounds, then over the possible numbers of pulls).

As before, assume without loss of generality that arm 1 is optimal.

Defining good events: fixed horizon versus asymptotic

Fixed horizon	**Asymptotic**
$G = G_1 \cap G_i$, where	$G_t = G_{1,t} \cap G_{i,t}$, where
$G_1 = \left\{\mu_1 < \min_{t\in[n]} \hat{\mu}_1(t) + \sqrt{\frac{2}{T_1(t)} \ln \frac{1}{\delta}}\right\}$	$G_{1,t} = \left\{\mu_1 < \hat{\mu}_1(t) + \sqrt{\frac{8}{T_1(t)} \ln t}\right\}$
$G_i = \left\{\hat{\mu}_{iu_i} + \sqrt{\frac{2}{u_i} \ln \frac{1}{\delta}} < \mu_1\right\}$	$G_{i,t} = \left\{\mu_i > \hat{\mu}_i(t) - \sqrt{\frac{8}{T_i(t)} \ln t}\right\}$
Conditional on G: $T_i(n) \le \frac{8\ln(1/\delta)}{\Delta_i^2} =: u_i$.	Conditional on G_t: $T_i(t) \le \frac{32\ln t}{\Delta_i^2} =: u_i(t)$ **or** arm i is not pulled in round t.

Interpretation of the asymptotic good events:

- $G_{1,t}$: the UCB of arm 1 is above its true mean at time t;
- $G_{i,t}$: the LCB of arm i is below its true mean at time t.

The fixed-horizon good event is a single event covering the whole run; the asymptotic one is a family of events, one per round, and only requires the two confidence intervals to be correct *at round* t.

Remark 21 (Indices)**.**
As in the slides, we write all quantities "at time t" to lighten notation. Strictly speaking, the decision at round t uses $\hat{\mu}_i(t-1)$ and $T_i(t-1)$ (with the bonus $\sqrt{8\ln t / T_i(t-1)}$); the argument is identical.

Step 1: under G_t, a suboptimal arm with many pulls is not chosen

We need to show

$$G_t \cap \{A_t = i\} \Rightarrow T_i(t) \le \frac{32\ln t}{\Delta_i^2}, \qquad \text{equivalently} \qquad G_t \cap \left\{T_i(t) > \frac{32\ln t}{\Delta_i^2}\right\} \Rightarrow A_t \neq i.$$

This uses elementary logic: $A \Rightarrow B \iff (B^c \Rightarrow A^c)$, and more generally $C \cap A \Rightarrow B \iff (C \cap B^c \Rightarrow A^c)$. (Also: $A \Rightarrow B \cup C \iff (A \cap B^c \Rightarrow C)$, which gives the "$T_i(t) \le u_i(t)$ *or* arm i not pulled" form of the table.)

Proof. Assume that both G_t and $T_i(t) > \frac{32\ln t}{\Delta_i^2}$ occur.

- From $T_i(t) > \frac{32\ln t}{\Delta_i^2}$: $\Delta_i^2 > \frac{32\ln t}{T_i(t)} = 4 \cdot \frac{8\ln t}{T_i(t)}$, so
$$\Delta_i > 2\sqrt{\frac{8\ln t}{T_i(t)}}.$$
- From $G_{i,t}$ and $G_{1,t}$, using $\mu_1 = \mu_i + \Delta_i$:
$$\underbrace{\hat{\mu}_i(t) - \sqrt{\frac{8\ln t}{T_i(t)}}}_{LCB_i(t)} + \Delta_i < \mu_i + \Delta_i = \mu_1 < \underbrace{\hat{\mu}_1(t) + \sqrt{\frac{8\ln t}{T_1(t)}}}_{UCB_1(t)}.$$
- Combining, since $\Delta_i > 2\sqrt{8\ln t / T_i(t)}$:
$$UCB_i(t) = \hat{\mu}_i(t) + \sqrt{\frac{8\ln t}{T_i(t)}} = \hat{\mu}_i(t) - \sqrt{\frac{8\ln t}{T_i(t)}} + 2\sqrt{\frac{8\ln t}{T_i(t)}} < \hat{\mu}_i(t) - \sqrt{\frac{8\ln t}{T_i(t)}} + \Delta_i < UCB_1(t).$$

So $UCB_i(t) < UCB_1(t)$, and arm i is not pulled in round t. □

Graphically, on the good event: $\cdots < LCB_i(t) < \mu_i < \cdots < \mu_1 < UCB_1(t)$, and once $T_i(t)$ is large, the interval of arm i is narrower than $\Delta_i/2$ on each side, so $UCB_i(t) < \mu_1 < UCB_1(t)$.

Step 2: bounding $\mathbb{E}[T_i(n)]$

Fixed horizon (recap): $\mathbb{E}[T_i(n)] \le \mathbb{E}[\mathbb{I}\{G\}T_i(n)] + \mathbb{E}[\mathbb{I}\{G^c\}T_i(n)] \le u_i + \mathbb{P}(G^c)\,n$, with $\mathbb{P}(G_i^c) \le n\delta + \exp\left(-\frac{u_i c^2 \Delta_i^2}{2}\right)$; put together, this gives the desired result. This approach is **too loose for the asymptotic version** ("$n = \infty$"): the term $\mathbb{P}(G^c)\,n$ requires $\mathbb{P}(G^c) \lesssim 1/n$ for a known n.

Asymptotic: we decompose *round by round*. Since $\mathbb{I}\{G_t\} + \mathbb{I}\{G_t^c\} \equiv 1$,

$$\begin{aligned}\mathbb{E}\big[\mathbb{I}\{A_t = i\}\big] &= \mathbb{E}\big[\mathbb{I}\{G_t\}\,\mathbb{I}\{A_t = i\}\big] + \mathbb{E}\big[\mathbb{I}\{G_t^c\}\,\mathbb{I}\{A_t = i\}\big] \\ &\le \mathbb{E}\big[\mathbb{I}\{T_i(t) \le u_i(t)\}\,\mathbb{I}\{A_t = i\}\big] + \mathbb{E}\big[\mathbb{I}\{G_t^c\}\big],\end{aligned}$$

using step 1 for the first term ($G_t \cap \{A_t = i\} \subseteq \{T_i(t) \le u_i(t)\} \cap \{A_t = i\}$) and $\mathbb{I}\{A_t = i\} \le 1$ for the second. Summing over rounds, and using $u_i(t) \le u_i(n)$ for $t \le n$:

$$\begin{aligned}\mathbb{E}[T_i(n)] = \sum_{t\in[n]} \mathbb{E}\big[\mathbb{I}\{A_t = i\}\big] &\le \sum_{t\in[n]} \mathbb{E}\big[\mathbb{I}\{T_i(t) \le u_i(n)\}\,\mathbb{I}\{A_t = i\}\big] + \sum_{t\in[n]} \mathbb{P}(G_t^c) \\ &\le u_i(n) + 1 + \sum_{t=1}^{\infty} \mathbb{P}(G_t^c) \\ &\le \frac{32\ln n}{\Delta_i^2} + 1 + \sum_{t=1}^{\infty} \frac{2}{t^3} \le \frac{32\ln n}{\Delta_i^2} + 4.\end{aligned}$$

Explanations:

- **Counting argument**: $\sum_t \mathbb{I}\{T_i(t) \le u_i(n)\}\,\mathbb{I}\{A_t = i\}$ counts the rounds in which arm i is pulled while its counter is at most $u_i(n)$. How many times can one increment a counter while it is less than u_i? At most $u_i + 1$ times (each pull increments T_i by one).
- $\mathbb{P}(G_t^c) \le 2t^{-3}$ (shown below), and $\sum_{t=1}^{\infty} \frac{1}{t^3} \le 1 + \int_1^{\infty} \frac{1}{t^3}\,dt = 1.5$, so $\sum_t \frac{2}{t^3} \le 3$. Crucially, this series **converges**: the total probability of bad events over an infinite horizon is finite. This is why δ_t must decrease fast enough ($\delta_t = t^{-4}$).

Step 3: bounding $\mathbb{P}(G_t^c)$

We need to show $\mathbb{P}(G_{1,t}^c) \le t^{-3}$; the same bound holds for $\mathbb{P}(G_{i,t}^c)$ (by symmetry, for the lower tail). Together, by the union bound,

$$\mathbb{P}(G_t^c) \le \mathbb{P}(G_{1,t}^c) + \mathbb{P}(G_{i,t}^c) \le 2t^{-3}.$$

As in the fixed-horizon proof, $T_1(t)$ is random, so we take a union over all its possible values $s \in [t]$ (if $T_1(t) = 0$, the UCB is infinite and $G_{1,t}$ holds), and work with $\hat{\mu}_{1,s}$, the average of the first s samples of arm 1:

$$G_{1,t}^c = \left\{\mu_1 \ge \hat{\mu}_1(t) + \sqrt{\frac{8\ln t}{T_1(t)}}\right\} \subseteq \bigcup_{s=1}^{t} \left\{\mu_1 \ge \hat{\mu}_{1,s} + \sqrt{\frac{8\ln t}{s}}\right\}.$$

Then, by the union bound and the Chernoff bound (for s i.i.d. 1-subgaussian samples, $\mathbb{P}\big(\mu \ge \hat{\mu}_s + \sqrt{2\ln(1/\delta)/s}\big) \le \delta$), with $\delta = t^{-4}$:

$$\mathbb{P}(G_{1,t}^c) \le \sum_{s=1}^{t} \mathbb{P}\left(\mu_1 \ge \hat{\mu}_{1,s} + \sqrt{\frac{8\ln t}{s}}\right) = \sum_{s=1}^{t} \mathbb{P}\left(\mu_1 \ge \hat{\mu}_{1,s} + \sqrt{\frac{2\ln(1/t^{-4})}{s}}\right) \le \sum_{s=1}^{t} t^{-4} = t^{-3}.$$

Remark 22 (Why t^{-4}?)**.**
The union over the t possible values of $T_1(t)$ costs a factor t, so the per-s error probability δ_t must satisfy that $\sum_t t\,\delta_t$ converges. With $\delta_t = t^{-4}$, we get $\sum_t 2t^{-3} < \infty$. (A choice like $\delta_t = t^{-2}$ would give $\sum_t t^{-1} = \infty$, which is not enough for this proof.)

Conclusion

Multiplying by Δ_i and summing over suboptimal arms gives $R_n \leq \sum_{i:\Delta_i>0} \left(4\Delta_i + \frac{32 \ln n}{\Delta_i}\right)$. Dividing by $\ln n$ and letting $n \to \infty$:

$$\limsup_{n\to\infty} \frac{R_n}{\ln n} \leq \sum_{i:\Delta_i>0} \frac{32}{\Delta_i}.$$

This is the **same order** as the result of last week ($\ln n/\Delta_i$), but now we **do not need to know** n **a priori**! The price is only a worse constant (32 instead of 16).

0.14.4 Asymptotically optimal UCB

The constant can be squeezed further. Algorithm 6 and Theorem 8.1 in Lattimore & Szepesvári give an asymptotically optimal result:

$$\limsup_{n\to\infty} \frac{R_n}{\ln n} \leq \sum_{i:\Delta_i>0} \frac{2}{\Delta_i}.$$

Differences with our version:

- the exploration bonus is $\sqrt{\frac{2 \ln f(t)}{T_i(t-1)}}$ with $f(t) = 1+t\ln^2(t)$, instead of $f(t) = t^4$ (i.e. $\sqrt{8 \ln t/T_i(t-1)}$): a much smaller bonus, hence less exploration;
- the good events measure the UCB crossing the threshold $\mu_1 - \varepsilon$ rather than μ_1 (a finer analysis).

"Asymptotically optimal" means that no reasonable algorithm can have a better constant: this matches the lower bound of Lai and Robbins (see below), which we do not study in this course.

0.14.5 UCB: a long history

- **1985**: Lai and Robbins give a clean formulation of the multi-armed bandit problem.
  - They proved the fundamental **logarithmic lower bound** (we do not study lower bounds; they serve to understand the best possible performance of any algorithm).
  - They also introduced the idea of **optimism** and of using confidence bounds.
- **1985–1995**: UCB algorithm proposed, asymptotic regret bounds.
- **2002**: Auer et al. proposed the **UCB1** algorithm and analysed its (finite-time) regret. This is what we saw in class.
- **2010–2017**: asymptotically optimal and minimax-optimal algorithms: MOSS, KL-UCB, etc.
- Coming up: Thompson Sampling, an algorithm from **1933**!

0.15 Thompson Sampling

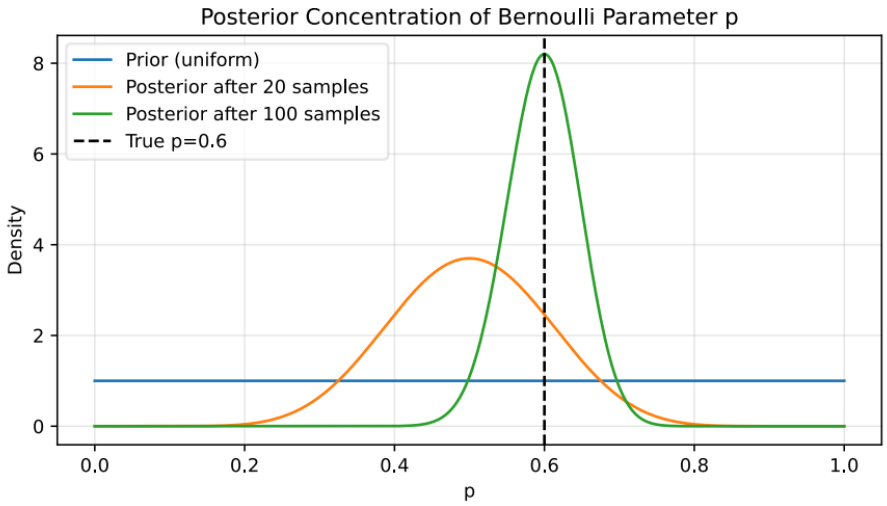
0.15.1 Prelude: Bayesian statistics

In Bayesian statistics, we maintain a **belief distribution** over unknown quantities.

- The belief distribution measures all we know about the unknown.
- Without any data, the belief distribution has a large variance.
- With more data, the belief distribution concentrates around the true parameter.

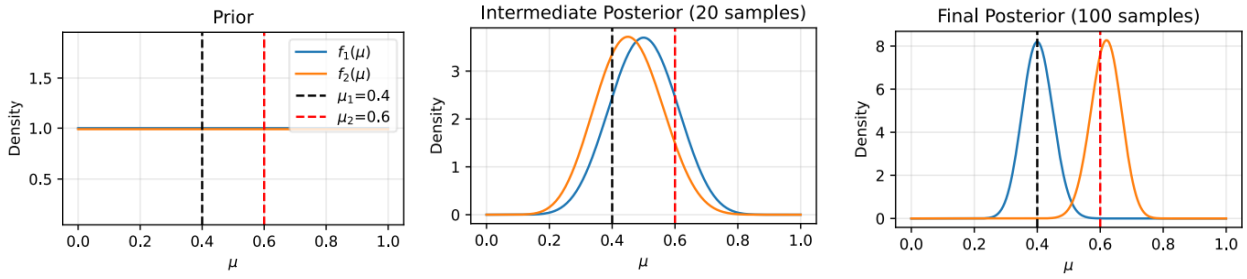
Formally, the belief is updated with Bayes' rule: starting from a **prior** $f_0(\mu)$, after observing data D we get the **posterior**

$$f(\mu \mid D) = \frac{\mathbb{P}(D \mid \mu)\, f_0(\mu)}{\int \mathbb{P}(D \mid \mu')\, f_0(\mu')\, d\mu'} \propto \mathbb{P}(D \mid \mu)\, f_0(\mu).$$



0.15.2 Bayesian view of bandits

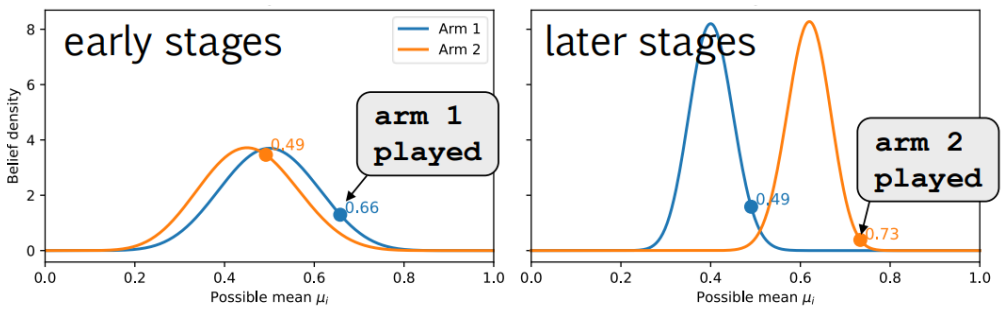
- In multi-armed bandits with $\{0, 1\}$ rewards, the unknown variables are the mean rewards of each arm: $\mu_1, \mu_2, \ldots, \mu_k$.
- We maintain a belief $f_{i,t}(\mu)$ for all arms i, at all times t, instead of $\hat{\mu}_i$ and a UCB.
- Initially, we know nothing about μ_i: any value in $[0, 1]$ is equally likely. The **prior** belief $f_{i,0}(\mu)$ for each arm i is the uniform distribution on $[0, 1]$.
- Over time, we pull arms and observe rewards, and the belief $f_{i,t}(\cdot)$ concentrates near μ_i.



0.15.3 Algorithm

Thompson Sampling is a **randomized** algorithm for multi-armed bandits.

- The algorithm maintains a belief distribution over the possible arm values at all times. Contrast: UCB maintains a point estimate and a confidence interval of the arm values.
- At each step, the algorithm:
  - samples values $\tilde{\mu}_1, \tilde{\mu}_2, \ldots, \tilde{\mu}_k$ according to the belief distributions;
  - plays the arm with the highest sampled value (a form of optimism);
  - observes the data and updates the belief of the played arm.



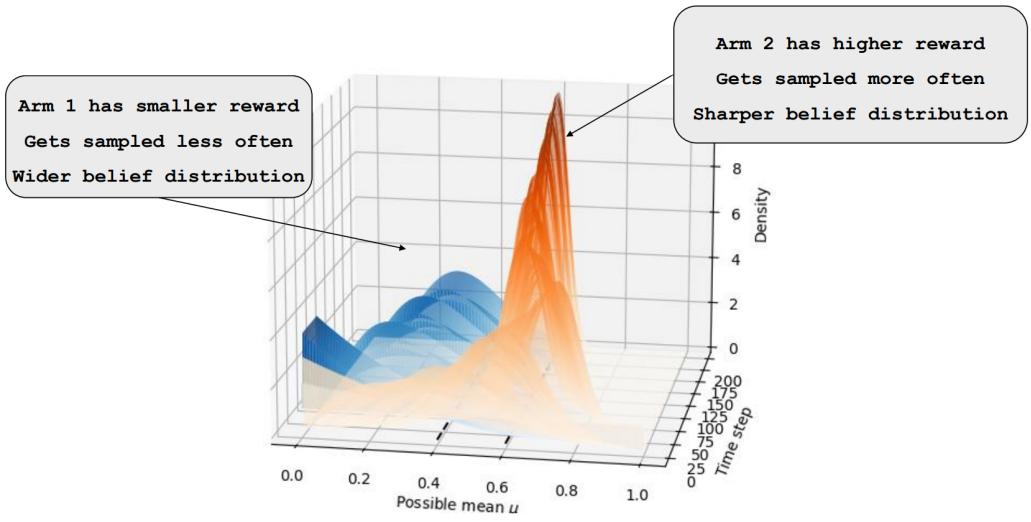
Definition 23.
(Thompson Sampling).

for $t = 1, 2, \ldots$ **do**:
  for $i = 1, 2, \ldots, k$ **do**: sample $\tilde{\mu}_i(t) \sim f_{i,t}(\cdot)$;
  choose $A_t = \arg\max_i \tilde{\mu}_i(t)$;
  observe X_t, update $f_{A_t,t}(\cdot)$.

0.15.4 Intuition

- Exploration happens through the **randomness in the sampling**.
- Initially, beliefs are weak ⇒ larger variance ⇒ samples are spread out, any arm can come out on top ⇒ more exploration.
- Over time, beliefs concentrate, samples are close to the true means ⇒ the arm with the highest mean almost always wins ⇒ exploitation.
- Smooth transition from exploration to exploitation, and the beliefs of the candidates for best arm concentrate more quickly (they are sampled more).
- Equivalently, Thompson Sampling plays each arm with the (posterior) *probability that it is the best arm*: an arm that is clearly worse has a tiny probability of producing the largest sample, so it is rarely played.

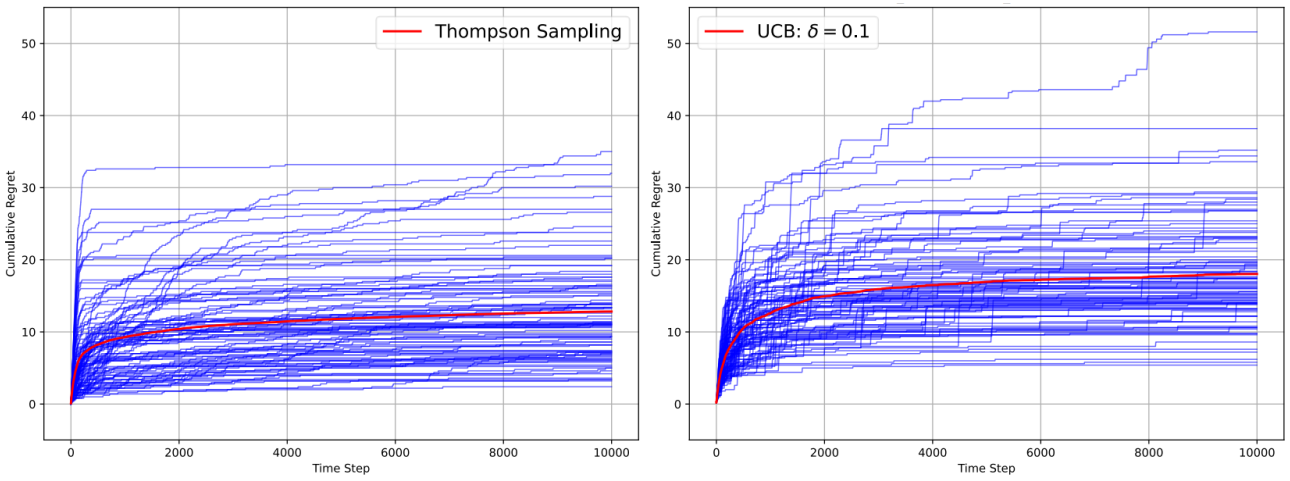
Evolution of the posterior



This is the Bayesian analogue of what we observed for UCB: worse arms have wider confidence intervals.

Thompson Sampling versus UCB

Setting: two-armed Bernoulli bandit with means $[0.4, 0.6]$.



- The regret curves of Thompson Sampling are similar to those of UCB.
- Smooth exploration-exploitation tradeoff.
- In practice, Thompson Sampling can be better than UCB.

0.15.5 Calculating posteriors: Beta-Bernoulli

In the Bernoulli bandit setting (rewards 0 or 1), we use a convenient **conjugate prior**: the **Beta distribution**.

Definition 24 (Beta distribution)**.**
The Beta distribution with parameters $\alpha, \beta > 0$ has density on $[0, 1]$

$$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}, \qquad B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}.$$

Its mean is $\frac{\alpha}{\alpha+\beta}$, and its variance $\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$ decreases as $\alpha+\beta$ grows. Beta$(1, 1)$ is the uniform distribution on $[0, 1]$.

- Belief for arm i: $\mu_i \sim \text{Beta}(\alpha_i, \beta_i)$, where α_i counts the past successes and β_i the past failures (starting from $\alpha_i = \beta_i = 1$ for the uniform prior, so $\alpha_i - 1$ successes and $\beta_i - 1$ failures have been observed).
- After observing a reward $r_t \in \{0, 1\}$ from arm i:

$$\alpha_i \leftarrow \alpha_i + r_t, \qquad \beta_i \leftarrow \beta_i + (1 - r_t).$$

- The posterior remains a Beta distribution (this is what *conjugate* means): updating beliefs just amounts to incrementing these two counters.
- More data $\Rightarrow$ the posterior becomes narrower (more confident).
- Successes (1) shift the density rightward; failures (0) shift it leftward.

Remark 23 (Why the posterior stays Beta)**.**
With a Beta(α, β) prior and one Bernoulli observation $r \in \{0, 1\}$ (likelihood $\mu^r(1-\mu)^{1-r}$), Bayes' rule gives

$$f(\mu \mid r) \propto \mu^r(1-\mu)^{1-r} \cdot \mu^{\alpha-1}(1-\mu)^{\beta-1} = \mu^{(\alpha+r)-1}(1-\mu)^{(\beta+1-r)-1},$$

which is the (unnormalized) density of Beta$(\alpha + r, \beta + 1 - r)$.

Remark 24 (Example)**.**
Arm i with a uniform prior, pulled 5 times with rewards $1, 0, 1, 1, 0$: the posterior is $\text{Beta}(1+3, 1+2) = \text{Beta}(4, 3)$, with mean $4/7$. Thompson Sampling draws $\tilde{\mu}_i \sim \text{Beta}(4, 3)$, compares it with the samples of the other arms, and plays the arm with the largest sample.

0.15.6 Some practical points

- Thompson Sampling has similar theoretical regret guarantees as UCB.
- In practice, it can sometimes achieve lower regret than UCB; it is usually competitive in performance. Benefits are seen more often in linear bandits (next lecture).
- Computing the posterior can be hard without conjugate priors! There are many techniques to sample from an un-normalized distribution (e.g. MCMC), and other techniques to approximate the posterior distribution.
- There is no hyperparameter to adjust the exploration-exploitation tradeoff (unlike δ in UCB). In practice, one can still:
  - speed up or slow down the update of the posterior;
  - or artificially inflate or reduce the variance of the belief distribution (more variance $\Rightarrow$ more exploration).
- Thompson Sampling, like UCB, is a method to minimize regret in bandits. But what if we are only interested in identifying the best arm?

0.16 Best-arm identification

0.16.1 Motivation

Suppose a company wants to test the efficacy of drugs on mice.

- There are a few different drug variants: these are the arms.
- We do not care about the outcome of the drug on the mice during the trial (no regret… poor mouse!).
- The only goal is to identify the best drug with high probability at the end of the trial.

How should the drugs be tried? (Source: Lattimore & Szepesvári, Chapter 33.)

0.16.2 Formulations

- **Basic formulation**: after n rounds of playing a multi-armed bandit, identify the best arm with high probability, i.e. return an arm i so as to minimize $\mathbb{P}(\mu^* - \mu_i > 0)$ (fixed-budget setting).
- **Slightly weaker requirement (PAC: probably approximately correct)**: return an arm i such that $\mathbb{P}(\mu^* - \mu_i > \epsilon) \leq \delta$. We allow an ϵ-suboptimal arm ("approximately correct"), with probability at least $1 - \delta$ ("probably").
- **Yet another alternative**: minimize the **simple regret** at time $n+1$, $\mathbb{E}[\mu^* - \mu_i]$ where i is the returned arm. This is different from cumulative regret minimization: only the quality of the final recommendation counts, not the rewards collected along the way.

0.16.3 Intuition: why is this different from regret minimization?

- Regret-minimizing algorithms play suboptimal arms only $O(\log n)$ times, with confidence intervals chosen such that the error probability is $\delta = O(n^{-2})$. Spending so few samples on suboptimal arms is good for regret, but not for distinguishing them precisely from the best arm.
- In best-arm identification, there is **no exploration-exploitation tradeoff**: it is **pure exploration**. Pulling a bad arm costs nothing in itself, so we pull suboptimal arms more often than in regret minimization.

Baseline: uniform exploration

Each arm is pulled n/k times, and we return the empirically best arm after round n. Assume arm 1 is the unique best arm. Using the union bound and then the Chernoff bound:

$$\mathbb{P}(\mu^* - \mu_{\hat{i}} > 0) \leq \sum_{i:\Delta_i>0} \mathbb{P}(\hat{\mu}_i \geq \hat{\mu}_1) \leq \sum_{i:\Delta_i>0} \exp\left(-\frac{(n/k)\Delta_i^2}{4}\right),$$

where $\hat{i}$ is the returned arm. Indeed, $(\hat{\mu}_i - \hat{\mu}_1) - (\mu_i - \mu_1)$ is the difference of two independent averages of n/k samples, hence $\sqrt{2k/n}$-subgaussian, so $\mathbb{P}(\hat{\mu}_i - \hat{\mu}_1 \geq 0) = \mathbb{P}\big((\hat{\mu}_i - \hat{\mu}_1) - (\mu_i - \mu_1) \geq \Delta_i\big) \leq \exp\left(-\frac{\Delta_i^2}{2\cdot 2k/n}\right)$.

Can we do better? Yes! There is no explore-exploit tradeoff, but there is still a benefit in **adaptively sampling some arms more than others**, based on the information collected so far: clearly bad arms can be discarded early, and the remaining budget spent on the hard comparisons between the top arms.

0.16.4 Sequential halving

Definition 25.
(Sequential Halving for best-arm identification). Input: number of arms k, time horizon (budget) n.

Set the number of phases $L = \lceil \log_2 k \rceil$, active set $\mathcal{A}_1 = [k]$.
for $\ell = 1, 2, \ldots, L$ **do**:
  let $T_\ell = \left\lfloor \frac{n}{L\,|\mathcal{A}_\ell|} \right\rfloor$;
  play each arm in $\mathcal{A}_\ell$ exactly T_ℓ times;
  for each arm $i \in \mathcal{A}_\ell$, compute $\hat{\mu}_i^\ell$: the empirical mean of its last T_ℓ samples;
  $\mathcal{A}_{\ell+1} \leftarrow$ the top $\lceil |\mathcal{A}_\ell|/2 \rceil$ arms in $\mathcal{A}_\ell$ (according to $\hat{\mu}_i^\ell$).
return the (single) arm in $\mathcal{A}_{L+1}$.

Remark 25.

- It **eliminates the worst half of the arms in each phase**, which makes a more efficient use of the experiment budget.
- Each phase uses (at most) the same budget n/L, shared among fewer and fewer arms: T_ℓ **increases exponentially** in ℓ (it roughly doubles each phase). The arms that survive the longest (the best ones) get the most samples, i.e. the most precise estimates, which is exactly where precision is needed.
- In each phase, only the samples of that phase are used ("last T_ℓ samples"): this keeps the samples of different phases independent, which simplifies the analysis.
- Compare with sequential elimination (lecture 4): both remove arms progressively, but sequential halving uses a fixed schedule (fixed budget, one halving per phase) rather than confidence intervals.

0.16.5 Performance of sequential halving

Theorem 13 (Sequential halving)**.**
The sequential halving algorithm, applied to a stochastic 1*-subgaussian bandit with* k *arms with mean vector* $\mu_1 > \mu_2 \geq \cdots \geq \mu_k$ *(arms sorted by decreasing mean) for* n *rounds, returns a suboptimal arm with probability at most*

$$3 \log_2(k) \exp\left(-\frac{n}{16\, H_2(\mu) \log_2 k}\right), \qquad H_2(\mu) = \max_{j:\Delta_j>0} \frac{j}{\Delta_j^2}.$$

Remark 26 (Interpretation of H_2)**.**
$H_2(\mu)$ measures the **difficulty** of the instance. The term j/Δ_j^2 is large when there are many arms (j arms at least as good as arm j) with small gaps. The error probability decays exponentially in $n/(H_2 \log_2 k)$: a budget of order $H_2(\mu) \log k$ samples suffices to identify the best arm with high probability.

Case 1: all suboptimal arms have the same gap

$\mu_1 = 1$, $\mu_2 = \mu_3 = \cdots = \mu_k = 1 - \Delta \Rightarrow H_2(\mu) = \max_j j/\Delta^2 = k/\Delta^2$.

- Error rate (exponent) for sequential halving: $\sim \frac{n\Delta^2}{k \log k}$.
- Uniform exploration is marginally better: exponent $\sim \frac{n\Delta^2}{k}$.

Here no arm can be discarded more easily than the others, so adaptivity does not help (and costs a $\log k$ factor).

Case 2: one arm is slightly worse, all others far worse

$\mu_1 = 1$, $\mu_2 = 1 - \Delta$, $\mu_3 = \cdots = \mu_k = 0 \Rightarrow H_2(\mu) = \max\left(2/\Delta^2, \max_{j\geq 3} j\right) = 2/\Delta^2$ (for small Δ).

- Error rate for sequential halving: $\sim \frac{n\Delta^2}{\log k}$.
- Uniform exploration: the hard comparison between arms 1 and 2 only gets n/k samples per arm, so its exponent is $\sim \frac{n\Delta^2}{k}$.
- Sequential halving is **much better than uniform exploration, especially for large** k: it devotes more samples to deciding between arms 1 and 2, after quickly discarding the obviously bad arms.

So far, we studied *unstructured* bandits: pulling one arm tells us nothing about the others, and every arm has to be explored on its own. This is why the regret grows with the number of arms k. In many applications (recommender systems being the canonical example), arms are *related*: they are described by features, and observing the reward of one arm tells us something about similar arms. This lecture introduces the **linear bandit** model, which captures this structure:

- each arm is a known **feature vector** $a \in \mathbb{R}^d$, and the expected reward is a **linear** function $a^\top \theta^*$ of an unknown parameter $\theta^* \in \mathbb{R}^d$;
- θ^* is estimated by (regularised) **least squares**, and its uncertainty is described by **confidence ellipsoids** instead of confidence intervals;
- the **LinUCB** algorithm extends the optimism principle to this setting, with a regret of order $d\sqrt{n}\log n$, **independent of the number of arms** k;
- Thompson Sampling and generalised linear bandits carry over naturally.

0.17 Recap

0.17.1 Multi-armed bandits

- A framework for online decision-making, with k possible actions (arms).
- At each round, play one arm $A_t \in [k]$ and get a reward X_t.
- Goal: maximise the cumulative reward $\sum_{t=1}^n X_t$.
- Each arm has its own reward distribution: $X_t \sim P_i$, $\mathbb{E}[X_t] = \mu_i$ when $A_t = i$.
- Challenge: initially, no information about the rewards (μ_i unknown) $\Rightarrow$ we must balance **exploration** (learning the μ_i) and **exploitation** (maximising X_t).

0.17.2 UCB and Thompson Sampling

UCB: with $\hat{\mu}_i(t)$ the empirical mean and $T_i(t)$ the number of pulls of arm i,

$$UCB_i(t,\delta) = \underbrace{\hat{\mu}_i(t)}_{\text{exploitation}} + \underbrace{\sqrt{\frac{2\ln(1/\delta)}{T_i(t)}}}_{\text{exploration}}, \qquad A_t = \arg\max_i UCB_i(t-1,\delta).$$

Thompson Sampling: a Bayesian algorithm that maintains a belief distribution $f_{i,t}(\cdot)$ for every arm; at each round, sample $\tilde{\mu}_i(t) \sim f_{i,t}(\cdot)$ for all i, play $A_t = \arg\max_i \tilde{\mu}_i(t)$, observe X_t and update $f_{A_t,t}$. Exploration comes from the randomness of the samples; over time, beliefs concentrate $\Rightarrow$ exploitation.

0.17.3 Regret bounds for multi-armed bandits

- Problem-dependent bound: $R_n \lesssim \sum_{i:\Delta_i>0} \frac{16\ln n}{\Delta_i}$.
- Problem-independent bound: $R_n \lesssim 8\sqrt{kn\ln n}$.
- In both cases, the **regret grows with the number of arms**: this is the cost of exploring each option separately.
- The reason is the **information structure**: pulling one arm tells nothing about the other arms.
- This is not always true in practice!

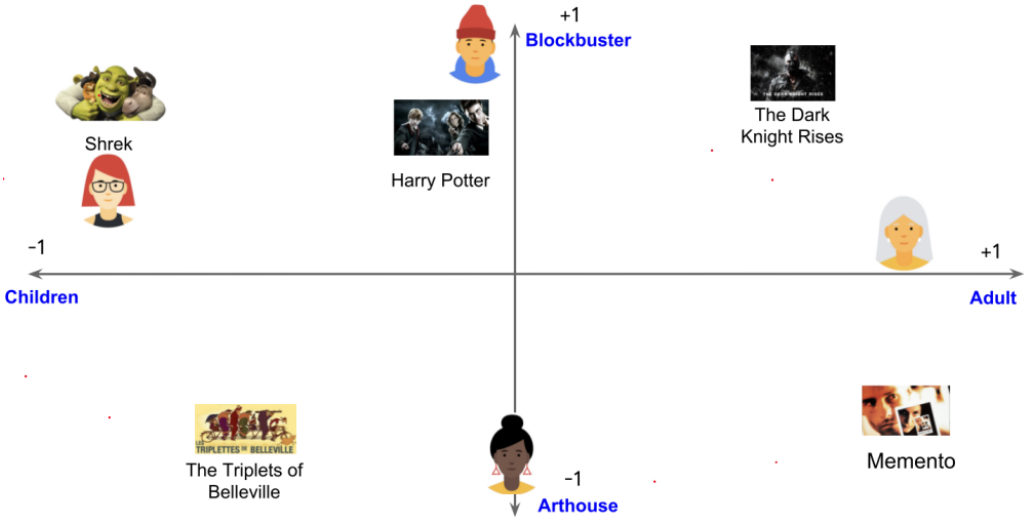
Remark 27 (Netflix scenario)**.**
A person does not like *Toy Story* and *Ratatouille* $\Rightarrow$ they probably do not like children's movies $\Rightarrow$ they probably do not like *Finding Nemo* either $\Rightarrow$ it is not worth recommending it (and regretting it!). An unstructured bandit would have to try *Finding Nemo* several times before learning this; a model that knows that these three movies are "similar" can skip it.

0.18 Recommender systems

0.18.1 Modeling recommender systems

Consider a recommender system like Netflix.

- **Items** (movies) have features: genre, language, actors, director, … Similar movies ⇒ similar features.
- **Users** have features too: preferences over genres, languages, actors, etc. Similar tastes ⇒ similar features.



Formally:

- user feature vector $\theta^* \in \mathbb{R}^d$;
- item feature vector $a \in \mathbb{R}^d$;
- **utility**: $\sum_{i=1}^{d} a_i \theta_i = a^\top \theta$, i.e. the sum over features of (feature a_i of the item) $\times$ (user's preference for feature i);
- **reward**: $a^\top \theta + \eta$, or more generally $g(a^\top \theta) + \eta$ for some function g. The reward is a **noisy signal of the utility**. It can be:
  - continuous: watch time;
  - discrete: rating given to the movie;
  - binary: watched / not watched.

0.18.2 Modeling assumptions

1. **Both features a and θ unknown**: collaborative filtering, matrix factorization.
   - Identify similar users and similar items; use the preferences of similar users to recommend new items.
   - Advantage: no need to handcraft features (they are learned).
   - **Cold start problem**: each new user needs to rate a few items to get good recommendations; analogously for new items.
2. **Only the user features θ unknown** (item features a known): this is the setting of this lecture.
   - Recommend diverse movies to start with.
   - With little data, we can already identify preferences: likes comedy, dislikes action, …
   - Use the data to learn the user features.

0.18.3 Recommender systems as bandits

Suppose a new user signs up on Netflix.

- Goal: recommend the best movies for this user over time, i.e. maximise the cumulative reward $\Leftrightarrow$ minimise the cumulative regret.
- Subgoal: learn the user preferences. We need to try out different movies, at the cost of showing a few bad movies.
- **Like multi-armed bandits**: it is an online learning problem with an exploration-exploitation tradeoff.
- **Unlike multi-armed bandits**: the reward from one arm (recommending a movie) gives information about the other arms!

0.19 The linear bandit model

0.19.1 Definition

Definition 26 (Linear bandit)**.**

- Arms are represented as a set of vectors $\mathcal{A} = \{a_1, a_2, \dots, a_k\}$, $a_i \in \mathbb{R}^d$.
- Arm played at time t: $A_t \in \mathcal{A}$.
- Reward at time t:
$$X_t = A_t^\top \theta^* + \eta_t = \langle \theta^*, A_t \rangle + \eta_t.$$
- Unknown parameter to be estimated: $\theta^* \in \mathbb{R}^d$.
- Noise model: the η_t are independent for all t and σ-subgaussian (we will mostly take $\sigma = 1$).
- Best arm: $a^* = \arg\max_{a \in \mathcal{A}} a^\top \theta^*$.
- Instantaneous regret:
$$r_t = a^{*\top} \theta^* - A_t^\top \theta^* = (a^* - A_t)^\top \theta^*.$$

The key difference with the unstructured bandit: the k unknown means $\mu_i = a_i^\top \theta^*$ are all determined by only d unknown numbers (the entries of θ^*). Observing X_t for arm A_t gives information about θ^* in the direction of A_t, hence about *every* arm that is not orthogonal to A_t.

Remark 28 (Reduction to the multi-armed bandit)**.**
Linear bandits generalise unstructured bandits. Take $d = k$ and let the arms form an orthonormal basis, $a_i = e_i = (0, \dots, 0, 1, 0, \dots, 0)$ (a 1 in position i). Then

$$X_t = a_i^\top \theta^* + \eta_t = \theta_i^* + \eta_t,$$

so θ_i^* plays the role of μ_i, and pulling arm i tells nothing about the other coordinates: we recover the standard k-armed bandit.

0.19.2 Flexibility of the linear bandit model

- The vector θ can include **known aspects of the user context**: age, gender, region, …; explicitly mentioned interests; behavioural statistics (number of comedy movies watched), etc.
- The **action set $\mathcal{A}_t = \{a_1^t, a_2^t, \dots, a_k^t\}$ can vary with time**, e.g.:
  - the same user's context can change (restaurants in a new city; recommending among some search results; …);
  - different users at different times;
  - new items appear, old ones disappear.

  The linear bandit model extends easily to a time-varying, arbitrary action set $\mathcal{A}_t$ (this is sometimes called a *contextual* linear bandit). In the unstructured setting, a brand new arm would have to

be explored from scratch; here, its feature vector immediately gives an estimate of its reward.
- **Interpreting θ^***: a **global weight vector over features**, and the expected reward of A_t is $A_t^\top \theta^*$. For instance, if the feature vector contains a crossed feature (young user, animation movie), a positive θ_i^* for this feature means that young people prefer animation movies.

0.19.3 Formulations of regret for linear bandits

- Instantaneous regret: $r_t = a^{*\top}\theta^* - A_t^\top \theta^* = (a^* - A_t)^\top \theta^*$. Note that it **does not depend on the noise** η_t.
- **Random cumulative regret**:

$$\hat{R}_n(A_1, \dots, A_n) = \sum_{t=1}^{n} \Big(\max_{a \in \mathcal{A}_t} \langle \theta^*, a \rangle - \langle \theta^*, A_t \rangle \Big) = \sum_{t=1}^{n} r_t.$$

  It is a random variable: it depends on the (random) choice of actions.
- **Regret**:

$$R_n = \mathbb{E}\big[\hat{R}_n\big] = \mathbb{E}\left[\sum_{t=1}^{n} \max_{a \in \mathcal{A}_t} \langle \theta^*, a \rangle - \sum_{t=1}^{n} X_t \right].$$

  Taking the expectation marginalises out the choice of actions. The two expressions coincide since $\mathbb{E}[X_t \mid A_t] = \langle \theta^*, A_t \rangle$ (the noise has zero mean).

Remark 29.
The analysis of LinUCB will first bound $\hat{R}_n$ *with high probability* (on a good event), and then convert it into a bound on R_n. This is the same "good event / bad event" strategy as for UCB.

0.20 Estimating θ^* via linear regression

0.20.1 Least squares

- Data: $(A_1, X_1, A_2, X_2, \dots, A_t, X_t)$.
- Model: $X_s = A_s^\top \theta^* + \eta_s$, with $\theta^* \in \mathbb{R}^d$ unknown, to be estimated, and $\eta_s \in \mathbb{R}$ a 1-subgaussian random variable.
- This is exactly **linear regression**. The least squares problem is

$$\min_{\theta} \sum_{s \in [t]} \big(X_s - A_s^\top \theta\big)^2.$$

Definition 27 (Least squares estimator, design matrix)**.**

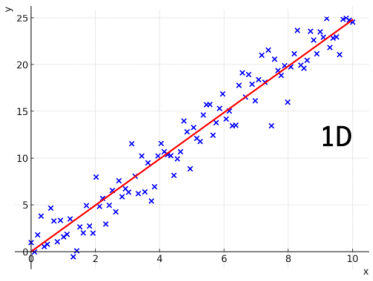
$$\hat{\theta}_t = V_t^{-1} \sum_{s \in [t]} A_s X_s, \qquad V_t = \sum_{s \in [t]} A_s A_s^\top \in \mathbb{R}^{d \times d}.$$

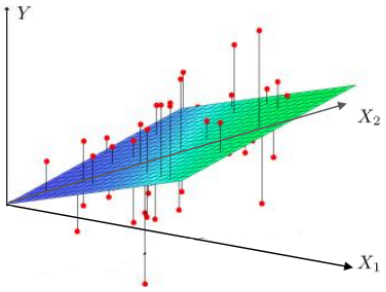
V_t is called the **design matrix**: it summarises which directions have been explored. It is symmetric and positive semi-definite.

Derivation. The objective $L(\theta) = \sum_s (X_s - A_s^\top \theta)^2$ is convex in θ, with gradient

$$\nabla L(\theta) = -2 \sum_s A_s \big(X_s - A_s^\top \theta\big).$$

Setting it to zero gives the normal equations $\big(\sum_s A_s A_s^\top\big)\theta = \sum_s A_s X_s$, i.e. $V_t \theta = \sum_s A_s X_s$, hence $\hat{\theta}_t = V_t^{-1} \sum_s A_s X_s$ when V_t is invertible. □





0.20.2 The estimator is unbiased

Substituting the model $X_s = A_s^\top \theta^* + \eta_s$ (and treating the A_s as fixed):

$$\hat{\theta}_t = V_t^{-1} \sum_{s \in [t]} A_s \left(A_s^\top \theta^* + \eta_s \right) = \underbrace{V_t^{-1} V_t}_{=I} \theta^* + V_t^{-1} \sum_{s \in [t]} A_s \eta_s = \theta^* + V_t^{-1} \sum_{s \in [t]} A_s \eta_s,$$

so, since $\mathbb{E}[\eta_s] = 0$,

$$\mathbb{E}\left[\hat{\theta}_t\right] = \mathbb{E}\left[V_t^{-1} \sum_{s \in [t]} \left(A_s A_s^\top \theta^* + A_s \eta_s \right) \right] = \theta^*.$$

Moreover, if the noise has variance σ^2, the covariance of the estimator is

$$\mathrm{Cov}\left(\hat{\theta}_t\right) = V_t^{-1} \left(\sum_s A_s A_s^\top \sigma^2 \right) V_t^{-1} = \sigma^2 V_t^{-1}.$$

This is the key fact: **the uncertainty of $\hat{\theta}_t$ is quantified by V_t^{-1}.**

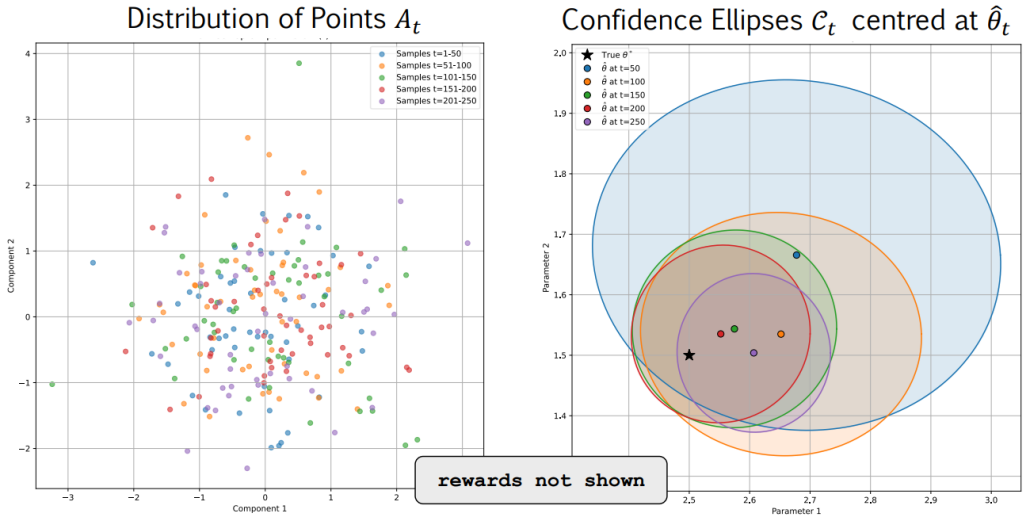
0.20.3 Uncertainty in linear regression: 1D case

In 1D, θ^* and $\hat{\theta}$ are scalars: $\hat{\theta}_t = \frac{\sum_s A_s X_s}{\sum_s A_s^2}$ and $\mathrm{Var}(\hat{\theta}_t) = \frac{\sigma^2}{\sum_s A_s^2}$. The uncertainty in $\hat{\theta}$ can be conveyed through **confidence intervals**:

- more noise in the data $\Rightarrow$ larger confidence intervals;
- more data points $\Rightarrow$ smaller confidence intervals (the width is proportional to $\sigma / \sqrt{\sum_s A_s^2}$).

0.20.4 Uncertainty in linear regression: 2D case

In dimension $d \geq 2$, confidence intervals become **confidence ellipses** (ellipsoids) $\mathcal{C}_t$ centred at $\hat{\theta}_t$.



The directions that have been better explored (many data points A_s along them) have less uncertainty (the ellipse is narrow in those directions); the less explored directions have more uncertainty (the ellipse is elongated).

0.20.5 Why ellipses?

- The collection of data points is represented by the design matrix $V_t = \sum_{s\in[t]} A_s A_s^\top$.
- More data points in a certain direction $a \Rightarrow$ the eigenvalues of V_t are larger in that direction ($a^\top V_t a = \sum_s \langle a, A_s\rangle^2$ grows).
- More data points in a certain direction $a \Rightarrow$ better estimate of the component of θ^* parallel to a.
- The uncertainty of θ^* is quantified by $V_t^{-1} \Rightarrow$ more uncertainty along the larger eigenvalues of V_t^{-1} (i.e. the smaller eigenvalues of V_t).

Definition 28 (Weighted norm, confidence set)**.**
For a positive definite matrix V, define the norm $\|x\|_V = \sqrt{x^\top V x}$. The confidence sets are

$$\mathcal{C}_t = \left\{\theta \in \mathbb{R}^d : \left\|\theta - \hat{\theta}_{t-1}\right\|^2_{V_{t-1}} \le \beta_t\right\},$$

where β_t is a radius chosen so that $\theta^* \in \mathcal{C}_t$ with high probability.

Remark 30 (Geometry)**.**
Write $V = \sum_j \lambda_j u_j u_j^\top$ (eigendecomposition). Then $\|\theta - \hat{\theta}\|_V^2 = \sum_j \lambda_j \langle \theta - \hat{\theta}, u_j\rangle^2 \le \beta$ is an ellipsoid whose axes are the eigenvectors u_j, with semi-axis lengths $\sqrt{\beta/\lambda_j}$. A large eigenvalue (well-explored direction) gives a short axis (little uncertainty), a small eigenvalue gives a long axis. In 1D, this reduces to the interval $|\theta - \hat{\theta}| \le \sqrt{\beta / \sum_s A_s^2}$.

0.20.6 Regularisation

What if V_t is not invertible? This happens for instance at the beginning ($t < d$), or whenever the played arms do not span $\mathbb{R}^d$: some directions have never been explored, and the least squares problem has infinitely many solutions.

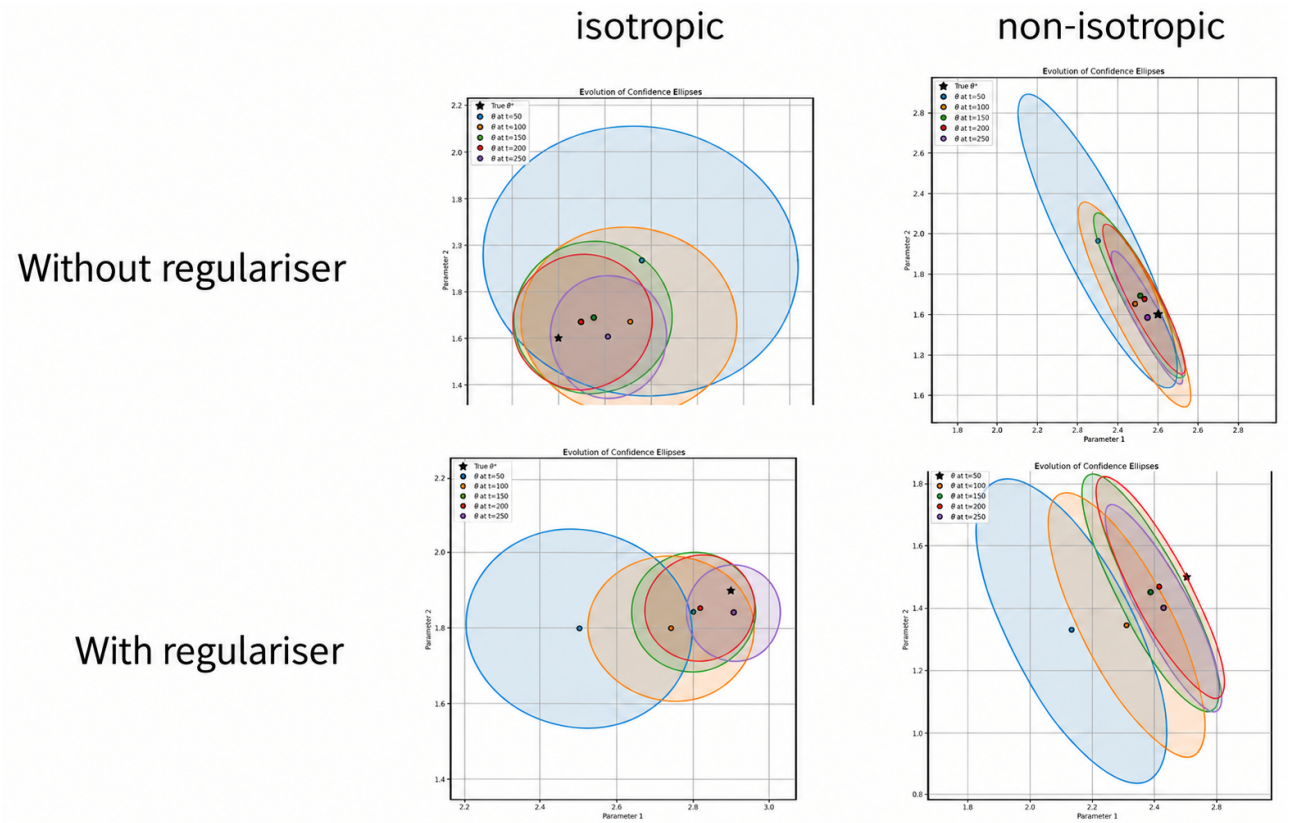
Simple fix: use a regulariser (ridge regression). Solve

$$\min_\theta \sum_{s\in[t]} \left(X_s - A_s^\top \theta\right)^2 + \lambda \|\theta\|_2^2,$$

whose solution is (same derivation, with the extra gradient term $2\lambda\theta$)

$$\hat{\theta}_t = V_t^{-1} \sum_{s\in[t]} A_s X_s, \qquad V_t = \lambda I + \sum_{s\in[t]} A_s A_s^\top.$$

For $\lambda > 0$, V_t is positive definite (all eigenvalues $\ge \lambda$), hence always invertible. The price is a small bias of $\hat{\theta}_t$ towards 0, which vanishes as data accumulates.



Impact on the confidence ellipses. Each panel shows the true θ^* (star), the estimates $\hat{\theta}_t$ for $t = 50, \ldots, 250$ and the corresponding confidence ellipses. In all cases, the ellipses shrink over time and keep containing θ^*.

- **Isotropic vs non-isotropic** refers to the distribution of the played vectors A_s: if they are spread evenly in all directions, V_t has similar eigenvalues and the confidence sets are nearly circles; if they are concentrated along one direction, the ellipses are narrow along that (well-explored) direction and elongated along the poorly explored one.
- **With a regulariser**, the estimate is biased towards 0 (penalty $\lambda\|\theta\|_2^2$): early estimates are visibly pulled towards the origin, then move towards θ^* as data accumulates, since λI becomes negligible compared to $\sum_s A_s A_s^\top$. In exchange, V_t is always invertible, and $V_0 = \lambda I$ makes the initial confidence set an isotropic ball.

0.21 LinUCB: UCB for linear bandits

0.21.1 Algorithm

Notation:

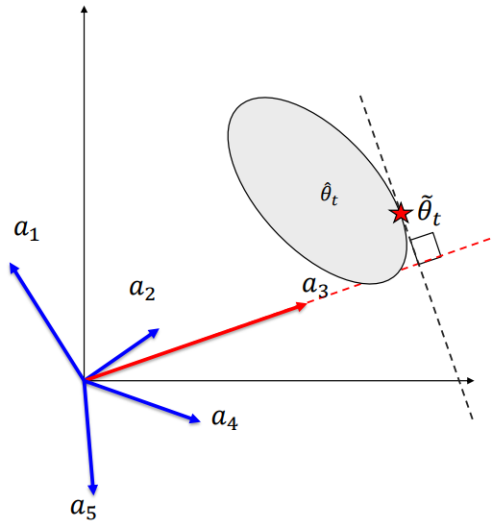
- $V_0 = \lambda I, \quad V_t = \lambda I + \sum_{s\in[t]} A_s A_s^\top$;
- $\hat{\theta}_t = V_t^{-1} \sum_{s\in[t]} A_s X_s$;
- $\mathcal{C}_t = \left\{\theta \in \mathbb{R}^d : \|\theta - \hat{\theta}_{t-1}\|^2_{V_{t-1}} \leq \beta_{t-1}\right\}$;
- $1 \leq \beta_1 \leq \beta_2 \leq \cdots \leq \beta_n$ (non-decreasing radii).

The principle is **optimism in the face of uncertainty**: the UCB of an arm a is the largest reward it could have under any parameter that is still plausible (i.e. in the confidence set).

Definition 29 (LinUCB, abstract form)**.**

for $t = 1, 2, \ldots, n$ **do**:

for all arms $a \in \mathcal{A}_t$: compute $UCB_t(a) = \max_{\theta \in \mathcal{C}_t} a^\top \theta$;
play $A_t = \arg\max_{a \in \mathcal{A}_t} UCB_t(a)$;
observe X_t and update V_t, $\hat{\theta}_t$, $\mathcal{C}_t$.



0.21.2 Computing the UCB

Combining the two maximisations,

$$A_t = \arg\max_{a \in \mathcal{A}_t} UCB_t(a), \quad UCB_t(a) = \max_{\theta \in \mathcal{C}_t} a^\top \theta \quad \Rightarrow \quad (A_t, \tilde{\theta}_t) = \arg \max_{(a,\theta) \in \mathcal{A}_t \times \mathcal{C}_t} a^\top \theta.$$

So LinUCB jointly picks an arm A_t and an **optimistic parameter** $\tilde{\theta}_t \in \mathcal{C}_t$ (the point of the ellipsoid that makes A_t look best).

Theorem 14 (Closed form of the UCB)**.**
If $\mathcal{C}_t = \left\{\theta : \|\theta - \hat{\theta}_{t-1}\|^2_{V_{t-1}} \leq \beta_t\right\}$, *then*

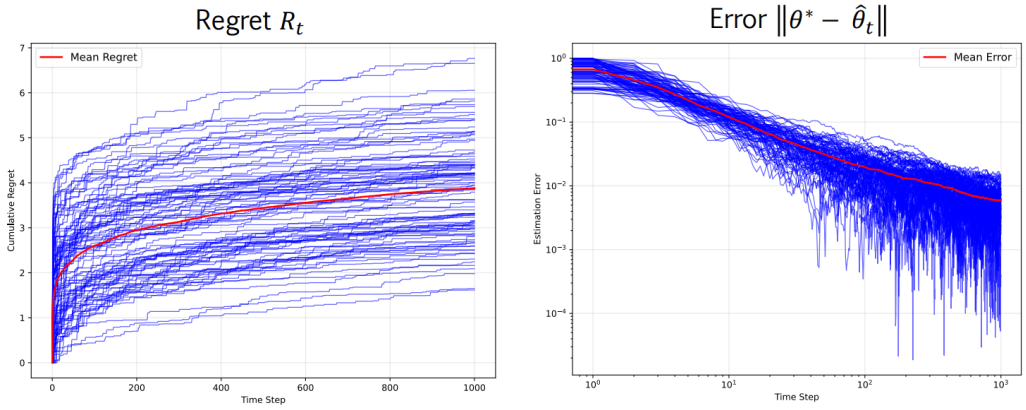
$$UCB_t(a) = \max_{\theta \in \mathcal{C}_t} a^\top \theta = \underbrace{a^\top \hat{\theta}_{t-1}}_{\textit{exploitation}} + \underbrace{\sqrt{\beta_t}\, \|a\|_{V_{t-1}^{-1}}}_{\textit{optimistic bonus}}.$$

Definition 30 (LinUCB, explicit form)**.**

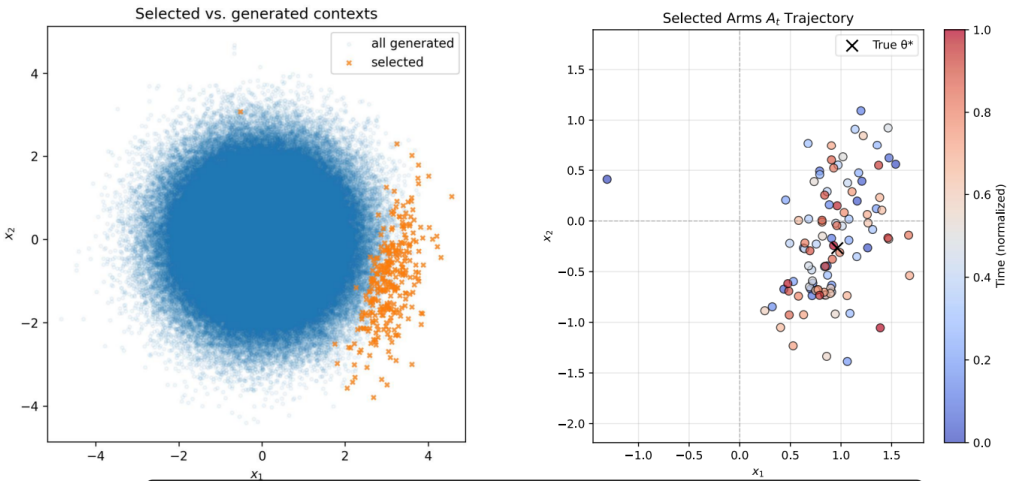
for $t = 1, 2, \ldots, n$ **do**:
play $A_t = \arg\max_{a \in \mathcal{A}_t} \left(a^\top \hat{\theta}_{t-1} + \sqrt{\beta_t}\, \|a\|_{V_{t-1}^{-1}}\right)$;
observe X_t and update V_t, $\hat{\theta}_t$, $\mathcal{C}_t$.

0.21.3 LinUCB in practice

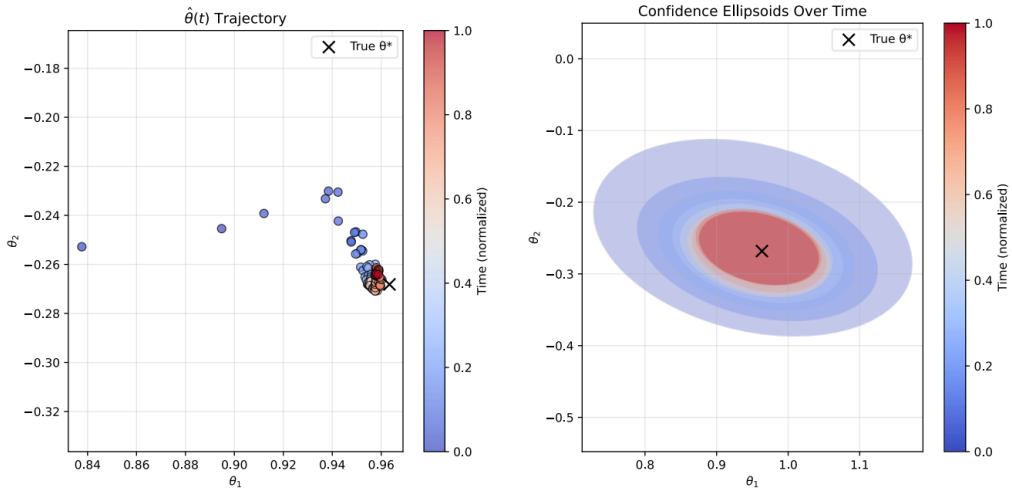
Setting: linear bandit with $k = 100$ arms and $d = 5$ dimensional features.



- **Regret** $R_t = O(\sqrt{t})$ (up to log factors), similar to unstructured bandits (problem-independent rate).
- **Error** $\|\theta^* - \hat{\theta}_t\| = O(1/\sqrt{t})$, similar to linear regression.



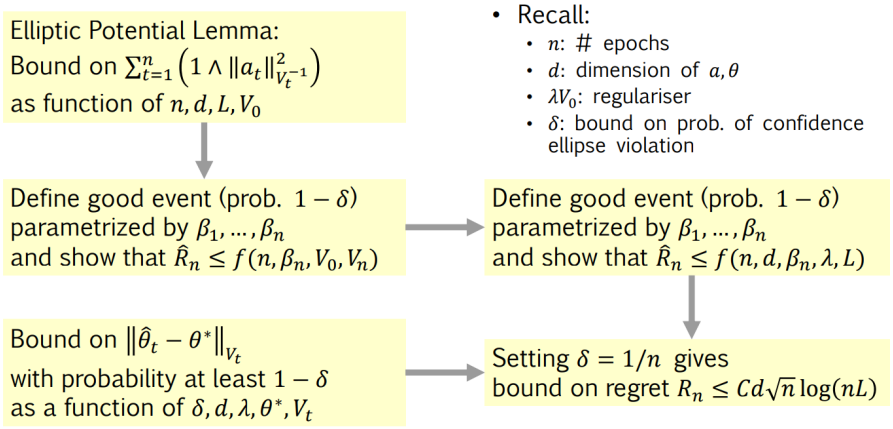
Actions over time: the selected arms align in the direction of θ^* (the best arms are those with the largest projection on θ^*). Consequently, θ^* is estimated precisely in that direction, and less precisely in the orthogonal directions: this is enough, since precision is only needed to rank the near-optimal arms.



Confidence ellipsoids with time: they shrink around θ^*, faster in the directions that are played often.

0.22 Analysis of LinUCB

0.22.1 High-level view



0.22.2 Intuition for the concentration bound (single arm)

We first prove a simple concentration inequality for a single, fixed arm a (with $\lambda = 0$, and treating the A_s as fixed).
Recall $\hat{\theta}_t = V_t^{-1}\sum_{s\in[t]} A_s X_s$ and $X_s = A_s^\top \theta^* + \eta_s$. Then

$$\left\langle a, \hat{\theta}_t - \theta^* \right\rangle = \left\langle a, V_t^{-1}\sum_{s\in[t]} A_s\left(A_s^\top\theta^* + \eta_s\right) - \theta^* \right\rangle.$$

Observe that $\sum_s A_s A_s^\top \theta^* = V_t\theta^*$, so $V_t^{-1}\sum_s A_s A_s^\top\theta^* = \theta^*$. Hence

$$\left\langle a, \hat{\theta}_t - \theta^* \right\rangle = \left\langle a, V_t^{-1}\sum_{s\in[t]} A_s\eta_s \right\rangle = \sum_{s\in[t]} \left\langle a, V_t^{-1}A_s\right\rangle \eta_s.$$

This is a **linear combination of independent, 1-subgaussian random variables** $(\eta_s)_{s\in[t]}$. Recall the properties of subgaussian random variables:

- η is 1-subgaussian $\Rightarrow$ $\alpha\eta$ is $|\alpha|$-subgaussian;
- η_1, η_2 independent, σ_1- and σ_2-subgaussian $\Rightarrow$ $\eta_1 + \eta_2$ is $\sqrt{\sigma_1^2 + \sigma_2^2}$-subgaussian.

Therefore $\sum_s \langle a, V_t^{-1}A_s\rangle\eta_s$ is subgaussian, with variance proxy

$$\sum_{s\in[t]} \left\langle a, V_t^{-1}A_s\right\rangle^2 = a^\top V_t^{-1}\left(\sum_{s\in[t]} A_s A_s^\top\right) V_t^{-1}a = a^\top V_t^{-1}V_t V_t^{-1}a = a^\top V_t^{-1}a = \|a\|^2_{V_t^{-1}}.$$

Applying the concentration inequality for subgaussians ($\mathbb{P}(Z \geq \sqrt{2\sigma^2\ln(1/\delta)}) \leq \delta$ for Z σ-subgaussian):

$$\mathbb{P}\left(\left\langle a, \hat{\theta}_t - \theta^*\right\rangle \geq \sqrt{2\|a\|^2_{V_t^{-1}}\ln(1/\delta)}\right) \leq \delta.$$

This is the exact analogue of the UCB confidence interval: $a^\top\hat{\theta}_t$ estimates $a^\top\theta^*$ with an error of order $\|a\|_{V_t^{-1}}\sqrt{2\ln(1/\delta)}$, and in the MAB case $\|e_i\|^2_{V_t^{-1}} = 1/T_i(t)$.

0.22.3 The simple bound is not sufficient

The bound above

$$\mathbb{P}\left(\left\langle a, \hat{\theta}_t - \theta^* \right\rangle \geq \sqrt{2\ln(1/\delta)\sum_{s\in[t]}\left\langle a, V_t^{-1}A_s\right\rangle^2}\right) \leq \delta$$

holds only for a **single arm**. For regret guarantees, we need such a bound for **all arms** simultaneously.

- One option: a union bound over all k arms. If the action set $\mathcal{A} = \{a_1, \dots, a_k\}$ is small, this is a reasonable strategy, and the regret bound will scale with $\log k$.
- For large (or infinite, or time-varying) action sets, we want a bound **independent of** k. We then need a bound on $\mathbb{P}\big(\|\hat{\theta}_t - \theta^*\|_{V_t} \geq \beta_t\big)$ instead of on $\langle a, \hat{\theta}_t - \theta^*\rangle$: a bound on the whole vector, valid for all directions at once.
- Moreover, in a bandit, A_s depends on past observations (hence on the past noise), so the A_s are not fixed and the argument above is not rigorous as such. See L&S, Chapter 20.

0.22.4 Key concentration result

Theorem 15 (Theorem 20.5 in L&S, not proven in class)**.**
Suppose we have a sequence of data points $(A_1, X_1, A_2, X_2, \dots)$ *satisfying*

- $X_t = \langle \theta^*, A_t\rangle + \eta_t$*, with* η_t *independent and* 1*-subgaussian;*
- *each* A_t *may depend on the past* $(A_1, X_1, \dots, A_{t-1}, X_{t-1})$ *(any bandit algorithm).*

Let $\hat{\theta}_t$ *be the least-squares estimator with regulariser weight* λ*. Fix* $\delta \in (0,1)$*. Then, with probability at least* $1-\delta$*, it holds* ***for all*** $t \in \mathbb{N}$*:*

$$\left\|\hat{\theta}_t - \theta^*\right\|_{V_t} < \sqrt{\lambda}\,\|\theta^*\|_2 + \sqrt{2\ln\frac{1}{\delta} + \ln\frac{\det V_t}{\lambda^d}}.$$

0.22.5 Understanding the good event

- Good event: $\forall t$, $\theta^* \in \mathcal{C}_t = \left\{\theta \in \mathbb{R}^d : \|\theta - \hat{\theta}_{t-1}\|^2_{V_{t-1}} \leq \beta_t\right\}$.
- It holds with probability $1-\delta$ (by the theorem above, for a suitable β_t).
- Intuitively:
  - $\|\theta^* - \hat{\theta}_{t-1}\|^2_{V_{t-1}} \leq \beta_t$ is a concentration event, and needs $\beta_t = O(\log t)$;
  - we expect the eigenvalues of V_{t-1} to grow proportionally to $t-1$ (if all directions keep being explored);
  - since $\|x\|_2^2 \leq \|x\|_V^2/\lambda_{\min}(V)$, put together we get

$$\left\|\theta^* - \hat{\theta}_t\right\|_2 = O\left(\sqrt{\frac{\log t}{t}}\right).$$

- With more data, the estimate $\hat{\theta}_t$ gets closer to θ^*.

0.22.6 Regret guarantees for LinUCB

Theorem 16 (Regret of LinUCB)**.**
Assume that:

- *rewards are bounded:* $\max_{t\in[n]} \sup_{a,b\in\mathcal{A}_t} (a-b)^\top \theta^* \leq 1$;
- *arm features are bounded:* $\|a\|_2 \leq L$ *for all* $a \in \bigcup_{t=1}^n \mathcal{A}_t$;
- $\beta_1, \beta_2, \ldots$ *are chosen such that, with probability* $\geq 1-\delta$ *where* $\delta = 1/n$,

$$\forall t, \quad \theta^* \in \mathcal{C}_t = \left\{\theta \in \mathbb{R}^d : \|\theta - \hat{\theta}_{t-1}\|^2_{V_{t-1}} \leq \beta_t\right\};$$

- $\beta_t = O\left(\ln \frac{1}{\delta} + d \ln \frac{d\lambda + nL^2}{d\lambda}\right)$ *suffices.*

Then

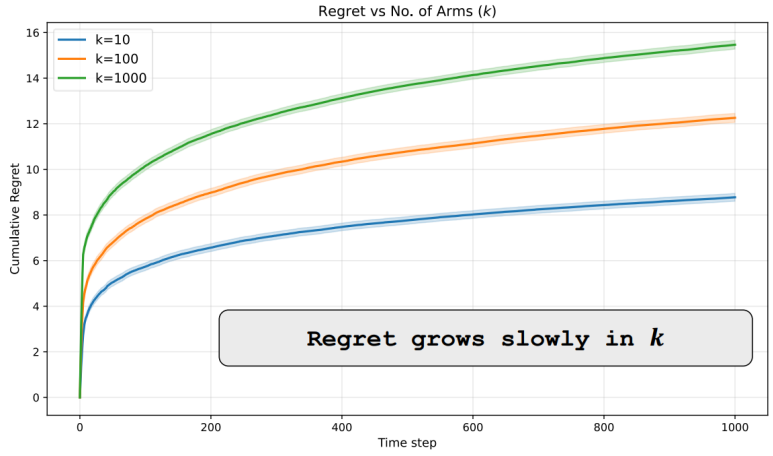
$$R_n = O\big(d\sqrt{n}\,\log(nL)\big).$$

Remark 31.

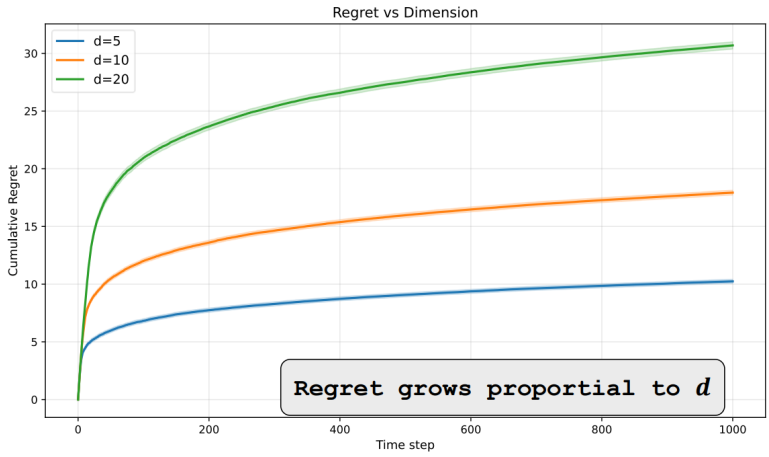
- The bound is **independent of the number of arms** k! It depends on the dimension d instead. The information structure (pulling one arm informs about the others) is what makes this possible.
- Compare with the unstructured bound $O(\sqrt{kn \ln n})$: for $k \gg d^2$ (many arms, few features), linear bandits are much better. In the MAB reduction ($d = k$), this bound gives $O(k\sqrt{n}\log n)$, which is worse than $O(\sqrt{kn \log n})$ by a $\sqrt{k}$ factor: the price of a general analysis that does not exploit the orthogonality of the arms.
- The rate in n is $\sqrt{n}$ (up to logs), as for the problem-independent bound of UCB.

0.22.7 Regret: variation with k and d

Setting: linear bandit with arm features $a_i^t \sim \mathcal{N}(0, I_d)$.



Variation with k ($d = 5$): the regret **grows slowly in** k. (More arms mean the best arm is a bit better aligned with θ^*, but no extra exploration per arm is needed.)



Variation with d ($k = 100$): the regret **grows proportionally to** d, as predicted by the bound.

0.23 Thompson Sampling for linear bandits

Key ideas in Thompson Sampling:

- maintain a belief distribution over the unknown parameters;
- at each time t: sample parameters from the belief distribution (exploration), then play assuming that the sample is the true parameter (exploitation).

In linear bandits, the unknown parameter is θ^*:

- maintain a **multivariate Gaussian belief** $f_t(\theta^*) \sim \mathcal{N}(\mu_t, \Sigma_t)$;
- after sampling $\tilde{\theta}_t \sim f_t$, play
$$A_t = \arg\max_{a \in \mathcal{A}_t} a^\top \tilde{\theta}_t;$$
- this is **computationally faster** than LinUCB's $A_t = \arg\max_{a \in \mathcal{A}_t} \left(a^\top \hat{\theta}_{t-1} + \sqrt{\beta_t} \|a\|_{V_{t-1}^{-1}} \right)$
- heuristic choice of the belief:
$$\mu_t = \hat{\theta}_t = V_t^{-1} \sum_{s \in [t]} A_s X_s, \qquad \Sigma_t^{-1} = V_t = \sum_{s \in [t]} A_s A_s^\top \ \ (+\lambda I).$$

0.24 Generalised linear bandits

- Linear bandit reward model: $X_t = A_t^\top \theta^* + \eta_t$.
  - The reward can be unbounded.
  - Unrealistic for scenarios where the reward is binary: click or no click.
- Better model:
$$X_t = g\left(A_t^\top \theta^*\right) + \eta_t,$$
where $g : \mathbb{R} \to [0, 1]$ is a nonlinear **link function**, e.g. the sigmoid $g(z) = \frac{1}{1+e^{-z}}$.
- This captures: $X_t = 1$ with probability $g(A_t^\top \theta^*)$, $X_t = 0$ otherwise.
- Often used in practice, e.g. online advertisement placement (click-through rates).
- All principles of LinUCB and Thompson Sampling carry over! The only change: $\hat{\theta}_t$ is estimated by **logistic regression** (maximum likelihood) instead of least squares. Since g is increasing, the best arm is still $\arg\max_a a^\top \theta^*$.
- Tight regret bounds have only been discovered recently (2020 onwards).

0.25 Towards reinforcement learning models

This lecture marks a turning point of the course.

- **So far: bandits.** Instantaneous reward after an action (the agent has immediate feedback); **memoryless environment** (the reward is only a function of the latest action, plus some noise); strong focus on "squeezing out" the lowest regret as a function of the system parameters.
- **Next: reinforcement learning (RL). Delayed reward** (the agent needs to craft *sequences* of actions to do well); **environment with memory** (it remembers what happened in the past, so the current situation depends on previous actions); strong focus on algorithms that learn efficiently, less analysis of asymptotic behaviour.
- We therefore need **probabilistic models for memory**, with mathematical properties that allow us to reason about them, yet rich enough to model real systems. **Markov chains** are the most important such model, with many applications beyond RL. In RL, the environment will be a Markov chain whose transitions are influenced by the agent's actions (Markov decision processes, next lectures).

In this lecture:

- we define (discrete-time, discrete-space) Markov chains, the transition matrix and the n-step transition probabilities (**Chapman-Kolmogorov** equations);
- we see that the evolution of a chain is entirely determined by its initial distribution $\boldsymbol{\mu}(0)$ and its transition matrix P;
- we **classify states**: transient / recurrent (null or positive), periodic / aperiodic, ergodic;
- we classify **chains**: communicating classes, closed and irreducible sets, and the **decomposition theorem**;
- we study the long-term behaviour: the **ergodic theorem** and the **stationary distribution** π;
- we apply everything to **random walks** on $\mathbb{Z}$, $\mathbb{Z}^d$ and on general graphs.

0.26 Definition and basic properties

0.26.1 The Markov condition

Throughout, we study **discrete-time, discrete-space** Markov chains only:

- discrete time: $t \in \mathbb{N}_0 = \{0, 1, 2, \dots\}$;
- discrete space: the **state space** $\mathcal{S}$ is a countable set (later, we will sometimes restrict $\mathcal{S}$ to be finite).

Definition 31 (Markov chain)**.**
The process $(X_0, X_1, X_2, \dots)$ with values in $\mathcal{S}$ is a **Markov chain** if it satisfies the **Markov condition**:

$$\mathbb{P}\big[X_n = x \mid X_0 = x_0, X_1 = x_1, \dots, X_{n-1} = x_{n-1}\big] = \mathbb{P}\big[X_n = x \mid X_{n-1} = x_{n-1}\big]$$

for all $n \geq 1$ and all $x, x_0, x_1, \dots, x_{n-1} \in \mathcal{S}$.

Intuitively: **to predict the future, no knowledge of the past is useful beyond the current state**. Equivalently, conditionally on the present, the past and the future are independent. Markov chains are "only one small step beyond independence": an i.i.d. sequence is a (trivial) Markov chain where even the current state is useless to predict the future.

0.26.2 Equivalent Markov conditions

The Markov condition can be stated in several equivalent ways:

- Conditioning on any finite set of past times, only the most recent one matters:

$$\mathbb{P}\big[X_{n+1} = x \mid X_{n_1} = x_{n_1}, X_{n_2} = x_{n_2}, \ldots, X_{n_k} = x_{n_k}\big] = \mathbb{P}\big[X_{n+1} = x \mid X_{n_k} = x_{n_k}\big]$$

  for all $n_1 < n_2 < \cdots < n_k \leq n$.
- Predicting further in the future, only the current state matters:

$$\mathbb{P}\big[X_{n+m} = x \mid X_0 = x_0, X_1 = x_1, \ldots, X_n = x_n\big] = \mathbb{P}\big[X_{n+m} = x \mid X_n = x_n\big]$$

  for all $m, n \geq 0$.

0.26.3 Homogeneous chains and the transition matrix

Definition 32 (Homogeneous chain)**.**
The chain X is **homogeneous** (time-shift-invariant) if

$$\mathbb{P}\big[X_{n+1} = j \mid X_n = i\big] = \mathbb{P}\big[X_1 = j \mid X_0 = i\big] \quad \text{for all } n, i, j.$$

Homogeneity is the analogue of "identically distributed" for i.i.d. random variables: the transition mechanism does not change over time. **From now on, unless otherwise specified, we assume homogeneous chains.**

Definition 33 (Transition matrix)**.**
The **transition matrix** P is the $|\mathcal{S}| \times |\mathcal{S}|$ matrix of transition probabilities

$$p_{ij} = \mathbb{P}\big[X_{n+1} = j \mid X_n = i\big].$$

Row i is the distribution of the next state when the current state is i.

P is a **stochastic matrix**:

(a) entries are non-negative: $p_{ij} \geq 0$;
(b) row sums are equal to one: $\sum_j p_{ij} = 1$ (from state i, we must go somewhere).

A Markov chain is conveniently drawn as a directed graph: one node per state, and an arrow $i \to j$ labelled p_{ij} whenever $p_{ij} > 0$.

0.26.4 n-step transition matrix

Definition 34 (n-step transition probabilities)**.**

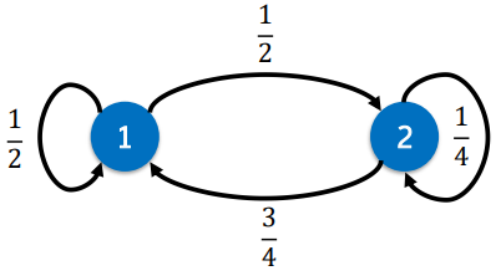
$$P_n: \quad p_{ij}(n) = \mathbb{P}\big[X_{m+n} = j \mid X_m = i\big] \qquad \text{(independent of } m \text{ by homogeneity)},$$

with $P_1 = P$ and $P_0 = I$, i.e. $p_{ij}(0) = \delta_{ij}$.

Remark 32 (Example: a two-state chain)**.**
Let $\mathcal{S} = \{1, 2\}$ with

$$P_1 = P = \begin{pmatrix} 1/2 & 1/2 \\ 3/4 & 1/4 \end{pmatrix}, \quad P_2 = \begin{pmatrix} 0.625 & 0.375 \\ 0.563 & 0.438 \end{pmatrix}, \quad P_3 = \begin{pmatrix} 0.594 & 0.406 \\ 0.609 & 0.391 \end{pmatrix}, \quad P_{10} = \begin{pmatrix} 0.600 & 0.400 \\ 0.600 & 0.400 \end{pmatrix}.$$

After 10 steps, we end up in state 1 or 2 with pretty much the same probability **regardless of where we started**: both rows of P_{10} are (almost) identical. The chain "forgets" its initial state. This is **ergodicity**, to which we return at the end of the lecture.



0.26.5 Chapman-Kolmogorov equations

Theorem 17 (Chapman-Kolmogorov)**.**
For all $m, n \geq 0$ and all $i, j \in \mathcal{S}$:

$$p_{ij}(m+n) = \sum_k p_{ik}(m)\, p_{kj}(n).$$

In matrix form: $P_{m+n} = P_m P_n$. Hence $P_n = P^n$: the n-step transition matrix is the n-th power of the transition matrix.

Proof. To go from i to j in $m+n$ steps, we must be in some state k at time m:

$$\begin{aligned}
p_{ij}(m+n) &= \mathbb{P}\big[X_{m+n} = j \mid X_0 = i\big] \\
&= \sum_k \mathbb{P}\big[X_{m+n} = j, X_m = k \mid X_0 = i\big] && \text{(law of total probability)} \\
&= \sum_k \mathbb{P}\big[X_{m+n} = j \mid X_m = k, X_0 = i\big]\, \mathbb{P}\big[X_m = k \mid X_0 = i\big] && \big(\mathbb{P}[A, B \mid C] = \mathbb{P}[A \mid B, C]\, \mathbb{P}[B \mid C]\big) \\
&= \sum_k \mathbb{P}\big[X_{m+n} = j \mid X_m = k\big]\, \mathbb{P}\big[X_m = k \mid X_0 = i\big] && \text{(Markov property)} \\
&= \sum_k p_{kj}(n)\, p_{ik}(m).
\end{aligned}$$

By induction, $P_n = P_{n-1} P_1 = \cdots = P^n$. □

0.26.6 Evolution of the marginal distribution of X_n

Definition 35 (Marginal distribution)**.**
$\mu_i(n) = \mathbb{P}[X_n = i]$, and $\boldsymbol{\mu}(n) = [\mu_1(n), \mu_2(n), \dots]$ is the distribution of X_n, written as a **row vector**.

Theorem 18 (Lemma)**.**
$\boldsymbol{\mu}(m+n) = \boldsymbol{\mu}(m) P^n$. *In particular,* $\boldsymbol{\mu}(n) = \boldsymbol{\mu}(0) P^n$.

Proof. By the law of total probability,

$$\mu_j(m+n) = \mathbb{P}\big[X_{m+n} = j\big] = \sum_i \mathbb{P}\big[X_{m+n} = j \mid X_m = i\big]\, \mathbb{P}\big[X_m = i\big] = \sum_i \mu_i(m)\, p_{ij}(n) = \big(\boldsymbol{\mu}(m) P_n\big)_j,$$

and $P_n = P^n$ by Chapman-Kolmogorov. □

Key point: the random evolution of a Markov chain is entirely determined by

- $\boldsymbol{\mu}(0)$: the **initial conditions**;

- P: the **dynamics**.

In fact, the whole joint law is determined: $\mathbb{P}[X_0 = x_0, X_1 = x_1, \dots, X_n = x_n] = \mu_{x_0}(0)\, p_{x_0x_1} p_{x_1x_2} \cdots p_{x_{n-1}x_n}$ (chain rule + Markov property).

0.26.7 Example: the simple random walk

Definition 36 (Simple random walk on $\mathbb{Z}$)**.**
$\mathcal{S} = \{0, \pm1, \pm2, \pm3, \dots\} = \mathbb{Z}$, and

$$p_{ij} = \begin{cases} p & j = i+1, \\ q = 1-p & j = i-1, \\ 0 & \text{otherwise.} \end{cases}$$

At each step, we move one unit to the right with probability p and one unit to the left with probability q.

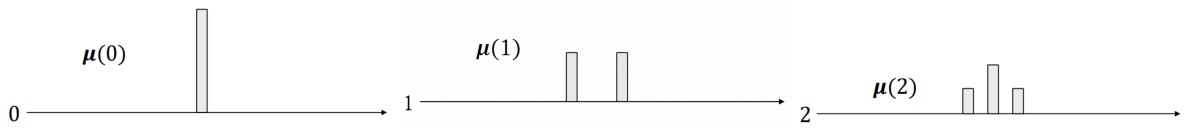
n-step transition probabilities. To go from i to j in n steps, we need to make $j - i$ more steps to the right than to the left. With r right steps and l left steps:

$$r + l = n, \qquad r - l = j - i \quad \Rightarrow \quad r = \tfrac{1}{2}(n + j - i),$$

which must be an integer. There are $\binom{n}{r}$ sequences of length n with r right steps and $l = n - r$ left steps, each with probability $p^r q^l$. Hence

$$p_{ij}(n) = \begin{cases} \binom{n}{\frac{1}{2}(n+j-i)}\, p^{\frac{1}{2}(n+j-i)}\, q^{\frac{1}{2}(n-j+i)} & n + j - i \text{ even}, \\ 0 & \text{otherwise.} \end{cases}$$

Starting from 0, the distribution $\boldsymbol{\mu}(n)$ spreads out over time (and drifts to the right if $p > q$); note the parity constraint: at even times, the walk is at an even position.



0.26.8 Chapman-Kolmogorov $\not\Rightarrow$ Markov

Chapman-Kolmogorov is a **necessary but not sufficient** condition for being a Markov chain. Counterexample:

- Let $Y_1, Y_3, Y_5, \dots$ be an i.i.d. sequence of random variables with values ±1, each with probability $1/2$: $\mathbb{P}[Y_{2k+1} = -1] = \mathbb{P}[Y_{2k+1} = +1] = 1/2$.
- Define the even terms as products of their neighbours: $Y_{2k} = Y_{2k-1}\, Y_{2k+1}$.
- The sequence $Y_2, Y_4, Y_6, \dots$ is also i.i.d. with the same distribution, and adjacent pairs Y_{2k}, Y_{2k+1} are also independent (knowing Y_{2k+1}, the product $Y_{2k-1}Y_{2k+1}$ is still a fair ±1). Therefore the whole sequence is **pairwise independent**.
- Hence, for all $n \geq 1$, $p_{ij}(n) = \mathbb{P}[Y_{m+n} = j \mid Y_m = i] = \frac{1}{2}$: all n-step matrices equal the matrix with all entries $\frac{1}{2}$, which trivially satisfies $P_{m+n} = P_m P_n$. Chapman-Kolmogorov holds.
- But Y is **not** a Markov chain:

$$\mathbb{P}\big[Y_{2k+1} = 1 \mid Y_{2k} = -1\big] = \tfrac{1}{2}, \qquad \text{but} \qquad \mathbb{P}\big[Y_{2k+1} = 1 \mid Y_{2k} = -1, Y_{2k-1} = 1\big] = 0,$$

since $Y_{2k+1} = Y_{2k}/Y_{2k-1} = -1$. Knowing one more past value changes the prediction.

0.27 Classification of states

0.27.1 Recurrence and transience

Definition 37 (Recurrent and transient states)**.**
State i is **recurrent** (or *persistent*) if

$$\mathbb{P}\big[X_n = i \text{ for some } n \geq 1 \mid X_0 = i\big] = 1,$$

i.e. starting from i, the chain returns to i with certainty. State i is **transient** otherwise, i.e. if

$$\mathbb{P}\big[X_n = i \text{ for some } n \geq 1 \mid X_0 = i\big] < 1.$$

Remark 33.
Starting at i is important in the definition: we may never visit state i starting from some $j \neq i$, even though i is recurrent!
By the Markov property, each return to i "restarts" the chain. Hence a recurrent state is visited **infinitely often** (with probability 1), whereas a transient state is visited only a finite number of times (a geometric number of returns, with success probability $1 - f_{ii}$ below): it is eventually abandoned.

Definition 38 (First passage probabilities)**.**
The probability of a **first visit** to state j, starting at state i, in exactly n steps:

$$f_{ij}(n) = \mathbb{P}\big[X_1 \neq j, X_2 \neq j, \ldots, X_{n-1} \neq j, X_n = j \mid X_0 = i\big], \qquad f_{ij}(0) = 0.$$

The probability of **ever** visiting j after i:

$$f_{ij} = \sum_{n=1}^{\infty} f_{ij}(n).$$

The events {first visit at time n} are disjoint, which is why we can sum them. Note:

$$j \text{ is recurrent} \iff f_{jj} = 1.$$

0.27.2 Linking transition probabilities and first passage times

We now relate $p_{ij}(n)$ (easy to compute: powers of P) to $f_{ij}(n)$ (which defines recurrence). The tool is **probability generating functions**.

Definition 39 (Generating functions)**.**
For $|s| < 1$:

$$P_{ij}(s) = \sum_{n=0}^{\infty} s^n p_{ij}(n) \quad (p_{ij}(0) = \delta_{ij}), \qquad F_{ij}(s) = \sum_{n=0}^{\infty} s^n f_{ij}(n) \quad (f_{ij}(0) = 0).$$

Theorem 19.

(a) $P_{ii}(s) = 1 + F_{ii}(s)\, P_{ii}(s)$*;*
(b) $P_{ij}(s) = F_{ij}(s)\, P_{jj}(s)$ *for* $i \neq j$*.*

In one formula: $P_{ij}(s) = F_{ij}(s)\, P_{jj}(s) + \delta_{ij}$ *for any* i, j*.*

Proof. Fix two states i, j, and define for $m \geq 1$ the events

$$A_m = \{X_m = j\}, \qquad B_m = \{X_k \neq j \text{ for } 1 \leq k < m,\ X_m = j\} \quad \text{(first visit to } j \text{ happens at time } m\text{)}.$$

The B_m are disjoint, and $B_m \Rightarrow A_m$. If A_m occurs, the first visit to j happened at some time $r \in \{1, \dots, m\}$, so

$$p_{ij}(m) = \mathbb{P}\big[A_m \mid X_0 = i\big] = \sum_{r=1}^{m} \mathbb{P}\big[A_m, B_r \mid X_0 = i\big].$$

Moreover,

$$\mathbb{P}\big[A_m, B_r \mid X_0 = i\big] = \mathbb{P}\big[A_m \mid B_r, X_0 = i\big]\, \mathbb{P}\big[B_r \mid X_0 = i\big] = \underbrace{\mathbb{P}\big[A_m \mid X_r = j\big]}_{p_{jj}(m-r)}\ \underbrace{\mathbb{P}\big[B_r \mid X_0 = i\big]}_{f_{ij}(r)},$$

where $\mathbb{P}[A_m \mid B_r, X_0 = i] = \mathbb{P}[A_m \mid X_r = j]$ follows from the Markov property: A_m depends only on $X_r = j$, not on what happened before. Hence

$$p_{ij}(m) = \sum_{r=1}^{m} f_{ij}(r)\, p_{jj}(m-r) \qquad (m = 1, 2, \dots).$$

This is a **convolution**, which becomes a **multiplication in the transform domain**:

$$\begin{aligned} P_{ij}(s) &= \sum_{m=1}^{\infty} s^m p_{ij}(m) + \delta_{ij} = \sum_{m=1}^{\infty} s^m \sum_{r=1}^{m} f_{ij}(r)\, p_{jj}(m-r) + \delta_{ij} \\ &= \sum_{m=1}^{\infty} \sum_{r=1}^{m} s^r f_{ij}(r)\ s^{m-r} p_{jj}(m-r) + \delta_{ij} = \sum_{r=1}^{\infty} s^r f_{ij}(r) \sum_{m=r}^{\infty} s^{m-r} p_{jj}(m-r) + \delta_{ij} \\ &= F_{ij}(s)\, P_{jj}(s) + \delta_{ij}, \end{aligned}$$

where we exchanged the two sums (all terms are non-negative). □

0.27.3 Recurrence/transience in terms of n-step transition probabilities

Theorem 20.

- $\sum_n p_{jj}(n) = \infty \iff$ *j is recurrent; in that case* $\sum_n p_{ij}(n) = \infty$ *for all i such that* $f_{ij} > 0$.
- $\sum_n p_{jj}(n) < \infty \iff$ *j is transient; in that case* $\sum_n p_{ij}(n) < \infty$ *for all i.*

In words: the infinite series of n-step *return* probabilities is infinite/finite exactly when the series of n-step transition probabilities from any other state (that can reach j) is infinite/finite. This is the practical criterion: recurrence is defined through f_{jj}, but checked through $p_{jj}(n)$, which can be computed.

Remark 34 (Interpretation)**.**
$\sum_{n\geq 1} p_{jj}(n) = \sum_{n\geq 1} \mathbb{E}\big[\mathbf{1}\{X_n = j\} \mid X_0 = j\big] = \mathbb{E}[\text{number of returns to } j \mid X_0 = j]$. So the theorem says: j is recurrent iff the expected number of returns is infinite, consistent with "a recurrent state is visited infinitely often".

0.27.4 Classification as a function of the recurrence time

Definition 40 (First visit time)**.**

$$T_j = \min\{n \geq 1 : X_n = j\}, \qquad T_j = \infty \text{ if the chain never visits } j.$$

Note: $\mathbb{P}[T_i = \infty \mid X_0 = i] > 0$ iff i is transient, in which case $\mathbb{E}[T_i \mid X_0 = i] = \infty$.

Definition 41 (Mean recurrence time)**.**

$$\mu_i = \mathbb{E}\big[T_i \mid X_0 = i\big] = \begin{cases} \sum_n n f_{ii}(n) & \text{if } i \text{ is recurrent,} \\ \infty & \text{if } i \text{ is transient.} \end{cases}$$

(Careful: the mean recurrence time μ_i is a number; do not confuse it with the marginal distribution $\mu_i(n)$.)

Definition 42 (Null and positive recurrence)**.**
A recurrent state i is called

- **null-recurrent** if $\mu_i = \infty$;
- **positive-recurrent** if $\mu_i < \infty$.

A null-recurrent state is "on the edge": we are sure to come back, but the expected time to come back is infinite.

Theorem 21.
A recurrent state i is null $\iff$ $p_{ii}(n) \to 0$ as $n \to \infty$.

0.27.5 Periodicity

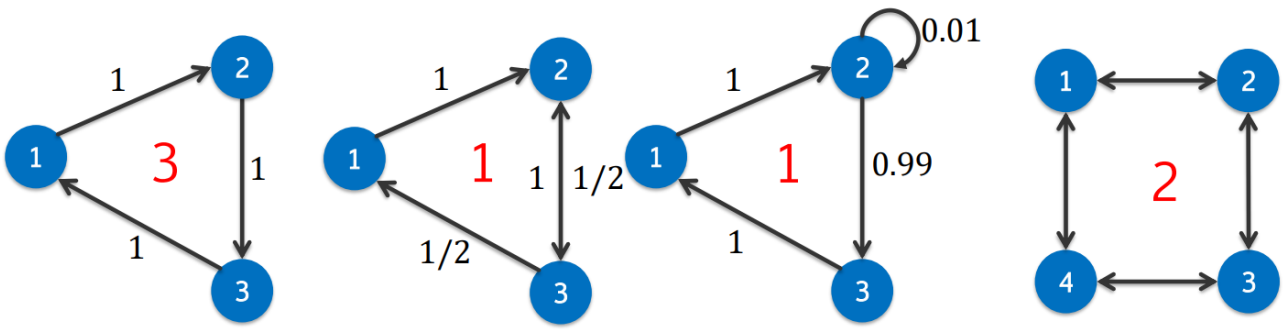
Definition 43 (Period)**.**
The **period** d_i of state i is

$$d_i = \gcd\{n \geq 1 : p_{ii}(n) > 0\}.$$

State i is **periodic** if $d_i > 1$, and **aperiodic** otherwise ($d_i = 1$).

In words: returns to i are only possible at times that are multiples of d_i. Note that $p_{ii}(d_i)$ itself need not be positive: what matters is the gcd of all possible return times. A self-loop ($p_{ii} > 0$) immediately makes i aperiodic.

Definition 44 (Ergodic state)**.**
A state i is **ergodic** if it is positive-recurrent and aperiodic.



Examples (slide 18, from left to right):

1. Deterministic cycle $1 \to 2 \to 3 \to 1$: returns to 1 only at times $3, 6, 9, \ldots \Rightarrow$ period 3.
2. $1 \to 2 \to 3$ with probability 1, then from 3 back to 1 or to 2 with probability $1/2$ each: return times to 1 include 3 ($1 \to 2 \to 3 \to 1$) and 5 ($1 \to 2 \to 3 \to 2 \to 3 \to 1$), and $\gcd(3, 5) = 1 \Rightarrow$ aperiodic.
3. Cycle $1 \to 2 \to 3 \to 1$ with a self-loop of probability 0.01 at state 2: return times 3 and 4, $\gcd(3, 4) = 1 \Rightarrow$ aperiodic. Even a tiny self-loop destroys periodicity (the exact value of the probability does not matter).
4. Square $1 \leftrightarrow 2 \leftrightarrow 3 \leftrightarrow 4 \leftrightarrow 1$ (bidirectional edges): every return takes an even number of steps $\Rightarrow$ period 2.

0.28 Classification of chains

0.28.1 Communicating states

We now look at the relationships of states to each other.

Definition 45 (Communication)**.**

- i **communicates with** j (written $i \to j$) if $p_{ij}(m) > 0$ for some $m \geq 0$ (j is reachable from i).
- i **intercommunicates with** j (written $i \leftrightarrow j$) if $i \to j$ and $j \to i$.

Properties:

(a) For $i \neq j$: $i \to j \iff f_{ij} > 0$.
(b) For $i = j$: $i \to i$ always, because $p_{ii}(0) = 1$.
(c) $i \leftrightarrow j$ and $j \leftrightarrow k \Rightarrow i \leftrightarrow k$ (by Chapman-Kolmogorov, $p_{ik}(m + n) \geq p_{ij}(m)p_{jk}(n) > 0$).

Hence "$\leftrightarrow$" is reflexive, symmetric and transitive: it is an **equivalence relation**, which **partitions $\mathcal{S}$ into equivalence classes** of states (the communicating classes, i.e. the strongly connected components of the transition graph).

0.28.2 Properties of classes of states

Theorem 22.
If $i \leftrightarrow j$, then:
(a) i and j have the same period;
(b) i is transient $\iff$ j is transient;
(c) i is null-recurrent $\iff$ j is null-recurrent;
(d) i is positive-recurrent $\iff$ j is positive-recurrent.
*In other words, periodicity, transience, null-recurrence and positive recurrence are **class properties**.*

Proof of (b). If $i \leftrightarrow j$, there exist $m, n \geq 0$ such that

$$\alpha = p_{ij}(m)\, p_{ji}(n) > 0.$$

By Chapman-Kolmogorov (keeping only one term of the sum: go from i to j in m steps, loop from j to j in r steps, come back to i in n steps),

$$p_{ii}(m + r + n) \geq p_{ij}(m)\, p_{jj}(r)\, p_{ji}(n) = \alpha\, p_{jj}(r) \qquad \text{for any } r \geq 0.$$

Summing over r:

$$\sum_r p_{ii}(r) < \infty \;\Rightarrow\; \sum_r p_{jj}(r) \leq \frac{1}{\alpha} \sum_r p_{ii}(m + r + n) < \infty.$$

In other words, i transient implies j transient. By symmetry, j transient implies i transient. □

0.28.3 Closed and irreducible sets of states

Definition 46 (Closed and irreducible sets)**.**
A set $C \subset \mathcal{S}$ of states is called

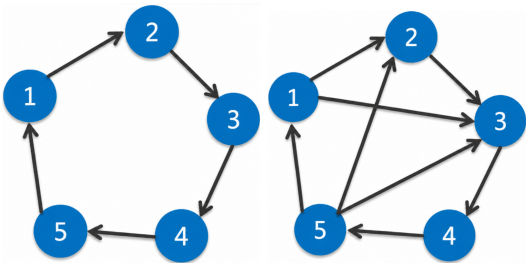
(a) **closed** (or **absorbing**) if $p_{ij} = 0$ for all $i \in C$, $j \notin C$: "we can never leave";
(b) **irreducible** if $i \leftrightarrow j$ for all $i, j \in C$: "we can move freely inside C".

Because of the preceding theorem, all states in an irreducible set share the same properties (periodic/aperiodic, and transient/null-recurrent/positive-recurrent). If the whole $\mathcal{S}$ is irreducible, we say that the **chain** has these shared properties (e.g. "an irreducible, aperiodic, positive-recurrent chain").

Remark 35 (How to classify states in practice)**.**
Arrows represent any positive probability: the exact values do not matter to classify states (except to distinguish null from positive recurrence in infinite chains, and recurrence from transience in infinite chains).

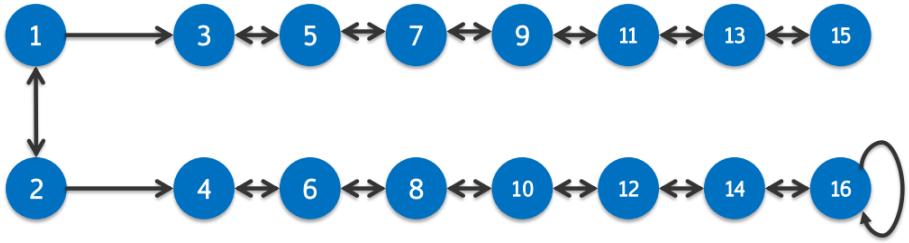
1. Draw the transition graph and find the communicating classes (groups of states that can reach each other).
2. A class from which an arrow leaves is **not closed**; in a finite chain, it is **transient**. A closed finite class is positive-recurrent.
3. For each class, compute the period: gcd of the lengths of the cycles through one of its states (a self-loop $\Rightarrow$ aperiodic).

0.28.4 Examples



- **Left**: the cycle $1 \to 2 \to 3 \to 4 \to 5 \to 1$. The entire $\mathcal{S}$ is one closed irreducible finite class $\Rightarrow$ positive-recurrent; every return takes a multiple of 5 steps $\Rightarrow$ **periodic with period** 5.
- **Right**: the same cycle with extra arrows ($1 \to 3$, $5 \to 2$, $5 \to 3$, …). Still irreducible and finite $\Rightarrow$ positive-recurrent; but now there are cycles of different lengths whose lengths have gcd 1 (e.g. $1 \to 3 \to 4 \to 5 \to 1$ of length 4 and the original cycle of length 5) $\Rightarrow$ **aperiodic**.

Exercise: what are the irreducible classes of the following chain?



- $\{1, 2\}$: **transient**, **2-periodic**. It is not closed (we can leave to 3 or 4 and never come back); returns to 1 only via $1 \to 2 \to 1$, i.e. at even times.
- $\{3, 5, \ldots, 15\}$: **recurrent**, **2-periodic**, **absorbing**. A line with bidirectional arrows is bipartite (odd/even positions along the line), so returns take an even number of steps.
- $\{4, 6, \ldots, 16\}$: **recurrent**, **aperiodic**, **absorbing**. Same structure, but the self-loop at 16 breaks the periodicity.

0.28.5 Examples: random walks

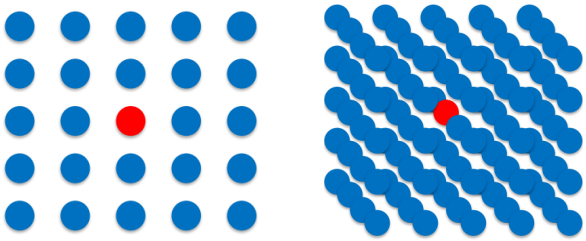
- **Simple random walk on** $\mathbb{Z}$ (irreducible, period 2):
  - for $p = q = \frac{1}{2}$ (unbiased): **null-recurrent** (even though the mode of $\mathbb{P}[X_n = i]$ is always at 0!);
  - for $p \neq q$: **transient** (the walk drifts to $\pm\infty$).



- **Simple random walk with a (reflecting) barrier at** 0 ($\mathcal{S} = \{0, 1, 2, \dots\}$):
  - for $p = q = \frac{1}{2}$ (unbiased): **null-recurrent**;
  - for $p > q$: **transient** (drift away from the barrier, to $+\infty$);
  - for $p < q$: **positive-recurrent** (drift towards the barrier, the walk keeps coming back quickly).



- **Unbiased random walk on** $\mathbb{Z}^2$ (move to one of the 4 neighbours uniformly): **null-recurrent**.
- **Unbiased random walk on** $\mathbb{Z}^3$ (one of the 6 neighbours): **transient**.



Remark 36 (Why? (not on the slides))**.**
Use the criterion $\sum_n p_{00}(n) = \infty \iff$ *null* recurrent. On $\mathbb{Z}$, returns happen only at even times and $p_{00}(2n) = \binom{2n}{n}(pq)^n$. By Stirling, $\binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi n}}$, so

$$p_{00}(2n) \sim \frac{(4pq)^n}{\sqrt{\pi n}}.$$

- If $p = q = \frac{1}{2}$: $4pq = 1$ and $p_{00}(2n) \sim \frac{1}{\sqrt{\pi n}}$, whose sum diverges $\Rightarrow$ recurrent; but $p_{00}(2n) \to 0$ $\Rightarrow$ null-recurrent (by the theorem "null $\iff$ $p_{ii}(n) \to 0$").
- If $p \neq q$: $4pq < 1$, the terms decay geometrically, the sum converges $\Rightarrow$ transient.

On $\mathbb{Z}^d$ (unbiased), $p_{00}(2n) \asymp n^{-d/2}$: the sum diverges for $d = 1, 2$ (recurrent, and null since the terms go to 0) and converges for $d \geq 3$ (transient). A drunk man finds his way home, a drunk bird may not (Pólya).

0.28.6 Decomposition theorem

Theorem 23 (Decomposition theorem)**.**
The state space $\mathcal{S}$ can be partitioned uniquely as

$$\mathcal{S} = T \cup C_1 \cup C_2 \cup \cdots$$

where T is the set of ***transient*** *states, and the C_k are* ***irreducible closed sets of recurrent states****.*

Note: we do not assert that T is irreducible! T may be made of several classes (or of states that do not intercommunicate at all).

Proof. Let $C_1, C_2, \ldots$ be the recurrent equivalence classes of $\leftrightarrow$ (each class is either entirely recurrent or entirely transient, by the class-property theorem). They are irreducible by definition; we only need to show that they are all closed.

Suppose the contrary: there exist $i \in C_r$ and $j \notin C_r$ such that $p_{ij} > 0$. Then $j \not\to i$ (otherwise i and j would intercommunicate and would be in the same class). So once the chain jumps from i to j, it can never come back to i:

$$\mathbb{P}\big[X_n \neq i \text{ for all } n \geq 1 \mid X_0 = i\big] \geq \mathbb{P}\big[X_1 = j \mid X_0 = i\big] = p_{ij} > 0,$$

which contradicts the assumption that i is recurrent. □

So there are two possibilities for the trajectory of the chain:

- either the chain stays in T forever,
- or it is (eventually, or from the beginning) in one of the C_r, and stays there forever.

For finite $\mathcal{S}$, only the second scenario is possible (a finite set of transient states, each visited finitely often, cannot hold the chain forever).

0.28.7 Decomposition for finite $\mathcal{S}$

If $\mathcal{S}$ is finite:

- there are **no null-recurrent states** (null recurrence only happens in infinite state spaces);
- **at least one** state is positive-recurrent;
- a class of states is positive-recurrent $\iff$ the class is absorbing (closed);
- a class of states is transient $\iff$ the class is non-absorbing;
- there exists at least one class of absorbing states;
- if the whole chain is irreducible, it is positive-recurrent.

0.29 Limit behaviour

0.29.1 Ergodic theorem

Periodicity is problematic for convergence. Example: $\mathcal{S} = \{1, 2\}$ with $p_{12} = p_{21} = 1$. Then

$$p_{11}(n) = p_{22}(n) = \begin{cases} 0 & n \text{ odd}, \\ 1 & n \text{ even}, \end{cases}$$

which does not converge!

Theorem 24 (Ergodic theorem)**.**
For an ***irreducible aperiodic*** *chain (not necessarily finite),*

$$p_{ij}(n) \xrightarrow[n\to\infty]{} \frac{1}{\mu_j} =: \pi_j \qquad \text{for all } i \text{ and } j,$$

where μ_j is the mean recurrence time of j (with $1/\infty = 0$).

Interpretation: the long-term probability $\pi_j = p_{ij}(\infty)$ of being in state j "from anywhere" (it does not depend on i: the chain forgets where it started) converges to the **rate of revisiting that state**. If, on average, we come back to j every μ_j steps, then in the long run we spend a fraction $1/\mu_j$ of the time in j.

- If the chain is positive-recurrent (ergodic), $\pi_j > 0$ for all j, and π is a probability distribution.
- If the chain is null-recurrent or transient, $\mu_j = \infty$ and $p_{ij}(n) \to 0$ for all i, j: the probability mass "escapes to infinity" (only possible in infinite state spaces).

Can we compute that rate? Yes!

0.29.2 Stationary distribution π

Definition 47 (Stationary distribution)**.**
A distribution π (row vector, $\pi_j \geq 0$, $\sum_j \pi_j = 1$) is **stationary** if

$$\pi = \pi P.$$

If $\boldsymbol{\mu}(0) = \pi$, then $\boldsymbol{\mu}(n) = \pi P^n = \pi$ for all n: the distribution of X_n no longer changes.

The limit of the ergodic theorem is stationary: letting $n \to \infty$ in $P^{n+1} = P^n P$, every row π of P^∞ satisfies $\pi = \pi P$.

Theorem 25 (Spectral view, finite chains)**.**
For an irreducible aperiodic finite Markov chain, the transition matrix $P \in \mathbb{R}^{d\times d}$:

- *has a single dominant eigenvalue $\lambda_1 = 1$; its associated (left) eigenvector is (proportional to) the stationary distribution π;*
- *all other eigenvalues satisfy $|\lambda_2|, |\lambda_3|, \ldots, |\lambda_d| < 1$.*

Note that 1 is always an eigenvalue of a stochastic matrix: $P\mathbf{1}^\top = \mathbf{1}^\top$ (rows sum to one), so it is a right eigenvector; π is the corresponding **left** eigenvector. The other eigenvalues having modulus < 1 is what makes P^n converge (the components along them decay like $|\lambda_k|^n$; the speed of convergence is governed by $|\lambda_2|$).

Computing the unique stationary distribution:

- solve the linear system $\pi = \pi P$ together with $\pi\mathbf{1}^\top = 1$ (normalisation; the system $\pi = \pi P$ alone only determines π up to a constant);
- for large systems, use the **power method**: take a vector $e = [1, 1, \ldots, 1]/d$ (for example), and repeatedly multiply by P: $eP^n \to \pi$. This is exactly simulating the evolution of $\boldsymbol{\mu}(n)$ from $\boldsymbol{\mu}(0) = e$.

Remark 37 (Back to the two-state example)**.**
Recall $P = \begin{pmatrix} 1/2 & 1/2 \\ 3/4 & 1/4 \end{pmatrix}$ and $P^{10} \approx \begin{pmatrix} 0.600 & 0.400 \\ 0.600 & 0.400 \end{pmatrix}$. Solving $\pi = \pi P$: $\pi_1 = \frac{1}{2}\pi_1 + \frac{3}{4}\pi_2 \Rightarrow \pi_1 = \frac{3}{2}\pi_2$, and with $\pi_1 + \pi_2 = 1$:

$$\pi = \big[3/5,\ 2/5\big],$$

which matches the rows of P^{10}. The eigenvalues of P are 1 and $\mathrm{tr}(P) - 1 = -\frac{1}{4}$, so P^n converges to its limit like $(1/4)^n$: very fast. Consistently with the ergodic theorem, the mean recurrence times are $\mu_1 = 5/3$ and $\mu_2 = 5/2$.

0.30 Random walks on general graphs

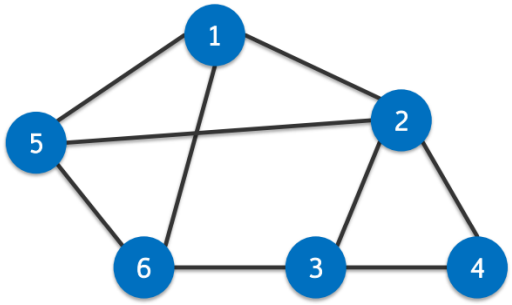
0.30.1 Random walk as a Markov chain

Definition 48 (Random walk on a graph)**.**
Let $G(V, E)$ be an undirected graph, assumed connected (otherwise, take G to be the giant component of the actual network). The **random walk** on G is the Markov chain with state space $\mathcal{S} = V$:

- $X_n \in V$ is the node at time n;
- at each time step, go to a neighbour of X_n chosen uniformly at random $\to X_{n+1}$.

It is a random memoryless exploration of the graph. With d_i the degree of node i, the transition matrix is

$$p_{ij} = \begin{cases} 1/d_i & (i,j) \in E, \\ 0 & \text{otherwise.} \end{cases}$$



If $G(V, E)$ is **undirected, connected and non-bipartite**, then $\{X_n\}$ is an **ergodic** (irreducible, aperiodic) Markov chain:

- connected $\Rightarrow$ irreducible (every node can reach every other node);
- non-bipartite $\Rightarrow$ aperiodic: on an undirected graph, $i \to j \to i$ gives returns of length 2; a bipartite graph has only even cycles (period 2), whereas a non-bipartite graph has an odd cycle, giving an odd return length and hence gcd = 1;
- finite and irreducible $\Rightarrow$ positive-recurrent.

Ergodicity then gives a stationary distribution π with

$$p_{ij}(n) = \mathbb{P}(\text{at vertex } j \text{ after } n \text{ steps} \mid \text{starting at vertex } i) \xrightarrow[n\to\infty]{} \pi_j \qquad \text{for all } i, j \in V :$$

the random walk "forgets" its starting point i.

0.30.2 Stationary distribution of a random walk

Theorem 26 (Lemma)**.**
For the random walk on an undirected connected graph,

$$\pi \propto [d_1, d_2, \ldots, d_n], \qquad \textit{i.e.} \qquad \pi_j = \frac{d_j}{\sum_i d_i} = \frac{d_j}{2|E|}.$$

Proof. By definition, π is stationary iff $\pi = \pi P$. Let $x = [d_1, d_2, \ldots, d_n]\, P$. Then

$$x_j = \sum_i d_i\, p_{ij} = \sum_i d_i \cdot \frac{1}{d_i} \mathbf{1}\{(i,j) \in E\} = \sum_i \mathbf{1}\{(i,j) \in E\} = d_j.$$

So $[d_1, \ldots, d_n]$ is a left eigenvector of P with eigenvalue 1, hence $\pi \propto [d_1, \ldots, d_n]$ (by uniqueness of the stationary distribution of an ergodic chain); normalising uses $\sum_i d_i = 2|E|$. □

Intuition:

- in the stationary regime, the random walk "sees" uniformly random edges (each of the $2|E|$ directed edges is traversed with the same frequency);
- the nodes visited by the random walk are therefore biased by their number of edges, i.e. their degree: high-degree nodes are visited more often.

Remark 38.
This compact characterisation is specific to **undirected** graphs: no such simple formula for π exists for directed graphs (in general, π must be computed numerically, e.g. with the power method). Sampling nodes through a random walk is a common way to explore large networks, but it gives a degree-biased sample.